

# Notes of Masterclass on the proof of Geometric Langlands Correspondence

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# 1 Day 1 Fun(ctor) Day

## 1.1 All you need to know about $\text{Bun}_G$ (Justin Campbell)

Throughout the lectures, we consider the base field  $\mathbf{k}$  to be algebraically closed and of characteristic zero. We let  $G$  be a connected reductive group over  $\mathbf{k}$  and  $\mathbf{B}G = \text{pt}/G$  be the classifying stack of  $G$ . More concretely from Yoneda  $\text{pt}/G$  can be described by the following sheaf of groupoid

$$\text{Maps}(S, \text{pt}/G) = \langle (\text{étale locally trivial}) \text{ principal } G\text{-torsors on } S \rangle, \forall S \text{ affine scheme}$$

**Example 1.1.**  $\text{pt}/\text{GL}_n$  classifies rank  $n$  vector bundles.

Next input is  $X$ , a smooth connected projective curve over  $\mathbf{k}$ .

**Definition 1.1.** We define  $\text{Bun}_G$  on  $X$  to be  $\text{Maps}(X, \text{pt}/G)$ , where we use  $\text{Maps}$  to denote the mapping stack. More concretely,  $\text{Maps}(S, \text{Bun}_G) = \langle G\text{-bundles on } X_S \rangle$ .

Also we define  $\text{Bun}_G$  as a functor, it has nice properties:

**Proposition 1.1.**  $\text{Bun}_G$  is an algebraic stack locally of finite type over  $\mathbf{k}$ . More precisely:

- (i) The diagonal  $\Delta : \text{Bun}_G \rightarrow \text{Bun}_G \times \text{Bun}_G$  is affine.
- (ii) There exists scheme  $U$  locally of finite type over  $\mathbf{k}$  and a smooth surjection  $U \rightarrow \text{Bun}_G$  (i.e. an "atlas")

**Proposition 1.2.**  $\text{Bun}_G$  is smooth.

*Proof.* (Sketch) An algebraic stack which is locally of finite type admits a tangent complex, which (for nice stacks) is a perfect complex. Such a stack is smooth is equivalent to say its tangent complex is connective (i.e. concentrated in non-positive cohomological degree).

We can actually calculate the stalk of the tangent complex of  $\text{Bun}_G$  at a  $G$ -bundle  $\mathcal{P}_G \in \text{Bun}_G$ , which is

$$R\Gamma(X, \mathfrak{g} \times^G \mathcal{P}_G)[1]$$

However, we know that for any coherent sheaf on  $X$ ,  $R^i\Gamma(X, \mathcal{F}) = 0$  for  $\forall i > 1$  since  $\dim X = 1$ . ■

**Remark 1.1.** We can see none of the above proof uses the assumption that  $G$  is reductive, we can prove similar properties for affine group schemes.

Caveat is the following: for  $G$  a non-trivial reductive group,  $\text{Bun}_G$  is never quasi-compact. For example,  $\pi_0(\text{Bun}_{\text{GL}_m}) = \mathbb{Z} = \pi_1(\text{GL}_m)$  has infinitely many connected components. Also for  $G$  nonabelian, the connected components of  $\text{Bun}_G$  are not quasi-compact. For example, we can consider  $G = \text{GL}_2$ , then we have specializations

$$\mathcal{O}^{\oplus 2} \xrightarrow{\text{special.}} \mathcal{O}(-1) \oplus \mathcal{O}(1) \rightsquigarrow \mathcal{O}(-2) \oplus \mathcal{O}(2) \rightsquigarrow \dots$$

Next, we are going to talk about Hecke actions. We fix a point  $x \in X(\mathbf{k})$ . Let  $\hat{\mathcal{O}}_x$  be the complete local ring at  $x$  and  $\hat{K}_x := \text{Frac}(\hat{\mathcal{O}}_x)$  the fractional field. We also let  $D_{x,S} := \text{Spec}(\mathcal{O}_S \hat{\otimes} \hat{\mathcal{O}}_x)$  and similarly the punctured disc  $\mathring{D}_{x,S} := \text{Spec}(\mathcal{O}_S \hat{\otimes} \hat{K}_x) = D_{x,S} \setminus (S \times \{x\})$ , we have the following theorem:

**Theorem 1.1** (Beauville-Laszlo). *There is an equivalence of groupoids:*

$$\mathrm{Maps}(S, \mathrm{Bun}_G) \cong \left\langle (\mathcal{P}'_G, \mathcal{P}''_G, \alpha) \left| \begin{array}{l} \mathcal{P}'_G : G\text{-torsor on } D_{x,S}, \mathcal{P}''_G : G\text{-torsor on } S \times (X \setminus \{x\}) \\ \alpha \text{ an isomorphism of them on punctured disc } \mathring{D}_{x,S}. \end{array} \right. \right\rangle$$

Next, we can define the (local) Hecke stack, which is a stack that classifies Hecke modifications at point  $x$ .

**Definition 1.2.** Let  $\mathcal{H}_{G,x}$  be the local Hecke stack, which is:

$$\mathrm{Maps}(S, \mathcal{H}_{G,x}) = \left\langle (\mathcal{P}^1_G, \mathcal{P}^2_G, \alpha) \left| \begin{array}{l} \mathcal{P}^1_G, \mathcal{P}^2_G : G\text{-torsor on } D_{x,S} \\ \alpha : \mathcal{P}^1_G|_{\mathring{D}_{x,S}} \xrightarrow{\cong} \mathcal{P}^2_G|_{\mathring{D}_{x,S}}. \end{array} \right. \right\rangle$$

We can see easily from the definition that  $\mathcal{H}_{G,x}$  is a groupoid over  $\mathrm{Bun}_G(D_x)$ , since we can compose those  $\alpha$ 's. Combining Beauville-Laszlo theorem 1.1, we obtain an action of  $\mathcal{H}_{G,x}$  on  $\mathrm{Bun}_G$  through  $\mathrm{Bun}_G(D_x)$ . More precisely, we have  $\mathcal{H}_{G,x} \times_{\mathrm{Bun}_G(D_x)} \mathrm{Bun}_G \rightarrow \mathrm{Bun}_G$ .

We can use uniformization to write  $\mathcal{H}_{G,x}$  into a double quotient, let

$$\begin{aligned} \mathcal{L}_x G &= \text{loop group of } G \text{ at } x \supseteq \mathcal{L}_x^+ G = \text{arc group of } G \text{ at } x, \text{ where} \\ \mathrm{Maps}(S, \mathcal{L}_x G) &= \mathrm{Maps}(D_{x,S}, G); \quad \mathrm{Maps}(S, \mathcal{L}_x^+ G) = \mathrm{Maps}(\mathring{D}_{x,S}, G) \end{aligned}$$

**Proposition 1.3.** (i) *The arc group  $\mathcal{L}_x^+ G$  is an affine group scheme (but not of finite type).*

(ii) *The loop group  $\mathcal{L}_x G$  is an ind-affine group scheme.*

Moreover, an observation is that locally on  $S$ , any  $G$ -bundle on  $D_{x,S}$  descends to  $S$ . As a result, we can see that  $\mathrm{Bun}_G(D_x)$ , which is the stack to classifies  $G$ -bundle on the disc, is isomorphic to  $\mathrm{pt} / \mathcal{L}_x^+ G$ , where  $\mathcal{L}_x^+ G$  is the automorphism group. The same is also true for those  $G$ -bundles on punctured disc  $\mathring{D}_{x,S}$  which extend to  $D_{x,S}$ . As a result, according to the definition of  $\mathcal{H}_{G,x}$ , we can rewrite it as

$$\mathcal{H}_{G,x} = \mathrm{Bun}_G(D_x) \times_{\mathrm{Bun}_G(\mathring{D}_x)} \mathrm{Bun}_G(D_x) \cong \mathrm{pt} / \mathcal{L}_x^+ G \times_{\mathrm{pt} / \mathcal{L}_x G} \mathrm{pt} / \mathcal{L}_x^+ G \cong \mathcal{L}_x^+ G \backslash \mathcal{L}_x G / \mathcal{L}_x^+ G$$

Finally, we recall the definition of affine grassmannian:

$$\mathrm{Gr}_{G,x} := \mathcal{L}_x G / \mathcal{L}_x^+ G$$

This is an ind-proper ind-scheme, namely  $\mathrm{Gr}_{G,x} = \varinjlim_N X_N$  as a colimit of proper schemes. We can see

$$\mathcal{H}_{G,x} = \mathcal{L}_x^+ G \backslash \mathrm{Gr}_{G,x}$$

## 1.2 All you need to know about LocSys (Dima Arinkin)

We work over the same assumption before, instead of  $G$ -bundles, we define the stack of  $G$ -local systems  $\mathrm{LocSys}$  (we also use the notation  $\mathrm{LS}$  in the sequel). Roughly, it is

**Definition 1.3.**  $\mathrm{LocSys}_G = \langle G\text{-local systems on } X \rangle$

However, there are basically two ways to understand local systems. The first approach is: a local system is a  $G$ -bundle with an (integrable) connection.

**Example 1.2.** For example if  $G = \mathrm{GL}_n$ , we can look at the pair  $(\mathcal{E}, \nabla)$ , where  $\mathcal{E}$  is a rank  $n$  vector bundle on  $X$  and  $\nabla : \mathcal{E} \rightarrow \mathcal{E} \otimes \Omega_X$  satisfying  $\nabla(f \cdot s) = f\nabla(s) + s \otimes df$ .

We have the following properties of the stack of local systems  $\mathrm{LocSys}$  :

**Properties.** •  $\mathrm{LocSys}_G$  is a derived algebraic stack of finite type, which is quasi-smooth (=derived lci), in other words, its cotangent complex is concentrated in degree  $-1$  and above.

- Consider the forgetful map  $\mathrm{LocSys}_G \rightarrow \mathrm{Bun}_G : (\mathcal{E}, \nabla) \mapsto \mathcal{E}$ , for fixed  $\mathcal{E}$ , those  $\nabla$ 's form an affine space.

**Remark 1.2.** (i) One thing to notice is that from 1.1, we know  $\mathrm{Bun}_G$  is NOT quasi-compact, however  $\mathrm{LocSys}_G$  is of finite type, which in particular means it is quasi-compact. This tells us that not every  $G$ -bundle can be lifted to a  $G$ -local systems. There exists  $G$ -bundles do not have the capability to support a connection.

(ii) Why  $\mathrm{LocSys}_G$  is quasi-smooth, while  $\mathrm{Bun}_G$  is smooth? It's because if we compute fiberwise the tangent space, then  $H^i(X, \text{vector bundle}) = 0$  for  $i > 1$ , but  $H_{\mathrm{dR}}^i(X, \text{local system}) = 0$  for  $i > 2$ .

**Example 1.3.** We take for example  $G = \mathrm{GL}_2$  and the underlying curve  $X = \mathbb{P}^1$ , recall that we have the following specialization at connected components of  $\mathrm{Bun}_G$ :

$$\begin{aligned} \mathcal{O}^{\oplus 2} &\overset{\text{special.}}{\rightsquigarrow} \mathcal{O}(-1) \oplus \mathcal{O}(1) \rightsquigarrow \mathcal{O}(-2) \oplus \mathcal{O}(2) \rightsquigarrow \dots \\ \mathcal{O}(-1) \oplus \mathcal{O} &\rightsquigarrow \mathcal{O}(-2) \oplus \mathcal{O}(1) \rightsquigarrow \mathcal{O}(-3) \oplus \mathcal{O}(2) \dots \end{aligned}$$

We know that the infinite sequence of specializations cause the non-quasicompactness of  $\mathrm{Bun}_G$ , we can ask the following question: which vector bundles admit a connection? One way to cheat: because we consider  $\mathbb{P}^1$ , who is simply-connected, thus the only bundle with connections is the trivial bundle  $\mathcal{O}^{\oplus 2}$ . All other bundles are exiled from our discussion of  $G$ -local systems.

Then we can ask what should be the connections on  $\mathcal{O}^{\oplus 2}$ ? In coordinates, it should be in the form

$$d + \begin{pmatrix} a_{11}(x) & a_{12}(x) \\ a_{21}(x) & a_{22}(x) \end{pmatrix} dx$$

where  $a_{ij}(x)$  are polynomials written as  $a_{ij}(x) = a_{ij}^{(d)}x^d + a_{ij}^{(d-1)}x^{d-1} + \dots + a_{ij}^{(0)}$ . And we require that under the change of coordinates  $x \sim \frac{1}{z}$ , they have no poles. we can rewrite  $a_{ij}dx$  as

$$\left( -\frac{a_{ij}^{(d)}}{z^{d+2}} - \frac{a_{ij}^{(d-1)}}{z^{d+1}} - \dots - \frac{0}{z} \right) dz$$

However, the condition of no-poles forces  $a_{ij}^{(d)} = 0$  and gives us also a redundant equation  $0 = 0$ . This is the place that derived structure comes into play. The answer for description of the stack  $\mathrm{LocSys}_{\mathrm{GL}_2, \mathbb{P}^1}$ , or more generally  $\mathrm{LocSys}_{G, \mathbb{P}^1}$  is

$$\mathrm{LocSys}_{G, \mathbb{P}^1} = (\mathrm{pt} \times_{\mathfrak{g}} \mathrm{pt}) / G = \Omega \mathfrak{g} / G$$

The second approach to define the stack of local system as  $G$ -torsors on  $X_{\text{dR}}$  (recall  $\text{Bun}_G = \mathbf{Maps}(X, \mathbf{BG})$ ).

**Definition 1.4.** We define the Simpson shape  $X_{\text{dR}}$  and  $\text{LocSys}_G$  as the following:

$$\begin{aligned} X_{\text{dR}}(S) &= \text{Maps}(S^{\text{red}}, X); \\ \text{LocSys}_G &= \mathbf{Maps}(X_{\text{dR}}, \text{pt}/G). \end{aligned}$$

where  $X_{\text{dR}}$  formalizes the parallel transports. In other words, if we want to test the maps from  $S$  to  $X$ , the parallel transports allow us to transfer from the reduced part of  $S$  to non-reduced part. Next, we introduce the **Ran** space (of  $X$ ), which basically classifies those finite non-empty subsets of  $X_{\text{dR}}$

**Definition 1.5.** We define  $\text{Ran}(X)$  to be a presheaf such that

$$\text{Maps}(S, \text{Ran}(X)) = \{ \text{finite, non-empty subset of } \text{Maps}(S, X_{\text{dR}}) = \text{Maps}(S^{\text{red}}, X) \}$$

Less formally, we have maps  $X_{\text{dR}}^n \rightarrow \text{Ran}(X)$  given by  $X_{\text{dR}}^n \rightarrow X_{\text{dR}}^m$  for any surjective map  $\{1, \dots, m\} \rightarrow \{1, \dots, n\}$ .

**Proposition 1.4.** We have  $\text{Ran}(X) = \text{colim}_n X_{\text{dR}}^n$ .

We can ask, what is a quasi-coherent sheaf on  $\text{Ran}(X)$ ? First, we need data of a quasi-coherent sheaf on  $X_{\text{dR}}$ , which is a quasi-coherent sheaf on  $X$  with a connection, i.e. a  $\mathcal{D}$ -module on  $X$ .

Second, we need a  $\mathcal{D}_{X \times X}$ -module which is  $S_2$ -equivariant. Also, we want the restriction on the diagonal  $\Delta_X : X \rightarrow X \times X$  agrees with the  $\mathcal{D}_X$ -module before. Then we can go to further copies of  $X$ , and they should have nice compatibilities. Now we can define an important piece in GLC picture, recall  $\text{Rep}(G) \cong \text{QCoh}(\text{pt}/G)$ .

**Definition 1.6.** We define the Ran version of  $\text{Rep}(G)$  to be

$$\begin{aligned} \text{Rep}(G)_{\text{Ran}} &= \text{QCoh}((\text{pt}/G)_{\text{Ran}}), \text{ where} \\ (\text{pt}/G)_{\text{Ran}} &= \langle (\underline{x} \in \text{Ran}(X), G\text{-local system on } D_{\underline{x}}) \rangle \end{aligned}$$

### 1.3 Construction of (coarse) Langlands functor (Sam Raskin)

**Overview:** What we want is the following commutative diagram

$$\begin{array}{ccc} & & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \\ & \nearrow \exists \mathbb{L}_G & \downarrow \\ \mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G, \text{coarse}}} & \text{QCoh}(\text{LocSys}_{\check{G}}) \end{array}$$

The goal of geometric Langlands is to construct the functor  $\mathbb{L}_G$  makes the above diagram commute and such that  $\mathbb{L}_G$  is an equivalence. The two ingredients in the construction of  $\mathbb{L}_{G, \text{coarse}}$  are:

- ① A compact object  $\text{Poinc}_! = \text{Poinc}_!^{\text{vac}} \in \mathcal{D}\text{-mod}(\text{Bun}_G)$ , called the (vacuum) Poincaré sheaf.

② The spectral action of  $\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}) \circ \mathcal{D}\text{-mod}(\mathrm{Bun}_G)$ , which is related to the Hecke action.

We denote the functor  $\mathrm{Hom}(\mathrm{Poinc}_!, -)$  by  $\mathrm{coeff}(-) : \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \rightarrow \mathbf{Vect}$ , this is actually the Whittaker coefficient. To be a compact object means  $\mathrm{coeff}(-) = \mathrm{Hom}(\mathrm{Poinc}_!, -)$  commutes with all colimits.

Assume  $\mathcal{Y}$  is an algebraic stack with certain nice properties, then we have a variant on Yoneda Lemma for the category  $\mathrm{QCoh}(\mathcal{Y})$ . Let  $\mathcal{G} \in \mathrm{QCoh}(\mathcal{Y})$ , we can formulate a (continuous) functor

$$F_{\mathcal{G}} : \mathrm{QCoh}(\mathcal{Y}) \rightarrow \mathbf{Vect} \mathcal{H} \mapsto R\Gamma(\mathcal{Y}, \mathcal{G} \otimes_{\mathcal{O}_{\mathcal{Y}}} \mathcal{H})$$

And the claim (comes from Yoneda) is that this gives us an equivalence

$$\mathrm{QCoh}(\mathcal{Y}) \longrightarrow \mathrm{Func}_{\mathbf{k}\text{-linear}}^{\mathrm{cont}}(\mathrm{QCoh}(\mathcal{Y}), \mathbf{Vect}); \quad \mathcal{G} \mapsto F_{\mathcal{G}}$$

To be precise, the additional condition (very mild) put on  $\mathcal{Y}$  is that  $\mathcal{Y}$  quasi-compact with affine diagonal, making  $\mathrm{QCoh}(\mathcal{Y})$  dualizable, which is true for  $\mathrm{LocSys}_{\check{G}}$ .

To define the coarse Langlands functor  $\mathbb{L}_{G,\mathrm{coarse}}$ , we use the dual Yoneda:

$$R\Gamma(\mathrm{LocSys}_{\check{G}}, \mathcal{G} \otimes \mathbb{L}_{G,\mathrm{coarse}}(F)) = \mathrm{coeff}(\mathcal{G} \star F)$$

where  $\star$  is the spectral action in point (2) above. One of the basic axioms of geometric Langlands correspondence is that  $\mathbb{L}_{G,\mathrm{coarse}} : \mathrm{Poinc}_! \mapsto \mathcal{O}_{\mathrm{LocSys}_{\check{G}}}$ . The construction of  $\mathbb{L}_{G,\mathrm{coarse}}$  is forced by this requirement:

**Proposition 1.5.** *The functor  $\mathbb{L}_{G,\mathrm{coarse}}$  is the unique functor that makes the following diagram commutative:*

$$\begin{array}{ccc} \mathcal{D}\text{-mod}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}) \\ & \searrow \mathrm{coeff} & \swarrow R\Gamma(\mathrm{LocSys}_{\check{G}}, -) \\ & \mathbf{Vect} & \end{array}$$

Now we construct the Poincaré sheaf. We fix the Borel  $B = TN \subseteq G$  with unipotent subgroup  $N$ . We have projection map  $\mathrm{Bun}_B \rightarrow \mathrm{Bun}_T$  and we denote by  $\mathrm{Bun}_N^{\Omega_X}$  the fiber at  $\rho(\Omega_X) := 2\rho(\Omega_X^{\frac{1}{2}})$ , where  $2\rho$  is the sum of positive coroots. This involves a choice of square root of  $\Omega_X$ .

**Example 1.4.** If we consider  $G = \mathrm{SL}_2$ , then  $\mathrm{Bun}_B$  classifies exact sequences  $0 \rightarrow \mathcal{L} \rightarrow \mathcal{E} \rightarrow \mathcal{L}^{-1} \rightarrow 0$  and

$$\begin{aligned} \mathrm{Bun}_N^{\Omega_X} &= \left\langle 0 \rightarrow \Omega_X^{\frac{1}{2}} \rightarrow \mathcal{E} \rightarrow \Omega_X^{-\frac{1}{2}} \rightarrow 0 \right\rangle \\ &= R\Gamma(X, \Omega_X)[1] \rightarrow H^1(X, \Omega_X) = \mathbb{A}^1 \end{aligned}$$

For general  $G$ , give a non-degenerate character  $\psi : N \rightarrow \mathbb{G}_a$ , we can get a map (also denoted by  $\psi$ )

$$\psi : \mathrm{Bun}_N^{\Omega_X} \rightarrow \mathrm{Bun}_{N/[N,N]}^{\Omega_X} \rightarrow \prod_{i: \text{simple roots}} \mathrm{Bun}_{\mathbb{G}_a}^{\Omega_x} = \prod_i R\Gamma(X, \Omega_X) \rightarrow \mathbb{A}^{\mathrm{rk} G} \xrightarrow{\text{sum}} \mathbb{A}^1$$

**Definition 1.7.** We define the Poincaré sheaf to be

$$\text{Poinc}_! := (\mathfrak{p}_N)_! (\psi^*(\text{exp}))$$

where  $\mathfrak{p}_N$  is the composition  $\text{Bun}_N^\Omega \rightarrow \text{Bun}_B \rightarrow \text{Bun}_G$  and  $\text{exp} = (\mathcal{O}_{\mathbb{A}^1}, d - dt) \in \mathcal{D}\text{-mod}(\mathbb{A}^1)$ .

Accordingly, we have

$$\begin{aligned} \text{coeff}(\mathcal{F}) &= \text{Hom}((\mathfrak{p}_N)_! (\psi^*(\text{exp})), \mathcal{F}) = \text{Hom}((\psi^*(\text{exp})), (\mathfrak{p}_N)^!(\mathcal{F})) \\ &= \text{C}_{\text{dR}}^\bullet \left( (\mathfrak{p}_N)^!(\mathcal{F}) \otimes^! \psi^*(\text{exp}) \right) \quad (\text{de Rham cohomology}) \end{aligned}$$

**Example 1.5.** When  $G = \mathbb{G}_m$ , then we can write  $\text{Poinc}_!$  as  $\delta_{\text{triv},!} \in \mathcal{D}\text{-mod}(\text{Bun}_G)$ , where  $\text{triv} : \text{pt} \rightarrow \text{Bun}_{\mathbb{G}_m}$  is the trivial bundle, and  $\delta_{\text{triv},!}$  means the  $!$ -pushforward of the constant sheaf along the map  $\text{triv}$ .

We accomplished the first point, now we move to the second, the spectral action of  $\text{QCoh}(\text{LocSys}_{\check{G}})$  on  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ . Recall in Definition 1.2, we define the local Hecke stack. We have the celebrated geometric Satake equivalence

**Theorem 1.2** (Geometric Satake). *There is a monoidal equivalence of categories*

$$\text{Sat}^{\text{nv}} : \text{Rep}(\check{G}) \xrightarrow{\cong} \mathcal{D}\text{-mod}(\mathcal{H}_{G,x}); \quad \mathcal{V}^\lambda \mapsto \text{IC}_{\mathcal{H}_{G,x}^\lambda}[\text{shift}]$$

We can formulate a global Hecke stack  $\mathcal{H}_{G,x}^{\text{glob}}$  that classifies the data of

$$\mathcal{H}_{G,x}^{\text{glob}}(\mathbf{k}) = \left\langle (\mathcal{P}_G, \mathcal{P}'_G, \alpha) \left| \begin{array}{l} \mathcal{P}_G, \mathcal{P}'_G \in \text{Bun}_G \\ \alpha : \mathcal{P}_G|_{X \setminus x} \xrightarrow{\cong} \mathcal{P}'_G|_{X \setminus x} \end{array} \right. \right\rangle$$

we have maps

$$\begin{array}{ccccc} & & \mathcal{H}_{G,x}^{\text{glob}} & & \\ & \swarrow h^\leftarrow & \downarrow s_x & \searrow h^\rightarrow & \\ \text{Bun}_G & & \mathcal{H}_{G,x}^{\text{local}} & & \text{Bun}_G \end{array} \quad \begin{array}{l} (h^\leftarrow, h^\rightarrow) : (\mathcal{P}_G, \mathcal{P}'_G, \alpha) \mapsto (\mathcal{P}_G, \mathcal{P}'_G) \\ s_x : (\mathcal{P}_G, \mathcal{P}'_G, \alpha) \mapsto (\mathcal{P}_G|_{D_x}, \mathcal{P}'_G|_{D_x}, \alpha|_{D_x}) \end{array}$$

Thus for  $\mathcal{V} \in \text{Rep}(\check{G})$ , we can associate it with a Hecke functor

$$H_{\mathcal{V},x} : \mathcal{D}\text{-mod}(\text{Bun}_G) \longrightarrow \mathcal{D}\text{-mod}(\text{Bun}_G); \mathcal{F} \mapsto h_!^\rightarrow \left( (h^\leftarrow)^*(\mathcal{F}) \otimes s_x^*(\text{Sat}^{\text{nv}}(\mathcal{V})) \right)$$

There is a variant that do not need fix  $x \in X$  a priori. For  $\mathcal{V} \in \text{Rep}(\check{G})$ , we can define a Hecke functor

$$H_{\mathcal{V}} : \mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G \times X)$$

such that its composition with  $(\text{id} \times x)^! : \mathcal{D}\text{-mod}(\text{Bun}_G \times X) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$  is  $H_{\mathcal{V},x}$ . Based on that, another variant is if we have two representations  $\mathcal{V}, \mathcal{W} \in \text{Rep}(\check{G})$  (or more generally a family of representations  $\{\mathcal{V}_i\}_{i \in I}$ ),



we can formulate the composition  $H_{\mathcal{V}} \circ H_{\mathcal{W}} : \mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G \times X^2)$ . Since geometric Satake functor  $\text{Sat}^{\text{mv}}$  is monoidal, we have:  $\text{swap}_{X^2} \circ H_{\mathcal{V}} \circ H_{\mathcal{W}} = H_{\mathcal{W}} \circ H_{\mathcal{V}}$ , also restricting  $H_{\mathcal{V}} \circ H_{\mathcal{W}}$  along the diagonal  $X \subseteq X \times X$ , we obtain  $H_{\mathcal{V} \otimes \mathcal{W}}$ .

The ultimate variant is the action of  $\text{Rep}(\check{G})_{\text{Ran}}$  on  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ . We can see the data of  $\text{Rep}(\check{G})_{\text{Ran}}$  is given by a family of points  $\underline{x} = \{x_i\}_{i \in I} \in \text{Ran}(X)$  and representations  $\{\mathcal{V}_i\}$  attached to each point  $x_i$ . The Hecke action of such object in  $\text{Rep}(\check{G})_{\text{Ran}}$  is given by

$$H_{\mathcal{V}_1, x_1} \circ H_{\mathcal{V}_2, x_2} \circ \dots$$

We want to extract the spectral action from  $\text{Rep}(\check{G})_{\text{Ran}} \curvearrowright \mathcal{D}\text{-mod}(\text{Bun}_G)$ . For  $\underline{x} \in \text{Ran}(X)$ , we have evaluation map  $\text{ev}_{\underline{x}} : \text{LocSys}_{\check{G}} \rightarrow (\text{pt}/G)_{\text{Ran}}|_{\underline{x}}$ . Pullback along  $\text{ev}$  gives  $\text{Loc}^{\text{spec}} : \text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{QCoh}(\text{LocSys}_{\check{G}})$ .

**Theorem 1.3** (Gaitsgory-Rozenblyum, [GLC-IV]). *The functor  $\text{Loc}^{\text{spec}} : \text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{QCoh}(\text{LocSys}_{\check{G}})$  is a quotient functor.*

To finish, we have the following theorem of Drinfeld-Gaitsgory:

**Theorem 1.4** ([GLC-II] and [GLC-IV]). *The action of  $\text{Rep}(\check{G})_{\text{Ran}}$  on  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  factors through  $\text{QCoh}(\text{LocSys}_{\check{G}})$ . In other words, for any  $\mathcal{V} \in \text{Ker}(\text{Loc}^{\text{spec}})$ , it acts by zero on  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ .*

We refer the Theorem 1.4 as the general vanishing theorem.

## 1.4 Ind-coherent sheaves and singular supports (Dima Arinkin)

To motivate, the notion of singular support (of coherent sheaves) is a refinement of support to measure directions of non-vanishing. Let us consider coherent sheaves on a quasi-smooth derived scheme  $X$ . Locally,  $X$  is of the form  $(\text{pt} \times_{\text{smooth scheme}} \text{smooth scheme})$ , i.e. the fiber of a map from smooth scheme to another smooth one. We define  $\text{Sing}(X) := H^{-1}(T^*X) = \text{Spec Sym}^* H^1(TX)$ . We want to associate  $\mathcal{F} \in \text{Coh}(X)$  a set  $\text{SingSupp}(\mathcal{F})$  that is included in  $\text{Sing}(X)$ .

**Example 1.6.** Let us consider  $X \subseteq \mathbb{A}^n$  cut out by  $\{f_1 = \dots = f_k = 0\}$ . Then

$$\text{Sing}(X) = \left\{ (x, a) \in X \times \mathbb{A}^k \mid a_1 df_1(x) + \dots + a_k df_k(x) = 0 \right\}$$

We can see on the smooth locus of  $X$ , those  $df_1, \dots, df_k$  are linearly independent, thus the fiber over  $x \in X^{\text{sm}}$  is only one point. However, over singular locus, we can see the fiber  $\text{Sing}(X) \rightarrow X$  could be more complicated.

Before we get into the definition of  $\text{SingSupp}(\mathcal{F})$ , there are some expected properties:

**Properties.** ① If  $\mathcal{F}$  has bounded tor-amplitude, namely  $\mathcal{F} \in \text{Perf}(X)$ , then  $\text{SingSupp}(\mathcal{F})$  is included in zero section.

② If we have a exact sequence  $1 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 1$ , then

$$\text{SingSupp}(\mathcal{F}_2) \subseteq \text{SingSupp}(\mathcal{F}_1) \cup \text{SingSupp}(\mathcal{F}_3)$$

- ③ The functoriality with respect to  $*$ -pullback of smooth morphism, if  $f : Y \rightarrow X$  smooth, then  $\text{Sing}(Y) = \text{Sing}(X) \times_X Y$ .

**Theorem 1.5** (Gulliksen).  $\text{Ext}^*(\mathcal{F}, \mathcal{F})$  is finitely generated as a  $\text{Sym}^*(H^1(TX))$ -module.

Thanks to Theorem 1.5, we can make a definition of **singular support**:

**Definition 1.8.** We define  $\text{SingSupp}(\mathcal{F}) \subseteq \text{Sing}(X)$  to be the support of the module  $\text{Ext}^*(\mathcal{F}, \mathcal{F})$ .

We draw attention to the application for  $\text{LocSys}_{\check{G}}$ . Let  $\mathcal{X}$  be a quasi-smooth derived stack, we can use Property ③ to define  $\text{Sing}(\mathcal{X})$  using smooth covers.

**Example 1.7.** We know that  $\text{Sing}(\text{LocSys}_{\check{G}})$  can be described as

$$\left\{ (\sigma, \xi) \mid \sigma \in \text{LocSys}_{\check{G}}, \xi \in H^{-1}(T_{\mathcal{E}}^*(\text{LocSys}_{\check{G}})) \right\}$$

Moreover, the tangent complex  $T_{\sigma}(\text{LocSys}_{\check{G}}) = C_{\text{dR}}^*(X, \mathfrak{g}_{\sigma})[1]$ , so  $\xi \in (H^1 T_{\sigma}(\text{LocSys}_{\check{G}}))^* = H_{\text{dR}}^0(X, \mathfrak{g}_{\sigma})$ . This means roughly that  $\xi$  is in the infinitesimal symmetry of local system  $\mathcal{E}$ .

**Definition 1.9.** We define nilpotent singular support of local system to be

$$\text{Nilp} = \left\{ (\sigma, \xi) \in \text{Sing}(\text{LocSys}_{\check{G}}) \mid \xi \text{ is nilpotent} \right\} \subseteq \text{Sing}(\text{LocSys}_{\check{G}})$$

In particular, we find that the zero section  $\{0\} \subseteq \text{Nilp}$ , which reflects that  $\text{Perf}(\text{LocSys}_{\check{G}}) \subseteq \text{Coh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})$  from the property ①. In the case when  $\check{G} = \text{GL}_n$ ,  $\mathcal{E}$  is a vector bundle with a connection  $\nabla$  and  $\xi$  is a horizontal endomorphism of the vector bundle.

The last piece is to define the **ind-coherent sheaves**. On a derived scheme  $X$ , we shall get an inclusion  $\text{QCoh}(X) \subseteq \text{IndCoh}(X)$ . We first look at  $\text{QCoh}(X)$ , it is cocomplete (i.e. admits colimits), and the compact generators is  $(\text{QCoh}(X))^{\text{cpt}} = \text{Perf}(X)$ . Starting from  $\text{Perf}(X)$ , we can add formal colimits to get  $\text{QCoh}(X)$ . This procedure can be also understood as taking the category of functors  $\text{Perf}(X)^{\text{op}} \rightarrow \mathbf{Vect}$ . Now we consider an enlargement of  $\text{Perf}(X)$  by  $\text{Coh}(X)$ . And we get the following picture:

$$\begin{array}{ccc} \text{Perf}(X) & \xrightarrow{\text{ind-completion}} & \text{QCoh}(X) = \text{Ind}(\text{Perf}(X)) \\ \downarrow & & \downarrow \Xi_X \\ \text{Coh}(X) & \xrightarrow{\text{ind-completion}} & \text{IndCoh}(X) := \text{Ind}(\text{Coh}(X)) \end{array}$$

We assume that  $X$  is quasi-smooth. For any given conical Zariski-closed subset  $\mathcal{N} \subseteq \text{Sing}(X)$ , define

**Definition 1.10.** The ind-coherent sheaves on  $X$  with nilpotent singular support  $\mathcal{N}$  is

$$\text{IndCoh}_{\mathcal{N}}(X) := \text{Ind} \left( \left\{ \mathcal{F} \in \text{Coh}(X) \mid \text{SingSupp}(\mathcal{F}) \subseteq \mathcal{N} \right\} \right)$$

We observe that by definition,  $\text{QCoh}(X) \subseteq \text{IndCoh}_{\{0\}}(X)$ . (In fact, they are the same, see [AG15, Theorem 4.2.6])

**Remark 1.3.** This definition works locally for quasi-smooth stack  $\mathcal{X}$ , for  $\mathcal{N} \subseteq \text{SingSupp}(\mathcal{X})$ , we define

$$\text{IndCoh}_{\mathcal{N}}(\mathcal{X}) := \varprojlim_{S \rightarrow \mathcal{X} \text{ smooth}} \text{IndCoh}_{\mathcal{N} \times_{\mathcal{X}} S}(S)$$

## 1.5 The Langlands functor and the roadmap to proof (Dennis Gaitsgory)

Here are some recollections:

(1) Yoneda tells us  $\text{Maps}(\mathcal{Y}, \text{pt}/G) = \{\text{right } t\text{-exact symmetric monoidal functors } \text{Rep}(G) \rightarrow \text{QCoh}(\mathcal{Y})\}$ .

(1') Similarly we have

$$\text{Maps}(S, \text{LocSys}_{\check{G}}) = \{\text{right } t\text{-exact symmetric monoidal functors } \text{Rep}(G) \rightarrow \text{QCoh}(S) \otimes D\text{-mod}(X)\}$$

(2) We have forgetful map  $(\text{pt}/\check{G})_{\text{Ran}} \rightarrow \text{Ran}$  and we define  $\text{Rep}(\check{G})_{\text{Ran}} = \text{QCoh}((\text{pt}/\check{G})_{\text{Ran}})$ .

(2') For prestack  $\mathcal{Y}$  (eg:  $(\text{pt}/\check{G})_{\text{Ran}}$ ), we can define category of quasi-coherent sheaves by right Kan extension:

$$\text{QCoh}(\mathcal{Y}) = \varinjlim_{S \rightarrow \mathcal{Y}} \text{QCoh}(S)$$

(3) We want to construct a "localization functor"  $\text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{QCoh}(\text{LocSys}_{\check{G}})$ , consider the following diagram

$$\text{LocSys}_{\check{G}} \xleftarrow{p} \text{LocSys}_{\check{G}} \times \text{Ran} \xrightarrow{\text{ev}} (\text{pt}/\check{G})_{\text{Ran}}$$

We define  $\text{Loc}_{\check{G}}^{\text{spec}} := p_* \circ \text{ev}^* : \text{Rep}(\check{G})_{\text{Ran}} \longrightarrow \text{QCoh}(\text{LocSys}_{\check{G}})$ .

(3') Let us pick family of points  $x_1, \dots, x_n$  on  $X$  and representations  $\mathcal{V}_1, \dots, \mathcal{V}_n \in \text{Rep}(\check{G})$ , we can formulate an element  $\bigotimes_i \mathcal{V}_{i, x_i} \in \text{Rep}(\check{G})_{\text{Ran}}$ .

**Remark 1.4.** The monoidal structure in  $\text{Rep}(\check{G})_{\text{Ran}}$  is the following: for another  $\bigotimes_j \mathcal{W}_{j, y_j}$ , then  $(\bigotimes_i \mathcal{V}_{i, x_i}) \otimes (\bigotimes_j \mathcal{W}_{j, y_j})$  can be defined component-wise: if  $y_j \notin \{x_1, \dots, x_n\}$ , we can insert  $\bigotimes \mathcal{W}_{j, y_j}$ , if two points  $x_i, y_j$  coincide, then the corresponding tensor component would be  $\mathcal{V}_{i, x_i} \otimes \mathcal{W}_{j, y_j} = (\mathcal{V}_i \otimes \mathcal{W}_j)_{x_i}$ .

Let  $\sigma \in \text{LocSys}_{\check{G}}$  be a point, then the stalk  $\left( \text{Loc}_{\check{G}}^{\text{spec}} \left( \bigotimes_i \mathcal{V}_{i, x_i} \right) \right)_{\sigma}$  would be

$$\bigotimes_i ((\mathcal{V}_i)_{\sigma})_{x_i}, \quad \text{where } (\mathcal{V}_i)_{\sigma} \text{ is the underlying vector space of } \sigma \text{ twisted by } \mathcal{V}_i$$

(3'') Here comes the Theorem 1.3 that  $\text{Loc}_{\check{G}}^{\text{spec}}$  is a quotient.

The following diagrams, we have seen the left one before, we can upgrade it into Ran-version, as the right one:

$$\begin{array}{ccccc} & \mathcal{H}_{G,x}^{\text{glob}} & & \mathcal{H}_{G,\text{Ran}}^{\text{glob}} & \\ & \swarrow h^{\leftarrow} & \downarrow s_x & \swarrow h^{\leftarrow} & \downarrow s_{\text{Ran}} \\ \text{Bun}_G & & \mathcal{H}_{G,x}^{\text{local}} & & \mathcal{H}_{G,\text{Ran}}^{\text{local}} \\ & \searrow h^{\rightarrow} & & \searrow h^{\rightarrow} & \\ & \text{Bun}_G & & \text{Bun}_G & \end{array}$$

What (naïve) geometric gives us is the equivalence functor  $\text{Sat}_G^{\text{nv}} : \text{Rep}(\check{G})_{\text{Ran}} \rightarrow \mathcal{D}\text{-mod}(\mathcal{H}_{G,\text{Ran}}^{\text{local}})$ . We get the Hecke functor (binary operation)

$$\text{Hecke} : \text{Rep}(\check{G})_{\text{Ran}} \otimes \mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G); \mathcal{V} \otimes \mathcal{M} \mapsto h_*^{-1} \left( s_{\text{Ran}}^! (\text{Sat}^{\text{nv}}(\mathcal{V})) \otimes^L (h^\leftarrow)^!(\mathcal{M}) \right)$$

**Theorem 1.6.** *The above binary operator factors through the quotient*

$$\begin{array}{ccc} \text{Rep}(\check{G})_{\text{Ran}} \otimes \mathcal{D}\text{-mod}(\text{Bun}_G) & & \\ \downarrow & \searrow \text{Hecke} & \\ \text{QCoh}(\text{LocSys}_{\check{G}}) \otimes \mathcal{D}\text{-mod}(\text{Bun}_G) & \longrightarrow & \mathcal{D}\text{-mod}(\text{Bun}_G) \end{array}$$

The Theorem says that the Kernel of the binary action **Hecke** falls out of  $\text{QCoh}(\text{LocSys}_{\check{G}})$ . And based on that, we can define the spectral action  $\text{QCoh} \circ \mathcal{D}\text{-mod}(\text{Bun}_G)$  (we still have not yet other ways to define it).

Having the spectral action, we define the coarse Langlands functor  $\mathbb{L}_{G,\text{coarse}}$  by dual Yoneda. Let assume  $\mathbb{L}_{G,\text{coarse}}$  admits a left adjoint  $\mathbb{L}_{G,\text{coarse}}^L : \text{QCoh}(\text{LocSys}_{\check{G}}) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$ , then it goes from the acting category to the category acted, meaning we need to choose one object in  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ . One can expect that  $\mathbb{L}_{G,\text{coarse}}^L$  is defined in the unique way that  $\mathcal{O}_{\text{LocSys}_{\check{G}}}$  sends to  $\text{Poinc}_!^{\text{vac}}$ .

**Remark 1.5.**  $\text{QCoh}(\text{LocSys}_{\check{G}})$  is a rigid symmetric monoidal category, then the left adjoint  $\mathbb{L}_{G,\text{coarse}}^L$  will send compact objects to compact ones.

Upshot: we want to lift  $\mathbb{L}_{G,\text{coarse}}$  to a functor

$$\begin{array}{ccc} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) & \\ & \nearrow \exists \mathbb{L}_G & \downarrow \Psi \\ \mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\text{coarse}}} & \text{QCoh}(\text{LocSys}_{\check{G}}) \end{array}$$

For prestack  $\mathcal{Y}$ , we have seen how to construct  $\text{IndCoh}(\mathcal{Y})$  by gluing from affine schemes, now we want a functor  $\Psi_{\mathcal{Y}} : \text{IndCoh}(\mathcal{Y}) \rightarrow \text{QCoh}(\mathcal{Y})$ . Notice that  $\text{IndCoh}$  is ind-completion (which is a left adjoint) of  $\text{Coh}$ , thus  $\Psi_{\mathcal{Y}}$  is given by the ind-completion of the embedding  $\text{Coh}(\mathcal{Y}) \hookrightarrow \text{QCoh}(\mathcal{Y})$ .

A cheap fact is that  $\Psi_{\mathcal{Y}}$  is  $t$ -exact, where  $\text{IndCoh}(\mathcal{Y})$  carries a canonical  $t$ -structure that commutes with filtered colimit and makes ind-completion  $t$ -exact. A less evident fact is that  $\Psi_{\mathcal{Y}}$  is actually an equivalence on eventually coconnective part, i.e.  $\Psi_{\mathcal{Y}} : \text{IndCoh}(\mathcal{Y})^{>-\infty} \xrightarrow{\cong} \text{QCoh}(\mathcal{Y})^{>-\infty}$ .

**Remark 1.6.** Same properties hold for  $\text{IndCoh}_{\mathcal{N}}(\mathcal{Y}) \rightarrow \text{QCoh}(\mathcal{Y})$ , we denote this functor by  $\Psi_{\mathcal{N}}$  if there is no ambiguity on  $\mathcal{Y}$ .

This gives us a hint how to lift  $\mathbb{L}_{G,\text{coarse}}$  on certain part of  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ . Suppose we know that  $\mathbb{L}_{G,\text{coarse}}$  sends compact objects in  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  to eventually coconnective objects in  $\text{QCoh}(\text{LocSys}_{\check{G}})$ , then we can construct

$$\mathcal{D}\text{-mod}(\text{Bun}_G)^c \rightarrow \text{IndCoh}(\text{LocSys}_{\check{G}})^{>-\infty}$$

by composing the inverse of  $\Psi_{\text{LocSys}_{\check{G}}}$  on eventually coconnective part. We want to show:

**Theorem 1.7.**  $\mathbb{L}_{G,\text{coarse}}$  sends compact objects in  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  to eventually coconnective objects in  $\text{QCoh}(\text{LocSys}_{\check{G}})$ .

**Remark 1.7.** Here we can observe that the condition nilpotent singular support is redundant here. (Spoiler: Eisenstein series: we start with  $\text{Perf}(\text{LocSys}_{\check{M}})$ , then its essential image under special Eisenstein series is generator in  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})$ ).

**Theorem 1.8.**  $\mathcal{D}\text{-mod}(\text{Bun}_G)^c \subseteq \mathcal{D}\text{-mod}(\text{Bun}_G)^{>-\infty}$

**Theorem 1.9.** (1)  $\mathbb{L}_{G,\text{coarse}}$  is left  $t$ -exact after some [shift].

(2)  $\mathbb{L}_{G,\text{coarse}}$  is right  $t$ -exact after another [shift'].

**Remark 1.8.** The Langlands functor  $\mathbb{L}_G$  is however, not left  $t$ -exact, up to any shift. An important counter example would be the constant sheaf  $\underline{\mathbf{k}}_{\text{Bun}_G}$ . We need to combine two facts: the first one is that  $\mathbb{L}_{G,\text{coarse}}$  sends it to zero, and also  $\mathbb{L}_G$  is supposed to be an equivalence, thus  $\mathbb{L}_G(\underline{\mathbf{k}}_{\text{Bun}_G}) \neq 0$ . It means  $\Psi_{\text{Nilp}}$  carries  $\mathbb{L}_G(\underline{\mathbf{k}}_{\text{Bun}_G})$  to zero object in  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})^{>-\infty}$ , which means  $\mathbb{L}_G(\underline{\mathbf{k}}_{\text{Bun}_G}) \in \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})^{\leq n}, \forall n \in \mathbb{Z}$ . Thus  $\mathbb{L}_G$  does NOT have a left bounded cohomological amplitude.

However,  $\mathbb{L}_G$  is right  $t$ -exact after the same shift [shift'] as in Theorem 1.9(2), or we can say  $\mathbb{L}_G$  has right bounded cohomological amplitude by [shift']. It is because  $\Psi_{\text{Nilp}}$  is  $t$ -exact and induces equivalences on eventually connective part.

*Proof of Theorem 1.8.* Let  $\mathcal{Y}$  be a quasi-compact algebraic stack, then we can get an inclusion  $\mathcal{D}\text{-mod}(\mathcal{Y})^c \subseteq \mathcal{D}\text{-mod}(\mathcal{Y})^{>-\infty}$ . In fact, we can consider the pair of adjunction for right  $\mathcal{D}$ -modules

$$\text{IndCoh}(\mathcal{Y}) \begin{array}{c} \xrightarrow{\text{ind}^r} \\ \perp \\ \xleftarrow{\text{oblv}^r} \end{array} \mathcal{D}\text{-mod}(\mathcal{Y})$$

where  $\text{ind}^r$  is  $t$ -exact and the forgetful functor  $\text{oblv}^r$  is conservative. Then by definition  $\text{IndCoh}(\mathcal{Y})$  is compactly generated by  $\text{Coh}(\mathcal{Y})$  whose objects are cohomologically bounded. Since we know  $\text{ind}^r$  is  $t$ -exact, we get the inclusion we want. Now we move to  $\text{Bun}_G$ , which is not quasi-compact. However, every compact object in  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  is of the form  $j_!(\mathcal{F}_U)$ , where  $j : U \hookrightarrow \text{Bun}_G$  is an inclusion of quasi-compact open and  $\mathcal{F}_U \in \mathcal{D}\text{-mod}(U)^c$ . We need to be careful that the left adjoint  $j_!$  (hence right  $t$ -exact) is not always well-defined.

The stack  $\text{Bun}_G$  is however, more particular, that we have:

**Theorem 1.10.** [DG15]  $\text{Bun}_G$  is a union of quasi-compact  $U$ 's for which  $j_!$  is well-defined. ■

*Proof of Theorem 1.9.* We take  $\mathcal{M} \in \mathcal{D}\text{-mod}(\text{Bun}_G)^{>0}$  and let  $\mathcal{F} := \mathbb{L}_{G,\text{coarse}}(\mathcal{M})$ . Here comes the

**Big Beautiful Theorem 1.11.** [GLC-I, Proposition 2.3.2] Let  $\mathcal{Y}$  be an algebraic stack, assume  $\mathcal{Y}$  is eventually coconnective, let  $\mathcal{F} \in \text{QCoh}(\mathcal{Y})$  there exists  $d \in \mathbb{Z}$  (only depends on  $\mathcal{Y}$ ) such that if  $i_y^!(\mathcal{F}') \geq d$  for any  $i_y : \text{Spec } \mathbf{k}' \rightarrow \mathcal{Y}' = \mathcal{Y} \times_{\text{Spec } \mathbf{k}} \text{Spec } \mathbf{k}'$ , where  $\mathbf{k}' \supseteq \mathbf{k}$  arbitrary field extension and  $\mathcal{F}'$  pullback of  $\mathcal{F}$  to  $\mathcal{Y}'$ , then  $\mathcal{F} \geq 0$ .

Here  $\geq d$  means the cohomological degree concentrates in degree  $\geq d$ , similar for  $\geq 0$ , and  $\mathcal{Y}$  being eventually coconnective means its structure sheaf  $\mathcal{O}_{\mathcal{Y}}$  is.

There are counter examples for the failure of BBT: let  $\mathcal{Y} := \text{Spec } \mathbf{k}[\epsilon]$ ,  $\deg(\epsilon) = -2$  and take  $\mathcal{F} = \mathbf{k}[\epsilon^{\pm 1}]$ , which is obviously not coconnective, however,  $i_{\mathcal{Y}}^!(\mathcal{F})$  is zero for any point. The failure is due to  $\text{Spec } \mathbf{k}[\epsilon]$  is not eventually coconnective. Another counter example would be: We take  $\mathcal{Y} = \mathbf{A}^1$  and  $\mathcal{F} = \bigoplus_n \mathbf{k}(t)[n]$ . This counter example shows that it is not enough to ask just for  $\mathbf{k}$ -points on  $\mathcal{Y}$ .

We apply the **BBT 1.11** to  $\mathcal{Y} = \text{LocSys}_{\check{G}}$ . It remains to show that  $\forall \sigma \in \text{LocSys}_{\check{G}}(\mathbf{k}')$ ,  $(i_{\sigma})^! (\mathbb{L}_{G, \text{coarse}}(\mathcal{M})) \geq d$ , where  $i_{\sigma} : \text{Spec } \mathbf{k}' \rightarrow \text{LocSys}_{\check{G}}$ . To do this, we need the restricted local systems (definition will be given later in Definition 1.11)  $\iota^{\text{spec}} : \text{LocSys}_{\check{G}}^{\text{restr}} \subseteq \text{LocSys}_{\check{G}}$ , which is a disjoint union of connected components that are in bijection with semisimple local systems and each connected component is the formal neighborhood of the corresponding semisimple local system. In other words, the points in connected component are those local systems who semi-simplify to the same one. Moreover  $i_{\sigma}$  factors through  $i'_{\sigma} : \text{Spec } \mathbf{k}' \rightarrow \text{LocSys}_{\check{G}}^{\text{restr}}$ .

It is enough to show,  $(i'_{\sigma})^! (\iota^{\text{spec}})^! (\mathbb{L}_{G, \text{coarse}}(\mathcal{M})) \geq d$ . We can reformulate  $(\iota^{\text{spec}})^! \circ \mathbb{L}_{G, \text{coarse}}$ : in fact, we can put nilpotent singular support on  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  (condition for "classical" singular support for characteristic cycle map), we denote by  $\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)$  the corresponding full subcategory.

**Remark 1.9.** Here by semisimpleness of a local system  $\sigma$  we mean whenever it has a reduction to parabolic  $\check{P} = \check{M}\check{U}$ , it can be further reduced to  $\check{M}$ .

**Theorem 1.12.** [AGKRRV] *There is an equivalence of categories*

$$\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G) \cong \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{restr}}) \otimes_{\text{QCoh}(\text{LocSys}_{\check{G}})} \mathcal{D}\text{-mod}(\text{Bun}_G)$$

The above theorem gives us a functor  $\mathbb{L}_{G, \text{coarse}}^{\text{restr}} : \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{restr}})$  (called restricted Langlands functor) and the following diagram:

$$\begin{array}{ccc} \mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G, \text{coarse}}} & \text{QCoh}(\text{LocSys}_{\check{G}}) \\ \uparrow \iota \quad \downarrow \iota^R & & \uparrow (\iota^{\text{spec}})_* \quad \downarrow (\iota^{\text{spec}})^! \\ \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_{G, \text{coarse}}^{\text{restr}}} & \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{restr}}) \end{array}$$

So we reduce our problem to show  $(i'_{\sigma})^! \circ \mathbb{L}_{G, \text{coarse}}^{\text{restr}} \circ \iota^R(\mathcal{M}) \geq d$ . Both  $(i'_{\sigma})^!$  and  $\iota^R$  are left  $t$ -exact being right adjoint of  $t$ -exact functors. And we conclude the proof by a Theorem in [FR25] that  $\mathbb{L}_{G, \text{coarse}}^{\text{restr}}$  is  $t$ -exact. ■

## 1.6 Bonus A: Other versions of GLC(Joakim Færgeman)

We are going to take a look into other versions (*Betti/Restricted/Tempered*) of geomtric Langlands correspondence. In this section, we might change the base field or coefficient field.

## Betti GLC

We let  $X$  be smooth connected projective curve over  $\mathbb{C}$  and fix coefficient  $\mathbf{e} = \bar{\mathbf{e}}$ ,  $\text{char}(\mathbf{e}) = 0$ .

We defined before  $\text{LocSys}_{\check{G}}$  that classifies *de Rham* local systems, we now consider a Betti version  $\text{LocSys}_{\check{G}}^{\text{Betti}}$  which essentially classifies representations of  $\pi_1(X)$  in  $\check{G}$ .

$$\text{LocSys}_{\check{G}}^{\text{Betti}} := \mathbf{Maps}_{\text{Spc}}(X, \text{pt} / \check{G}) \quad \text{mapping space for homotopy types}$$

More concretely,  $\text{LocSys}_{\check{G}}^{\text{Betti}}(S) = \text{Maps}_{\text{Spc}}(X, \text{Maps}_{\text{dSt}}(S, \text{pt} / \check{G}))$ . Since we have adjunction  $\text{dSt} \rightleftharpoons \text{Spc}$ , it is equivalent to define  $\text{LocSys}_{\check{G}}^{\text{Betti}}$  as  $\mathbf{Maps}_{\text{dSt}}(X_{\text{B}}, \text{pt} / \check{G})$ , where  $X_{\text{B}}$  is the Betti shape of  $X$ .

Alternatively, let  $g = g(X)$  be the genus of  $X$ , then  $\pi_1(X) = \langle a_i, b_i \mid \prod_i [a_i, b_i] = 1 \rangle_{i=1, \dots, g}$ , we want  $\text{LocSys}_{\check{G}}^{\text{Betti}} \approx \text{Hom}(\pi_1(X), \check{G})$  and the presentation suggests that

$$\text{LocSys}_{\check{G}}^{\text{Betti}} \cong \check{G}^{2g} \times_{\check{G}} \{e\} / \check{G}$$

so that we attach each  $a_i, b_i$  an element in  $\check{G}$ , with one relation and then modulo the automorphism group.

**Example 1.8.** (1) When  $X = \mathbb{P}^1$ , by the above isomorphism, we can see

$$\text{LocSys}_{\check{G}}^{\text{Betti}} \cong (\text{pt} \times_{\check{G}} \text{pt} / \check{G}) = \Omega \check{\mathfrak{g}} / \check{G} \cong \text{LocSys}_{\check{G}}$$

(2) When we take  $\check{G} = \check{T}$ , then  $\text{LocSys}_{\check{T}}^{\text{Betti}} \cong \check{T}^{2g} \times \Omega \check{\mathfrak{t}} \times \mathbf{B}\check{T}$  since the map  $\check{T}^{2g} \rightarrow \check{T}$  is trivial.

On automorphic side, we consider for  $\mathcal{Y}$  algebraic stack over  $\mathbb{C}$ , the category  $\text{Shv}^{\text{all}}(\mathcal{Y})$  sheaves of  $\mathbf{e}$ -vector space on the underlying topological space  $\mathcal{Y}(\mathbb{C})^{\text{an}}$ , i.e.  $\text{Shv}^{\text{all}}(\mathcal{Y}) = \varinjlim_{S \rightarrow \mathcal{Y}} \text{Shv}^{\text{all}}(S)$ . If  $\Lambda \subseteq T^* \mathcal{Y}$  a conical closed subset, we may consider  $\text{Shv}_{\Lambda}^{\text{all}}(\mathcal{Y}) = \{F \in \text{Shv}^{\text{all}}(\mathcal{Y}) \mid \text{SS}(H^i(F)) \subseteq \Lambda\}$ . For  $\mathcal{Y} = \text{Bun}_G$ , recall

$$T^* \text{Bun}_G = \left\{ (\mathcal{P}_G, \varphi) \mid \mathcal{P}_G \in \text{Bun}_G, \varphi \in \Gamma(X, \mathfrak{g}_{\mathcal{P}_G}^* \otimes \Omega_X) \right\}$$

We take  $\text{Nilp} \subseteq T^* \text{Bun}_G$  under the condition that  $\phi$  is nilpotent. The Betti version of GLC is:

**Conjecture 1.1** (Betti GLC).  $\text{Shv}_{\text{Nilp}}^{\text{all}}(\text{Bun}_G) \cong \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}^{\text{Betti}})$

Two motivations for this equivalence (adding nilpotent singular support condition on automorphic side) are:

- (1) Laumon conjectured that Hecke eigensheaves should have nilpotent singular support (proved in [AGKRRV]).
- (2) The Eisenstein series  $\text{Eis}_! : \text{Shv}^{\text{all}}(\text{Bun}_M) \rightarrow \text{Shv}^{\text{all}}(\text{Bun}_G)$  preserves nilpotent singular support.

We can take a look at the case when  $G = T$ , in which case  $\text{Nilp} = \{0\}$ . On spectral side:

$$\begin{aligned} \text{QCoh}(\text{LocSys}_{\check{T}}^{\text{Betti}}) &\cong \text{QCoh}(\check{T}^{2g}) \otimes \text{QCoh}(\Omega \check{\mathfrak{t}}) \otimes \text{QCoh}(\mathbf{B}\check{T}) \\ &\cong \mathbf{k}[H_1(X) \otimes \check{\Lambda}]\text{-mod} \otimes \text{Sym}(\mathfrak{t}[1])\text{-mod} \otimes \mathbf{Vect}^{\check{\Lambda}} \end{aligned}$$

On automorphic side we have  $\text{Bun}_T \cong \text{Pic}_T^\circ(X) \times \mathbf{B}T \times \check{\Lambda}$ . Thus

$$\text{Shv}_{\{0\}}^{\text{all}}(\text{Bun}_T) \cong \mathbf{k}[\pi_1(\text{Pic}_T^\circ(X))]\text{-mod} \otimes H_*(T)\text{-mod} \otimes \mathbf{Vect}^{\check{\Lambda}}$$

We can deduce Conjecture 1.1 in this case since  $\pi_1(\text{Pic}_T^\circ(X)) \cong H_1(X) \otimes \check{\Lambda}$  and  $H_*(T) \cong \text{Sym}(\mathfrak{t}[1])$ .

## Restricted GLC

The original motivation for restricted version is to define a stack of étale local system on  $X/\mathbb{F}_q$ . For our purpose: for  $X$  curve over  $\mathbf{k} = \bar{\mathbf{k}}$ ,  $\text{char}(\mathbf{k}) = 0$ , the restricted GLC would imply both Betti and de Rham GLC. We have the following relations:

$$\text{Betti GLC} \iff \text{restricted GLC} \iff \text{de Rham GLC}$$

Recall we have  $\text{LocSys}_{\check{G}}(S) = \{\text{right } t\text{-exact symmetric monoidal functors } \text{Rep}(\check{G}) \rightarrow \text{QCoh}(S) \otimes \mathcal{D}\text{-mod}(X)\}$ .

**Definition 1.11.** We first define  $\text{QLisse}(X) \subseteq \mathcal{D}\text{-mod}(X)$  a subcategory of those  $F \in \mathcal{D}\text{-mod}(X)$  such that  $H^i(F)$  is a union of vector bundles with connection.

We define the stack of restricted (de Rham) local systems by Yoneda:

$$\text{Maps}(S, \text{LocSys}_{\check{G}}^{\text{restr}}) = \{\text{right } t\text{-exact symmetric monoidal functors } \text{Rep}(\check{G}) \rightarrow \text{QCoh}(S) \otimes \text{QLisse}(X)\}$$

And we obtained the inclusion that we used in the proof of **BBT 1.9**  $\iota^{\text{spec}} : \text{LocSys}_{\check{G}}^{\text{restr}} \hookrightarrow \text{LocSys}_{\check{G}}$ . Moreover, we have the following facts:

- (1)  $\iota^{\text{spec}}$  is bijective on  $\mathbf{k}$ -points: meaning symmetric monoidal functor  $\text{Rep}(\check{G}) \rightarrow \mathcal{D}\text{-mod}(X)$  factors through  $\text{QLisse}(X)$ . Moreover, for  $\sigma \in \text{LocSys}_{\check{G}}(\mathbf{k})$ , we have an isomorphism on the formal completion  $\text{LocSys}_{\check{G},\sigma}^{\text{restr},\wedge} \xrightarrow{\cong} \text{LocSys}_{\check{G},\sigma}^{\wedge}$ .
- (2) For two local systems  $\sigma_1, \sigma_2 \in \text{LocSys}_{\check{G}}(\mathbf{k})$ , they lie in the same connected component of  $\text{LocSys}_{\check{G}}^{\text{restr}}$  if and only if they have the same semi-simplification. The connected component containing irreducible local system  $\sigma$  is isomorphic to  $\text{LocSys}_{\check{G},\text{pt}/\text{Stab}(\sigma)}^{\wedge}$ , the formal completion at  $\text{pt}/\text{Stab}(\sigma)$ .

**Conjecture 1.2** (Restricted GLC).  $\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G) \cong \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}^{\text{restr}})$

Here are some motivations of restricted GLC 1.2:

- (1) Any object of  $\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)$  is regular holonomic. There is an analogous Betti restricted GLC, and they are equivalent because of Riemann-Hilbert.

**Remark 1.10.** For  $X$  curve over  $\bar{\mathbb{F}}_q$ , we can replace  $\mathcal{D}\text{-mod}_{\text{Nilp}}$  by constructible sheaves with nilpotent singular support  $\text{Shv}_{\text{Nilp}}(\text{Bun}_G)$  and take  $\mathbf{e} = \bar{\mathbb{Q}}_\ell$ . The right hand side for restricted GLC in this case remains the same.

- (2) Given a category  $\mathcal{C}$  acted on by  $\text{QCoh}(\text{LocSys}_{\check{G}}^{\text{restr}})$ . To have this action is equivalent to have functor  $\text{Rep}(\check{G})^{\otimes I} \otimes \mathcal{C} \longrightarrow \mathcal{C} \otimes \text{QLisse}(X)^{\otimes I}$  for any finite set  $I$ . Consider the Hecke action  $\text{Rep}(\check{G}) \circ \mathcal{D}\text{-mod}(\text{Bun}_G)$ :

**Theorem 1.13** (Nadler-Yun). *The restriction of Hecke action to  $\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)$  gives following diagram:*

$$\begin{array}{ccc} \text{Rep}(\check{G}) \otimes \mathcal{D}\text{-mod}(\text{Bun}_G) & \longrightarrow & \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \mathcal{D}\text{-mod}(X) \\ \uparrow & & \uparrow \\ \text{Rep}(\check{G}) \otimes \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G) & \longrightarrow & \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \text{QLisse}(X) \end{array}$$



The Theorem gives an action  $\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}^{\mathrm{restr}}) \circ \mathcal{D}\text{-mod}_{\mathrm{Nilp}}(\mathrm{Bun}_G)$ . In fact, we have used the following isomorphism in the proof of [BBT 1.9](#):

$$\mathcal{D}\text{-mod}_{\mathrm{Nilp}}(\mathrm{Bun}_G) \cong \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \otimes_{\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}})} \mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}^{\mathrm{restr}})$$

## Tempered GLC

Finally, we consider the tempered version. Since we expect for GLC that  $\mathcal{D}\text{-mod}(\mathrm{Bun}_G) \cong \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}})$ , we can ask what category should correspond to  $\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}})$ , the answer is so-called tempered  $\mathcal{D}$ -modules:  $\mathcal{D}\text{-mod}(\mathrm{Bun}_G)^{\mathrm{temp}}$ . We can first look at the spectral side, how do we characterize  $\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}})$ ? Answer:

$$\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}) \cong \mathrm{IndCoh}(\mathrm{LocSys}_{\check{G}}) \otimes_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\Omega_{\check{\mathfrak{g}}/\check{G}})} \mathrm{QCoh}(\Omega_{\check{\mathfrak{g}}/\check{G}})$$

for any choice of  $x \in X$ . The derived Satake equivalence says

$$\textbf{Theorem 1.14.} \quad \mathcal{D}\text{-mod}(\mathcal{L}_x^+ G \backslash \mathcal{L}_x G / \mathcal{L}_x^+ G) \cong \mathrm{IndCoh}_{\mathrm{Nilp}}(\Omega_{\check{\mathfrak{g}}/\check{G}})$$

so we can see that  $\mathrm{IndCoh}_{\mathrm{Nilp}}(\Omega_{\check{\mathfrak{g}}/\check{G}})$  acts on  $\mathcal{D}\text{-mod}(\mathrm{Bun}_G)$ , thus we can define the tempered category to be

$$\mathcal{D}\text{-mod}(\mathrm{Bun}_G)^{\mathrm{temp}} := \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \otimes_{\mathrm{IndCoh}_{\mathrm{Nilp}}(\Omega_{\check{\mathfrak{g}}/\check{G}})} \mathrm{QCoh}(\Omega_{\check{\mathfrak{g}}/\check{G}})$$

**Remark 1.11.** Here are some remarks from Day 5. We can ask, what is the relation between the above tempered category and classical notion of temperedness via Arthur packets. In fact, if we consider  $X$  curve positive characteristic, we have

$$\begin{array}{ccc} \mathrm{Tr}(\mathrm{Frob}, \mathrm{Shv}_{\mathrm{Nilp}}(\mathrm{Bun}_G)) & \xrightarrow{\cong} & \mathrm{Tr}(\mathrm{Frob}, \mathrm{icoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}})) \\ & \cong \downarrow \scriptstyle AGKRRV & \\ \mathrm{Func}_c(\mathrm{Bun}_G(\mathbb{F}_q), \bar{Q}_\ell) & & \end{array}$$

We obtain a mysterious filtration on  $\mathrm{Func}_c(\mathrm{Bun}_G(\mathbb{F}_q), \bar{Q}_\ell)$  indexed by nilpotent orbits in  $\mathfrak{g}^*$ . The conjectures are

**Conjecture 1.3.** (a) *The filtration is defined over  $\mathbb{Q}$ .*

(b)  $L^2(\mathrm{Bun}_G(\mathbb{F}_q)) \cong \bigoplus_{\mathcal{O}: \text{nilp orbits}} L^2(\mathrm{Bun}_G(\mathbb{F}_q))_{\mathcal{O}}$  and

$$\mathrm{Func}_c(\mathrm{Bun}_G(\mathbb{F}_q), \bar{Q}_\ell) \cap \left( \bigoplus_{\mathcal{O}' \leq \mathcal{O}} L^2(\mathrm{Bun}_G(\mathbb{F}_q))_{\mathcal{O}'} \right) = \mathrm{Func}_c^{\leq \mathcal{O}}(\mathrm{Bun}_G(\mathbb{F}_q), \bar{Q}_\ell)$$

(3) Consider  $\mathrm{gr}^{\mathcal{O}}(\mathrm{Func}_c(\mathrm{Bun}_G(\mathbb{F}_q), \bar{Q}_\ell))$  which embeds  $L^2(\mathrm{Bun}_G(\mathbb{F}_q))_{\mathcal{O}}$ , then under this inclusion,

$$\mathrm{gr}^{\mathcal{O}}(\mathrm{Func}_c(\mathrm{Bun}_G(\mathbb{F}_q), \bar{Q}_\ell))^{\mathrm{Hecke-finite}} \cong L^2(\mathrm{Bun}_G(\mathbb{F}_q))_{\mathcal{O}}^{\mathrm{disc}}$$

## 2 Day 2 (Kac-)Moody Day

### 2.1 Whittaker categories(Andreas Hayash)

The goal is to introduce the Whittaker coefficient functor and its left adjoint  $\text{Poinc}_!$ . The first step is introducing the Whittaker category on affine grassmannian.

For any point  $\underline{x} \in \text{Ran}(X)$ , the affine grassmannian on it can be taken as  $\mathcal{L}_{\underline{x}}G/\mathcal{L}_{\underline{x}}^+G$ . It carries an action by  $\mathcal{L}_{\underline{x}}G$  by left and in particular an action of subgroup  $\mathcal{L}_{\underline{x}}N \subseteq \mathcal{L}_{\underline{x}}G$ , where  $N$  is the unipotent radical of a fixed Borel  $B$ . Similar to a character  $\psi : \text{Bun}_N^{\Omega_X} \rightarrow \mathbb{G}_a$ , there is a (non-degenerate) character  $\mathcal{L}N \rightarrow \mathbb{G}_a$ , factors as

$$\mathcal{L}N \rightarrow \mathcal{L}(N/[N, N]) \cong \prod_{\text{isimple roots}} \mathcal{L}\mathbb{G}_a \xrightarrow{\text{Res}} \prod_{\text{isimple roots}} \mathbb{G}_a \xrightarrow{\chi_0} \mathbb{G}_a$$

where  $\chi_0$  can be chosen as any character that is non-trivial on each coordinate.

**Definition 2.1.** We define the Whittaker category on  $\underline{x}$  to be

$$\text{Whit}^!(G)_{\underline{x}} = \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})^{\mathcal{L}N,\chi}$$

Namely the objects in  $\text{Whit}^!(G)$  are  $\mathcal{D}$ -modules on  $\text{Gr}_{G,\underline{x}}$  that are  $\chi$ -equivariant under  $\mathcal{L}N$ -action. More precisely,  $c \in \text{Whit}^!(G)_{\underline{x}}$  then  $\text{act}^!(c) \cong \chi^!(\exp) \boxtimes c$ , for  $\text{act} : \mathcal{L}N \times \text{Gr}_{G,\underline{x}} \rightarrow \text{Gr}_{G,\underline{x}}$ .

**Remark 2.1.** Here  $\chi^!(\exp)$  is actually a character  $\mathcal{D}$ -module for  $\mathcal{L}N$ . In general, let  $(H, m)$  be a group scheme,  $F \in \mathcal{D}\text{-mod}(H)$  is called a *character  $\mathcal{D}$ -module* if  $m^!(F) \cong F \boxtimes F$  and  $\text{unit}^!(F) \cong \mathbf{k}$ .

The loop group  $\mathcal{L}N$  is an ind-pro group scheme, it can be written as  $\mathcal{L}N = \bigcup_{\alpha} N_{\alpha}$ , where each  $N_{\alpha}$  is a pro-group scheme. We can consider  $\mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})^{N_{\alpha},\chi} \xrightarrow{\text{oblv}} \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})^{N_{\beta},\chi} \rightarrow \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})$ , where the second arrow is fully-faithful. Then we have

$$\text{Whit}^!(G)_{\underline{x}} = \varinjlim_{\alpha} \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})^{N_{\alpha},\chi} = \bigcap_{\alpha} \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})^{N_{\alpha},\chi}$$

The limit is just the intersection due to the fully-faithfulness and there exists a fully-faithful forgetful map  $\text{oblv}_{\underline{x}} : \text{Whit}^!(G)_{\underline{x}} \rightarrow \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})$ , it admits a right adjoint functor

$$\mathbf{Av}_{\underline{x},*} = \mathbf{Av}_{\underline{x},*}^{\mathcal{L}N,\chi} : \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}}) \rightarrow \text{Whit}^!(G)_{\underline{x}}$$

We shall be careful that the averaging operator  $\mathbf{Av}$  is not continuous: In general for smooth group scheme  $H_{\alpha}$  of finite type,  $\mathbf{Av}^{H_{\alpha},\chi} : \mathcal{D}\text{-mod}(\mathcal{Y}) \rightarrow \mathcal{D}\text{-mod}(\mathcal{Y}/(H_{\alpha},\chi))$  is continuous. However taking limit would destroy the continuity, as in our case.

**Remark 2.2.** The category  $\text{Whit}^!(G)_{\underline{x}}$  does not depend on the choice of non-degenerate  $\chi$ .

There is a variant of Whittaker category, instead of taking invariant, we can take coinvariant

$$\text{Whit}_*(G)_{\underline{x}} := \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})_{\mathcal{L}N,\chi}$$

which is equipped with projection  $\mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}}) \longrightarrow \text{Whit}_*(G)_{\underline{x}}$ . However, there exists an isomorphism

$$\Theta : \text{Whit}_*(G)_{\underline{x}} \xrightarrow{\cong} \text{Whit}^!(G)_{\underline{x}}$$

Recall that for a representation  $V$  of group  $H$ , we can obtain the  $H$ -invariant representation by taking  $\text{Hom}_{H\text{-mod}}(\text{triv}, V)$ , the invariant category of category  $\mathcal{C}$  acted on by  $\mathcal{L}N$  could be obtained similarly by taking

$$\mathcal{C}^{\mathcal{L}N,\chi} = \text{Maps}_{\mathcal{D}\text{-mod}(\mathcal{L}N)\text{-mod}}(\mathbf{Vect}_{\chi}, \mathcal{C})$$

while the coinvariant category is obtained by  $\mathcal{C}_{\mathcal{L}N,\chi} = \mathbf{Vect}_{\chi} \otimes_{\mathcal{D}\text{-mod}(\mathcal{L}N)\text{-mod}} \mathcal{C}$ . Unlike the averaging functor, the projection functor  $\mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}}) \rightarrow \text{Whit}_*(G)_{\underline{x}}$  is continuous.

We denote  $\mathcal{D}\text{-mod}(\mathcal{H}_{G,\underline{x}}^{\text{local}}) = \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}})^{\mathcal{L}_{\underline{x}}^+ G}$  by  $\text{Sph}_{G,\underline{x}}$ . Then  $\text{Sph}_{G,\underline{x}}$  acts on  $\text{Whit}^!(\text{Gr}_{G,\underline{x}})$  as  $\mathcal{L}_{\underline{x}}^+ G \backslash \mathcal{L}_{\underline{x}} G / \mathcal{L}_{\underline{x}} G$  has a convolution action on  $\text{Gr}_{G,\underline{x}}$  that is  $\mathcal{L}N$ -invariant with respect to  $\chi$ . In particular, by naïve geometric Satake, we have an action  $\text{Rep}(\check{G})_{\underline{x}} \curvearrowright \text{Whit}^!(G)_{\underline{x}}$ .

We denote by  $\text{Vac}_{\text{Whit},\underline{x}} \in \text{Whit}^!(G)_{\underline{x}}$  the factorization unit. When  $\underline{x} = x$  is a point,  $\text{Vac}_{\text{Whit}}$  is the  $*$ -extension of  $\Omega_{\text{Gr}_N} \otimes \chi^!(\text{exp}) \in \mathcal{D}\text{-mod}(\text{Gr}_N)^{\mathcal{L}N,\chi}$  to  $\text{Gr}_G$  under the inclusion  $\text{Gr}_N \subseteq \text{Gr}_G$ , who lies in  $\text{Whit}^!(G)$ .

**Remark 2.3.** The above extension is clean, namely the  $*$ -extension equals to  $!$ -extension.

We have also a factorization unit in  $\text{Whit}_*(G)_{\underline{x}}$ , which is the projection of  $\delta_{1_{\text{Gr}_G}}$  along  $\mathcal{D}\text{-mod}(\text{Gr}_G) \rightarrow \text{Whit}_*(G)$ .

**Theorem 2.1** (Casselman-Shalika). *Consider the following composition of functors:*

$$\text{Rep}(\check{G})_{\underline{x}} \xrightarrow{\text{Sat}^{\text{nv}}} \text{Sph}_{G,\underline{x}} \xrightarrow{(-)^* \text{Vac}_{\text{Whit},\underline{x}}} \text{Whit}^!(G)_{\underline{x}}$$

*It is an equivalence, whose inverse is called Casselman-Shalika, denoted by  $\text{CS}_{G,\underline{x}} : \text{Whit}^!(G)_{\underline{x}} \rightarrow \text{Rep}(\check{G})_{\underline{x}}$ .*

We have forgetful map  $\pi_{\underline{x}} : \text{Gr}_{G,\underline{x}} \rightarrow \text{Bun}_G$ , define the **Whittaker coefficient functor** on  $\underline{x}$  to be the composition:

$$\text{coeff}_{\underline{x}} : \mathcal{D}\text{-mod}(\text{Bun}_G) \xrightarrow{\pi_{\underline{x}}^!} \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}}) \xrightarrow{\text{Av}_{\underline{x},*}} \text{Whit}^!(G)_{\underline{x}}$$

An important feature for  $\text{coeff}_{\underline{x}}$  is that it is continuous, even though  $\text{Av}_*$  is not. That is because there exists large enough  $N_{\alpha} \subseteq \mathcal{L}N$  such that after composing  $\pi_{\underline{x}}$ , the transition maps are equivalent for all  $N_{\beta} \supseteq N_{\alpha}$ .

**Definition 2.2.** We define the coefficient functor  $\mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \mathbf{Vect}$  by

$$\text{coeff}^{\text{Vac}} : \mathcal{D}\text{-mod}(\text{Bun}_G) \xrightarrow{\text{coeff}_{\underline{x}}} \text{Whit}^!(G)_{\underline{x}} \hookrightarrow \mathcal{D}\text{-mod}(\text{Gr}_{G,\underline{x}}) \xrightarrow{!-\text{fiber}} \mathbf{Vect}$$

**Remark 2.4.** Here the definition of  $\text{coeff}^{\text{Vac}}$  does not depend on the choice of  $\underline{x} \in \text{Ran}$ , as  $\text{Whit}^!(G)$  is an unital factorization category (whose Definition would come later in Definition 3.2) and we can make a Ran version Whittaker coefficient functor (which is unital)  $\text{coeff}_{\text{Ran}} : \mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \text{Whit}^!(G)_{\text{Ran}}$  such that the following

diagram commutes:

$$\begin{array}{ccc}
D\text{-mod}(\text{Bun}_G) & \xrightarrow{\text{coeff}^{\text{Vac}}} & \mathbf{Vect} \\
\downarrow \text{coeff}_{\text{Ran}} & & \uparrow \\
\text{Whit}^!(G)_{\text{Ran}} & \xrightarrow{\quad} D\text{-mod}(\text{Gr}_{G,\text{Ran}}) \xrightarrow{(1_{\text{Gr}_{G,\text{Ran}}})^!} & D\text{-mod}(\text{Ran})
\end{array}$$

**Definition 2.3.** We define the  $!$ -Poincaré functor  $\text{Poinc}_{!,\underline{x}}$  on  $\underline{x}$  to be the left adjoint of  $\text{coeff}_{\underline{x}}$ .

Notice that the functor  $\text{Poinc}_{!,\underline{x}}$  is not a priori well-defined, we know  $\mathbf{Av}_{\underline{x},*}$  has a left-adjoint  $\mathbf{oblv}_{\underline{x}}$ , however we also need a well-defined left adjoint  $(\pi_{\underline{x}})_!$  of  $\pi_{\underline{x}}^!$ . Unfortunately  $(\pi_{\underline{x}})_!$  is not globally defined, but it is defined on  $\text{Whit}^!(G)_{\underline{x}}$ , which is enough for our purpose. We have two observations:

**Obs 1:** If  $(\pi_{\underline{x}})_!(F)$  is well-defined, then so is  $(\pi_{\underline{x}})_!(S \star F)$  for any  $S \in \text{Sph}_{G,\underline{x}}$ , as  $(\pi_{\underline{x}})_!(S \star F) = S \star (\pi_{\underline{x}})_!(F)$ .

**Obs 2:**  $(\pi_{\underline{x}})_!(\text{Vac}_{\text{Whit},\underline{x}})$  is defined, this is because  $\text{Vac}_{\text{Whit},\underline{x}}$  is holonomic, on which  $!$ -pushforward is always defined. So combine the two observations with Casselman-Shalika 2.1, we are done.

## 2.2 Kac-Moody modules(Gurbir Dhillon)

The goal of introducing Kac-Moody modules is to construct objects in  $D\text{-mod}(\text{Bun}_G)$ . There are 3 basic moves:

① is Eisenstein series, ② is Poincaré series and ③ is Kac-Moody localization.

To motivate, let us look at classical case of modular form on number field. Let  $\mathbb{H} = \{x + iy \mid x, y \in \mathbb{R}, y > 0\}$  be the upper half plane. There is certain differential operator (Laplacian)  $\Delta \propto y^2(\partial_x^2 + \partial_y^2)$  that plays well with modular functions, for example if we take Eisenstein series  $E_s$ , then  $\Delta(E_s) = s(1-s)E_s$ . The group we take here is  $\text{SL}_2(\mathbb{R})$  and  $\mathbb{H} = \text{SO}_2(\mathbb{R}) \backslash \text{SL}_2(\mathbb{R})$  is obtained by modulo its compact subgroup. To get a differential operator on  $\mathbb{H}/\text{SL}_2(\mathbb{Z})$ , we consider  $\text{Diff}(G)^{G \times G}$ , which is exactly the center  $Z(U(\mathfrak{g})) = \mathbb{R}[\frac{1}{2}h^2 + ef + fe] \cong \mathbb{R}[\Delta]$ . And  $\Delta - \lambda$ 's are the  $D$ -modules to look at, the microsupport of its solution could be obtained via moment map.

Now back to our case, we consider  $\mathfrak{g}(K_x)$  to be the Lie algebra of  $\mathcal{L}_x G$ , which is isomorphic to  $\mathfrak{g} \otimes \mathbb{C}((t))$ . We want to describe its center  $Z(U(\mathfrak{g}(K_x)))$ , which should be big in comparison with  $Z(U(\mathfrak{g})) \cong \text{Fun}(\check{t}/W) \cong \text{Fun}(\check{\mathfrak{g}}/\check{G})$ .

**Example 2.1.** (1) For example if we take  $G = T$ , then  $\mathfrak{t}(K_x)$  is commutative, so

$$U(\mathfrak{t}(K_x)) \cong \text{Sym}(\mathfrak{t}(K_x)) \cong \text{Fun}(\mathfrak{t}(K_x)^*) \cong \text{Fun}(\check{t} \otimes \Omega_x^1) \cong \text{Fun}(\text{Conn}_{\check{T}}(\check{D}_x))$$

where the duality between  $K_x$  and differential form is given simply by taking the residue.

(2) Take  $G = \text{SL}_2$ , then  $Z(U(\mathfrak{sl}_2)) = \mathbb{k}[\Delta]$ , while  $Z(\mathfrak{sl}_2(K_x)) = \mathbb{k}[\Delta_n]$ , by  $x \in \mathfrak{g}$ , we use  $x_n$  to denote  $x \otimes t^n$ . Moreover, we know the Casimir operator  $\Delta = \frac{1}{2}h^2 + ef + fe$ , while

$$\Delta_n = \sum_{i+j=n} \frac{1}{2} : h_i h_j : + : e_i f_j : + : f_j e_i :$$

where  $: x_n y_m : = \begin{cases} x_n y_m & n < 0 \\ y_m x_n & n \geq 0 \end{cases}$  is the OPE product.

Next question is which modules to use? We need to add some integrable conditions:

**Definition 2.4.** We define the **Kazhdan-Lusztig category** to be  $\text{KL} := (\mathfrak{g}(K_x), G(\mathcal{O}_x))\text{-mod}$ , meaning those  $\mathfrak{g}(K_x)$ -modules on which  $\mathfrak{g}(\mathcal{O}_x)$  can be integrated.

One typical example of elements in KL is given by induction, let  $\mathcal{V}$  be  $G(\mathcal{O}_x)$ -module, then  $\text{ind}_{\mathfrak{g}(\mathcal{O}_x)}^{\mathfrak{g}(K_x)} \mathcal{V} \in \text{KL}$ . Moreover, KL is compactly generated by induction of compact generators in  $\text{Rep}(G(\mathcal{O}_x))$ . Also, for  $\lambda \in \Lambda^+$ , associated with  $\mathcal{V}^\lambda$  irreducible  $G$ -module, we have the induction diagram  $G \leftarrow G(\mathcal{O}_x) \hookrightarrow G(K_x)$  Doing the pull-push we get induced representation in KL, denoted by

$$\mathbb{V}^\lambda := \text{ind}_{\mathfrak{g}(\mathcal{O}_x)}^{\mathfrak{g}(K_x)} \mathcal{V}^\lambda$$

The special one  $\mathbb{V}^0 = \text{Vac}(G)$  that induced from trivial representation is called the **vacuum representation**.

**Example 2.2.** If we take  $G = \mathbb{G}_m$ , the differential operator  $t\partial t$  acts on one dimensional vector space by  $\lambda \in \mathbb{C}$  defining a representation of a Lie algebra, it does not integrate unless  $\lambda \in \mathbb{Z}$ .

**Lemma 2.1.** A representation  $\mathcal{V} \in \mathfrak{g}(K_x)\text{-mod}^\heartsuit$  belongs to KL if and only if

(i)  $X_n, n > 0$ 's act locally nilpotently and (ii) When  $G$  is semisimple, simply connected,  $U(\mathfrak{g}) \cdot v$  is finite dimensional or when  $G$  is a torus,  $V$  has integral eigenvalues as in the previous example.

A beautiful fact is, for  $\mathcal{V}, \mathcal{W}$  two finite dimensional  $G(\mathcal{O}_x)$ -modules, we have  $\text{Hom}(\mathcal{V}, \mathcal{W}) \cong \text{Hom}(\mathcal{W}^*, \mathcal{V}^*)$  and  $\text{Hom}\left(\text{ind}_{\mathfrak{g}(\mathcal{O}_x)}^{\mathfrak{g}(K_x)}(\mathcal{V}), \text{ind}_{\mathfrak{g}(\mathcal{O}_x)}^{\mathfrak{g}(K_x)}(\mathcal{W})\right) \cong \text{Hom}\left(\text{ind}_{\mathfrak{g}(\mathcal{O}_x)}^{\mathfrak{g}(K_x)}(\mathcal{W}^*), \text{ind}_{\mathfrak{g}(\mathcal{O}_x)}^{\mathfrak{g}(K_x)}(\mathcal{V}^*)\right)$ . This comes from a self duality of the category  $\text{KL} \cong \text{KL}^\vee$  (note that  $\text{ind}(\mathcal{V}), \text{ind}(\mathcal{W})$  are compact generators), which is given by

$$\text{KL} \boxtimes \text{KL} \longrightarrow \mathbf{Vect} \quad \mathcal{M} \boxtimes \mathcal{N} \mapsto \mathbb{C}^{\frac{\infty}{2}+*}(\mathfrak{g}(K_x), G(\mathcal{O}_x), \mathcal{M} \otimes \mathcal{N}) \text{ relative semi-infinite cohomology}$$

the informal definition of  $\mathbb{C}^{\frac{\infty}{2}+*}(\mathfrak{g}(K_x), G(\mathcal{O}_x), -)$  is that we can either regard it as (i) group cohomology along  $G(\mathcal{O}_x)$  or (ii) Lie algebra homology along  $\mathfrak{g}(K_x)/\mathfrak{g}(\mathcal{O}_x)$ .

Next topic is the *Drinfeld-Sokolov reduction*, rough idea about this is: the inclusion  $Z(U(\mathfrak{g})) \hookrightarrow U(\mathfrak{g})$  is also a "quotient" in the sense of Hamiltonian reduction. We can consider the projection  $\mathfrak{g}^* \rightarrow \mathfrak{g}^*/\check{G}$ , it admits the

$$\text{Kostant section } S : \mathfrak{g}^*/\check{G} \rightarrow \mathfrak{g}^*. \text{ For example if } G = \text{GL}_n, \text{ then the Kostant slice } S = \begin{pmatrix} 0 & 0 & * \\ 1 & 0 & * \\ 0 & 1 & * \end{pmatrix} \hookrightarrow f + \mathfrak{b},$$

where  $f$  corresponds to lower subdiagonal. In fact, after composing  $f + \mathfrak{b} \rightarrow f + \mathfrak{b}/N$ , we get an isomorphism  $S \xrightarrow{\cong} f + \mathfrak{b}/N$ . We can define  $C.(\mathfrak{n}; - \otimes \mathbf{k}_\psi) : \mathfrak{g}\text{-mod} \rightarrow \mathbf{Vect}$  by:

(i) First reduction to  $\mathfrak{n}$ , (ii) tensor with a non-degenerate character  $\psi$ , then (iii) take Lie algebra homology. Such operations make  $C.(\mathfrak{n}; - \otimes \mathbf{k}_\psi)$  equivalent to  $\text{Hom}(\text{ind}_{\mathfrak{n}}^{\mathfrak{g}}(\mathbf{k}_\psi), -)$ , and we can see it factors as

$$\mathfrak{g}\text{-mod} \longrightarrow \text{End}(\text{ind}_{\mathfrak{n}}^{\mathfrak{g}}(\mathbf{k}_\psi))\text{-mod} \longrightarrow \mathbf{Vect}$$

The fact is that  $\text{End}(\text{ind}_{\mathfrak{n}}^{\mathfrak{g}}(\mathbf{k}_\psi)) \cong Z(U(\mathfrak{g}))$ , so that we can regard  $\mathfrak{g}\text{-mod} \longrightarrow \text{End}(\text{ind}_{\mathfrak{n}}^{\mathfrak{g}}(\mathbf{k}_\psi))\text{-mod}$  as a quantum Hamiltonian picture of  $Z(U(\mathfrak{g}))$ . In the loop Lie algebra case, the similar picture gives the Drinfeld-Sokolov reduction.

**Definition 2.5.** Define the **Drinfeld-Sokolov reduction** to be the functor

$$\mathrm{DS}_G := \mathbb{C}^{\frac{\infty}{2}+*}(\mathfrak{n}(K_X); - \otimes \mathbf{k}_\psi) : \mathfrak{g}(K_X)\text{-mod} \longrightarrow \mathbf{Vect}$$

**Remark 2.5.** Like in the finite dimensional construction, the Drinfeld-Sokolov functor is composition of restriction functor with a functor  $\mathcal{L}\mathfrak{n}\text{-mod} \rightarrow \mathbf{Vect}$ , which is called *BRST reduction*.

### 2.3 (Monodromy-free) Opers(Dima Arinkin)

The Kazhdan-Lusztig category introduced before plays the role of local picture on the automorphic side. On the level of functions, analogous to differential operators on modular functions, the differential operators associated with loop group acts on functions on  $\mathrm{Bun}_G$ . On the level of categories, we are going to see that  $\mathrm{KL}_X$  carries a localization functor to  $\mathcal{D}\text{-mod}(\mathrm{Bun}_G)$ .

Correspondingly, on the spectral side, the role of local picture is played by *opers*. In principle, we should consider  $\check{G}$ -opers, but this section does not automorphic side, so we consider  $G$ -opers if there is no ambiguity. A  $G$ -oper, informally speaking, is a  $B$ -torsor  $\mathcal{P}_B$  on  $X$  and a connection  $\nabla$  on induced  $G$ -torsor  $\mathcal{P}_G = \mathcal{P}_B \overset{B}{\times} G$ , with some discrepancy between  $\mathcal{P}_B$  and  $\nabla$ .

More precisely, given a  $B$ -torsor  $\mathcal{P}_B$ , can consider the sheaf of locally defined connection on it, denoted by  $\underline{\mathrm{Conn}}(\mathcal{P}_B)$ , which is a torsor over  $\mathfrak{b}_{\mathcal{P}_B} \otimes \Omega_X$ . Similarly,  $\underline{\mathrm{Conn}}(\mathcal{P}_G)$  a torsor over  $\mathfrak{g}_{\mathcal{P}_G} \otimes \Omega_X = \left(\mathfrak{g} \overset{B}{\times} \mathcal{P}_B\right) \otimes \Omega_X$ . We have an exact sequence

$$\underline{\mathrm{Conn}}(\mathcal{P}_B) \longrightarrow \underline{\mathrm{Conn}}(\mathcal{P}_G) \xrightarrow{\delta} (\mathfrak{g}/\mathfrak{b})_{\mathcal{P}_B} \otimes \Omega_X$$

where the first map is extending the connection along (the tangent of)  $\mathcal{P}_B \rightarrow \mathcal{P}_G$ , so we can see the cofiber is exactly the third term. We recall on  $\mathfrak{g}$  there is a decreasing filtration:

$$\mathfrak{g}^{-\mathrm{rk}} = \mathfrak{g}, \dots, \mathfrak{g}^{i+1} := [\mathfrak{g}^i, \mathfrak{n}], \dots, \mathfrak{g}^0 = \mathfrak{b}, \mathfrak{g}^1 = \mathfrak{n}, \dots$$

To define opers, we put restriction on the image of  $\delta(\nabla)$ , more precisely:

**Definition 2.6.** A  $G$ -oper on  $X$  contains the data of  $\left(\mathcal{P}_B, \nabla \text{ connection on } \mathcal{P}_G = \mathcal{P}_B \overset{B}{\times} G\right)$  such that

- (i) The induced connection  $\delta(\nabla) \in (\mathfrak{g}^{-1}/\mathfrak{b})_{\mathcal{P}_B} \otimes \Omega_X \cong \left(\bigoplus_{\alpha \text{ simple root}} \mathfrak{g}_{-\alpha}\right)_{\mathcal{P}_B} \otimes \Omega_X$ .
- (ii) Each component in  $(\mathfrak{g}_{-\alpha})_{\mathcal{P}_B} \otimes \Omega_X$  is non-vanishing.

**Example 2.3.** We can take  $G = \mathrm{SL}_3$  and let  $\mathcal{P}_B$  be a point in  $\mathrm{Bun}_X^{\Omega_X}$ . Then  $\mathcal{P}_G$  is a rank 3 local system, with filtration  $0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \mathcal{E}_2 \subseteq \mathcal{E}_3 = \mathcal{P}_G$  such that  $\mathcal{E}_1/\mathcal{E}_0 = \Omega_X, \mathcal{E}_2/\mathcal{E}_1 = \mathcal{O}_X, \mathcal{E}_3/\mathcal{E}_2 = \Omega_X^{-1}$ . If it admits an oper structure, then condition (i) requires that  $\nabla(\mathcal{E}_i) \subseteq \mathcal{E}_{i+1} \otimes \Omega_X$  which induces morphisms  $\mathcal{E}_i/\mathcal{E}_{i+1} \rightarrow (\mathcal{E}_{i+1}/\mathcal{E}_i) \otimes \Omega_X$ , they are not just non-zero maps, but isomorphisms compatible with the flag structure.

Very informally, opers are geometrization of those locally of the form  $\frac{d}{dt} + \begin{pmatrix} * & * & * \\ 1 & * & * \\ 0 & 1 & * \end{pmatrix}$ , which we may call

”rational canonical form for connections”. Under some assumption on (for example semisimpleness) groups, the stack of opers embedding into stack of local systems with half dimension.

One should bear in mind that global  $\check{G}$ -opers have relation with  $D$ -modules on  $\text{Bun}_G$  while local opers has something to do with Kac-Moody modules. Ideally one would consider a Ran version of it, but now let us fix a point  $x \in X$ . On the disc  $D_x$ , we consider following spaces of opers (we take  $G = \text{GL}_3$  as example)

| Opers                                                             | Local coordinate                                                                                                                                                                                                                                      |                                                                                                                            |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| $\text{Op}^{\text{reg}}(D_x)$<br><i>regular opers</i>             | $\frac{d}{dx} + \begin{pmatrix} a_1(x) & a_2(x) & a_3(x) \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad a_1, a_2, a_3 \in \mathbb{C}[[x]]$<br>Equivalently $\left( \frac{d^3}{dx^3} + a_1(x) \frac{d^2}{dx^2} + a_2(x) \frac{d}{dx} + a_3(x) \right)$ | The space of regular opers is pro-affine scheme $\cong \mathbb{A}^\infty$ ,                                                |
| $\text{Op}^{\text{mer}}(D_x)$<br><i>meromorphic opers</i>         | We take coordinates $a_i \in \mathbb{C}((x)) = \bigcup_{n \rightarrow \infty} x^{-n} \mathbb{C}[[x]]$                                                                                                                                                 | The space of meromorphic opers is an ind scheme $= \bigcup \text{Op}^{\text{mer}, \leq m} \cong \bigcup \mathbb{A}^\infty$ |
| $\text{Op}^{\text{mon-free}}(D_x)$<br><i>monodromy-free opers</i> | We take $a_i(x) \in \mathbb{C}((x))$ such that there are 3 linearly independent solutions in $\mathbb{C}((x))$                                                                                                                                        | $\cong \text{Op}^{\text{mer}}(D_x) \times_{\text{LocSys}_G(\dot{D}_x)} \text{LS}_G(D_x)$                                   |

Table 1: Variation of spaces of opers

So we can see that regular opers  $\text{Op}^{\text{reg}}(D_x)$  is exactly we translate the Definition 2.6 to  $D_x$ , while  $\text{Op}^{\text{mer}}(D_x)$  are opers on the punctured disc. The space of monodromy-free opers is defined to be

$$\text{Op}^{\text{mer}}(D_x) \times_{\text{LocSys}_G(\dot{D}_x)} \text{LocSys}_G(D_x)$$

The space  $\text{Op}^{\text{mon-free}}(D_x)$  has some good properties: (i) It is formally smooth; (ii) It is a union of closed subschemes, each one is written as a product  $Y \times \mathbb{A}^\infty$ , where  $Y$  is of finite type.

The next question is, which sheaf category we are considering for  $Y \times \mathbb{A}^\infty$ ? Answer:

$$\text{IndCoh}^*(Y \times \mathbb{A}^\infty) := \varinjlim \text{IndCoh}(Y \times \mathbb{A}^n)$$

where the colimit is taken under  $*$ -pullback.

## 2.4 Introduction to factorization(Kevin Lin)

**Definition 2.7.** (1) A **factorization space** is a prestack  $\mathcal{Y}_{\text{Ran}}$  over  $\text{Ran}$ , equipped with isomorphisms

$$\mathcal{Y}_{\underline{x} \amalg \underline{x}'} \cong \mathcal{Y}_{\underline{x}} \times \mathcal{Y}_{\underline{x}'}$$

for any disjoint  $\underline{x}, \underline{x}' \in \text{Ran}$ .

(2) A **factorization category** is a sheaf of categories  $\underline{C}_{\text{Ran}}$  over  $\text{Ran}$  with equivalences

$$C_{\underline{x} \amalg \underline{x}'} \cong C_{\underline{x}} \otimes C_{\underline{x}'}$$

for any disjoint  $\underline{x}, \underline{x}' \in \text{Ran}$ .

**Example 2.4.** Let  $\mathcal{Y}_{\text{Ran}}$  be a factorization space, then  $\mathcal{D}\text{-mod}(\mathcal{Y}_{\text{Ran}}) = \mathcal{D}\text{-mod}(\mathcal{Y})_{\text{Ran}}$  is a factorization category. Also, we can take  $\text{IndCoh}^*(\mathcal{Y}_{\text{Ran}}) = \text{IndCoh}^*(\mathcal{Y})_{\text{Ran}}$ . If we take  $\mathcal{Y}_{\text{Ran}} = \text{Ran}$ , then we denoted the corresponding factorization category by  $\mathbf{Vect}_{\text{Ran}}$ .

**Definition 2.8.** Let  $\underline{\mathcal{C}}_{\text{Ran}}$  be a factorization category, a **factorization algebra** in  $\underline{\mathcal{C}}_{\text{Ran}}$  is a  $\mathcal{D}$ -module  $\mathcal{A}_{\text{Ran}}$  on  $\text{Ran}$  (i.e. in the global section of  $\underline{\mathcal{C}}_{\text{Ran}}$ ) such that for disjoint  $\underline{x}, \underline{x}' \in \text{Ran}$ :

$$\mathcal{A}_{\underline{x} \amalg \underline{x}'} \cong \mathcal{A}_{\underline{x}} \otimes \mathcal{A}_{\underline{x}'}$$

**Example 2.5.** The dualizing sheaf  $\omega_{\text{Ran}}$  is a factorization algebra.

**Definition 2.9.** For  $\underline{\mathcal{C}}_{\text{Ran}}, \underline{\mathcal{D}}_{\text{Ran}}$  two factorization categories, a functor  $F_{\text{Ran}} : \underline{\mathcal{C}}_{\text{Ran}} \rightarrow \underline{\mathcal{D}}_{\text{Ran}}$  is a factorization functor if for any disjoint  $\underline{x}, \underline{x}' \in \text{Ran}$ , the following diagram commutes.

$$\begin{array}{ccc} \underline{\mathcal{C}}_{\underline{x} \amalg \underline{x}'} & \xrightarrow{\cong} & \underline{\mathcal{C}}_{\underline{x}} \otimes \underline{\mathcal{C}}_{\underline{x}'} \\ \downarrow F_{\underline{x} \amalg \underline{x}'} & & \downarrow F_{\underline{x}} \otimes F_{\underline{x}'} \\ \underline{\mathcal{D}}_{\underline{x} \amalg \underline{x}'} & \xrightarrow{\cong} & \underline{\mathcal{D}}_{\underline{x}} \otimes \underline{\mathcal{D}}_{\underline{x}'} \end{array}$$

It is easy to see that a factorization functor  $F_{\text{Ran}}$  sends factorization algebra to factorization algebra.

Now let  $\underline{x}$  be an  $S$ -point of  $\text{Ran}$ , denote by  $\Gamma_{\underline{x}}$  the union of graphs of points in  $\underline{x}$ ,  $D_{\underline{x}}$  formal completion at  $\Gamma_{\underline{x}}$  and  $\mathring{D}_{\underline{x}} = D_{\underline{x}} \setminus \Gamma_{\underline{x}}$ . Let  $\mathcal{Y} \rightarrow X$  be a prestack equipped with a connection (i.e.  $\mathcal{Y}$  is pullback from some  $\mathcal{Y}_{\text{dR}} \rightarrow X_{\text{dR}}$ ). We associate it with a factorization space  $(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}} \rightarrow \text{Ran}$  such that for  $\underline{x} : S \rightarrow \text{Ran}$

$$(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\underline{x}} = \{ \text{horizontal section of } \mathcal{Y} \text{ over } D_{\underline{x}} \}$$

If  $x_1, \dots, x_n$  distinct  $\mathbf{k}$ -points on  $X$ , then  $(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{x_1 \amalg \dots \amalg x_n} \cong \prod_{i=1}^n \mathcal{Y}_{x_i}$ .

In the sequel, we will assume that  $\mathcal{Y}$  is affine over  $X$ , we define another factorization space  $(\mathcal{L}_{\nabla} \mathcal{Y})_{\text{Ran}} \rightarrow \text{Ran}$ :

$$(\mathcal{L}_{\nabla} \mathcal{Y})_{\underline{x}} = \{ \text{horizontal section of } \mathcal{Y} \text{ over } \mathring{D}_{\underline{x}} \}$$

Both factorization structures on  $(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}}$  or  $(\mathcal{L}_{\nabla} \mathcal{Y})_{\text{Ran}}$  come from the splitting property  $D_{\underline{x} \amalg \underline{x}'} \cong D_{\underline{x}} \amalg D_{\underline{x}'}$  (resp. the one for  $\mathring{D}_{\underline{x}}$ ). We know that both  $\text{IndCoh}^*(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}}$  and  $\text{IndCoh}^* \text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_{\text{Ran}}$  are factorization categories and inside of the former one,  $\mathcal{O}_{(\mathcal{L}_{\nabla} \mathcal{Y})_{\text{Ran}}}$  is a factorization algebra. If  $\mathcal{Y}$  is affine over  $X$ , then there is a canonical embedding:  $\iota : (\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}} \hookrightarrow (\mathcal{L}_{\nabla} \mathcal{Y})_{\text{Ran}}$  such that

$$\iota_* : \text{IndCoh}^*(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}} \rightarrow \text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_{\text{Ran}}$$

is a factorization functor and thus  $\iota_* \left( \mathcal{O}_{(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}}} \right)$  is also a factorization algebra.

**Why we need  $\mathcal{Y}$  affine over  $X$ ?**

**Remark 2.6.** In general,  $(\mathcal{L}_{\nabla} \mathcal{Y})_{\underline{x}}$  is an ind-scheme of ind-infinite type.



**Definition 2.10.** Let  $\underline{x} \in X$  be a point, we define  $\text{Ran}_{\underline{x}} \subseteq \text{Ran}$  whose points are those who contain  $\underline{x}$ . A **factorization module** for a factorization algebra  $\mathcal{A}_{\text{Ran}}$  (in a factorization category  $\mathcal{C}_{\text{Ran}}$ ) is a  $\mathcal{D}$ -module  $\mathcal{M}_{\text{Ran}_{\underline{x}}}$ , i.e. a global section of  $\mathcal{C}_{\text{Ran}_{\underline{x}}}$  such that for any  $\underline{y} \in \text{Ran}_{\underline{x}}$  and  $\underline{y}' \in \text{Ran}$  disjoint

$$\mathcal{M}_{\underline{y} \amalg \underline{y}'} \cong \mathcal{M}_{\underline{y}} \otimes \mathcal{A}_{\underline{y}'}$$

**Example 2.6.** Assume  $\mathcal{A}_{\text{Ran}}$  is factorization algebra, then  $\mathcal{A}_{\text{Ran}_{\underline{x}}}$  is a factorization module for  $\mathcal{A}_{\text{Ran}}$ .

**Claim:** There exists a lift  $\text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_x \rightarrow \mathcal{O}_{(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}}} \text{-mod}_x^{\text{fact}}$  such that the following diagram commutes:

$$\begin{array}{ccc} & & \mathcal{O}_{(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}}} \text{-mod}_x^{\text{fact}} \\ & \nearrow & \downarrow \text{fiber at } x \\ \text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_x & \xrightarrow{\Gamma} & \mathbf{Vect} \end{array}$$

It suffice to construct a functor  $\text{ins.vac} : \text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_x \rightarrow \text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_{\text{Ran}_{\underline{x}}}$  which fiberwise sends  $\mathcal{F} \mapsto \mathcal{F} \boxtimes \iota_* \left( \mathcal{O}_{\mathcal{L}_{\nabla}^+ \mathcal{Y}, \underline{y}_1} \right) \boxtimes \dots \boxtimes \iota_* \left( \mathcal{O}_{\mathcal{L}_{\nabla}^+ \mathcal{Y}, \underline{y}_n} \right)$ . The key input is: if  $\underline{y} \in \text{Ran}_{\underline{x}}$ , then we have inclusions  $\mathring{D}_x \hookrightarrow D_{\underline{y}} \setminus x \hookleftarrow \mathring{D}_{\underline{y}}$  which induces the correspondence  $(\mathcal{L}_{\nabla} \mathcal{Y})_x \leftarrow \text{Sect}_{\nabla} \left( D_{\underline{y}} \setminus x, \mathcal{Y} \right) \rightarrow (\mathcal{L}_{\nabla} \mathcal{Y})_{\underline{y}}$ . Doing pull-push for  $\text{IndCoh}$ , we obtain the functor  $\text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_x \rightarrow \text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_{\underline{y}}$  and we allow  $\underline{y}$  to vary. For example if we take  $\underline{y} = x \amalg \underline{y}_1 \amalg \dots \amalg \underline{y}_n$ , then the above inclusions become  $\mathring{D}_x \hookrightarrow \mathring{D}_x \amalg D_{\underline{y}_1} \amalg \dots \amalg D_{\underline{y}_n} \hookleftarrow \mathring{D}_x \amalg \mathring{D}_{\underline{y}_1} \amalg \dots \amalg \mathring{D}_{\underline{y}_n}$  which induce

$$(\mathcal{L}_{\nabla} \mathcal{Y})_x \leftarrow (\mathcal{L}_{\nabla} \mathcal{Y})_x \times (\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\underline{y}_1} \times \dots \times (\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\underline{y}_n} \rightarrow (\mathcal{L}_{\nabla} \mathcal{Y})_x \times (\mathcal{L}_{\nabla} \mathcal{Y})_{\underline{y}_1} \times \dots \times (\mathcal{L}_{\nabla} \mathcal{Y})_{\underline{y}_n}$$

which provides us the factorization module structure.

**Theorem 2.2.** The functor  $\text{IndCoh}^*(\mathcal{L}_{\nabla} \mathcal{Y})_x \rightarrow \left( \mathcal{O}_{(\mathcal{L}_{\nabla}^+ \mathcal{Y})_{\text{Ran}}} \right) \text{-mod}_x^{\text{fact}}$  is an equivalence of categories on the eventually coconnective part.

We say a unital structure on factorization category  $\mathcal{C}_{\text{Ran}}$  is a factorization algebra  $\text{unit}_{\mathcal{C}_{\text{Ran}}}$  (called factorization unit) with ability to assemble  $C_x \rightarrow C_x \otimes C_{\underline{y}_1} \otimes \dots \otimes C_{\underline{y}_n}; \mathcal{F} \mapsto \mathcal{F} \boxtimes \text{unit}_{C_{\underline{y}_1}} \boxtimes \dots \boxtimes \text{unit}_{C_{\underline{y}_n}}$  into a functor  $\text{ins.vac} : C_x \rightarrow \mathcal{C}_{\text{Ran}_{\underline{x}}}$ .

**Example 2.7.** Consider Kac-Moody modules  $\hat{\mathfrak{g}}_{\kappa} \text{-mod}$  (definition comes later in Definition 2.11), then it is a factorization category with factorization unit  $\text{Vac}(G)_{\kappa}$  and  $\hat{\mathfrak{g}}_{\kappa} \text{-mod} \xrightarrow{\text{oblv}} \mathbf{Vect}$  is a factorization functor. As a result, we know  $\mathbb{V}_{\mathfrak{g}, \kappa} := \text{oblv}(\text{Vac}(G)_{\kappa})$  is a factorization algebra. Similar to 2.2, we have following diagram

$$\begin{array}{ccc} & & \mathbb{V}_{\mathfrak{g}, \kappa} \text{-mod}_x^{\text{fact}} \\ & \nearrow & \downarrow \\ \hat{\mathfrak{g}}_{\kappa} \text{-mod}_x & \xrightarrow{\quad} & \mathbf{Vect} \end{array}$$

**Theorem 2.3.** The functor  $\hat{\mathfrak{g}}_{\kappa} \text{-mod}_x \rightarrow \mathbb{V}_{\mathfrak{g}, \kappa} \text{-mod}_x^{\text{fact}}$  is a factorization equivalence on eventually coconnective part.

## 2.5 Feigin-Frenkel isomorphism and FLE(Yifei Zhao)

The goal is to prove the following theorem:

**Theorem 2.4.** *There is a canonical equivalence, called **fundamental local equivalence (FLE)** at critical level, of unital factorization categories*

$$\mathbf{FLE}_{\text{crit}} : \mathbf{KL}(G)_{\text{crit}} \cong \mathbf{IndCoh}^* \left( \mathbf{Op}_G^{\text{mon-free}} \right)$$

Let  $\kappa$  be a  $G$ -invariant symmetric bilinear form  $\kappa : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbf{k}$ , we call it a *level* and *critical level* is  $-\frac{1}{2}$  killing.

**Definition 2.11.** Consider the central extension of  $\mathfrak{g}(K_x)$  by level  $\kappa$

$$0 \longrightarrow \mathbf{k} \cdot \mathbf{1} \longrightarrow \hat{\mathfrak{g}}_\kappa \longrightarrow \mathfrak{g}(K_x) \longrightarrow 0$$

The Kac-Moody Lie algebra  $\hat{\mathfrak{g}}_\kappa$  at level  $\kappa$  is defined by the cocycle  $(\xi \otimes f, \xi \otimes f') \mapsto \kappa(\xi, \xi') \text{Res}((df) \cdot f')$ .

**Definition 2.12.** We define the **Kazhdan-Lusztig category** at level  $\kappa$  to be  $\mathbf{KL}(G)_\kappa := (\hat{\mathfrak{g}}\text{-mod}_\kappa)^{\mathcal{L}^+G}$ . In particular, we get Kazhdan-Lusztig category at critical level  $\mathbf{KL}(G)_{\text{crit}}$ .

**Remark 2.7.** We are going to see there exists a localization functor  $\text{Loc} : \mathbf{KL}(G)_{\text{crit}} \longrightarrow \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_G)$ , where the later is the category of twisted  $\mathcal{D}$ -modules on  $\text{Bun}_G$  (in fact by  $\rho(\Omega_X)$ ), which is isomorphic to  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  due to construction called Pfaffian line bundle (cf. [BD]) which depends on the choice of  $\Omega_X^{\frac{1}{2}}$ .

Also on spectral side, there is naïve Satake functor  $\text{Sat}^{\text{nv}} : \text{Rep}(\check{G})^\heartsuit \xrightarrow{\cong} \mathcal{D}\text{-mod}_{\text{crit}}(\mathcal{H}_{G,x}^{\text{loc}})^\heartsuit$  which is symmetric monoidal, and there is further isomorphism  $\mathcal{D}\text{-mod}_{\text{crit}}(\mathcal{H}_{G,x}^{\text{loc}})^\heartsuit \cong \mathcal{D}\text{-mod}(\mathcal{H}_{G,x}^{\text{loc}})^\heartsuit$  but with modified commutativity constraint. So actually the twists are always there in the whole story.

Recall that locally over a disc  $D_x$ , we have

$$\begin{array}{ccccc} & & \iota^+ & & \\ & \nearrow & & \searrow & \\ \text{Op}_G^{\text{reg}} & \xrightarrow{\iota^+, \text{mon-free}} & \text{Op}_G^{\text{mon-free}} & \xrightarrow{\iota} & \text{Op}_G^{\text{mer}} \\ & & \downarrow & & \downarrow \\ \text{pt}/\check{G} \cong \text{LocSys}_{\check{G}}^{\text{reg}}(D_x) & \longrightarrow & & \longrightarrow & \text{LocSys}_{\check{G}}^{\text{mer}}(D_x) \cong \check{\mathfrak{g}}/\check{G} \end{array}$$

The FLE functor  $\mathbf{FLE}_{\text{crit}}$  is supposed to be unital. On the LHS, we have mentioned that the factorization unit is  $\text{Vac}(G)_{\text{crit}}$ , on the RHS, it should be  $(\iota^+, \text{mon-free})_*^{\text{IndCoh}}(\mathcal{O}_{\text{Op}_G^{\text{reg}}})$ . In particular at point  $x \in X$ , we have

$$\text{End}(\text{Vac}(G)_{\text{crit}}) \cong \text{End} \left( (\iota^+, \text{mon-free})_*^{\text{IndCoh}}(\mathcal{O}_{\text{Op}_G^{\text{reg}}}) \right)$$

We take  $H^0$  on both sides, then we shall get the **Feigin-Frenkel isomorphism** at  $x$ :

$$\text{ind}_{\hat{\mathfrak{g}}(\mathcal{O}_x)}^{\hat{\mathfrak{g}}_{\text{crit}}}(\mathbf{k})^{G(\mathcal{O}_x)} \cong \Gamma \left( \text{Op}_{G,x}^{\text{reg}}, \mathcal{O} \right)$$

The proof of Theorem 2.4 divided into the following steps.

**Step 1:** We establish the Feigin-Frenkel isomorphism. Let  $\mathfrak{Z}_{\mathfrak{g},\text{crit}}$  be the center of the factorization algebra  $\mathbf{V}_{\mathfrak{g},\kappa} = \text{oblv}(\text{Vac}(G)_{\text{crit}})$ , we will construct an isomorphism of  $\mathcal{D}_X$ -algebras  $\mathfrak{Z}_{\mathfrak{g},\text{crit}} \cong \mathcal{O}_{\text{Op}_{\check{G}}}$ , where  $\text{Op}_{\check{G}}$  is an affine  $\mathcal{D}$ -scheme over  $X$  (or we can say affine over  $X_{\text{dR}}$ ). So we can consider its local horizontal sections, which turn out to be

$$\mathcal{L}_{\check{V}}^+(\text{Op}_{\check{G}}) \cong \text{Op}_{\check{G}}^{\text{reg}}; \quad \mathcal{L}_{\check{V}}(\text{Op}_{\check{G}}) \cong \text{Op}_{\check{G}}^{\text{mer}}$$

**Step 2:** We want to construct a functor (called enhanced Drinfeld-Sokolov reduction)

$$\text{DS}_G^{\text{enh}} : \text{KL}(G)_{\text{crit}} \longrightarrow \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}})$$

**Step 3:** We want to show that  $\text{DS}_G^{\text{enh}}$  factors through an equivalence

$$\text{FLE}_{\text{crit}} : \text{KL}(G)_{\text{crit}} \xrightarrow{\cong} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}})$$

For **Step 1**, let us denote  $\hat{D}_0 := \text{Spf } \mathbf{k}[[t]]$  the formal disc,  $\text{Aut}$  the automorphism group of  $\hat{D}_0$  and  $\text{Aut}^+ \subseteq \text{Aut}$  that preserves the base point. We have that  $\text{Aut}$  is a group ind-scheme while  $\text{Aut}^+$  is a group scheme, the quotient  $\text{Aut} / \text{Aut}^+ \cong \hat{D}_0$ . There is a tautological  $\text{Aut}$ -torsor over  $X_{\text{dR}}$ , denoted by  $\hat{X}_{\text{dR}} = \{x \in X_{\text{dR}} \mid \hat{D}_x \cong \hat{D}_0\}$  so that pullback to  $X$  is induced from an  $\text{Aut}^+$ -torsor. We denote by  $\mathfrak{Z}_{\mathfrak{g},\text{crit},0} = \text{ind}_{\mathbf{k}[[t]]}^{\hat{\mathfrak{g}}_{\text{crit}}}(\mathbf{k})^{G[[t]]}$  the center at  $\hat{D}_0$ , which is acted on by  $\text{Aut}$ , and  $\text{Op}_{\check{G},0}$  the affine scheme parametrizing  $\check{G}$ -opers on  $\hat{D}_0$ , also acted on by  $\text{Aut}$ . (small caveat here: as the construction involves square root of  $\Omega$ , we actually need a double cover of  $\text{Aut}$ ).

As the rest are twists by  $\text{Aut}$ -torsor, it suffices to construct an  $\text{Aut}$ -equivariant isomorphism  $\text{Spec}(\mathfrak{Z}_{\mathfrak{g},\text{crit},0}) \cong \text{Op}_{\check{G},0}$  via so-called **Birth of Opers**. Note that an  $\text{Aut}$ -equivariant map  $\text{Spec}(\mathfrak{Z}_{\mathfrak{g},\text{crit},0}) \rightarrow \text{Op}_{\check{G},0}$  is equivalent to  $\check{G}$ -oper on  $\text{Spec}(\mathfrak{Z}_{\mathfrak{g},\text{crit},0}) \times \hat{D}_0$  with  $\text{Aut}$ -equivariance.

The general paradigm is if  $H^+ \subseteq H \curvearrowright Y$ , then  $Y \times^H H/H^+ \cong Y/H^+$  and they map to  $Y \times^H H/H \cong Y/H$ . Inspired by the paradigm, we know it suffices to: (i) construct an  $\text{Aut}^+$ -equivariant  $\check{B}$ -torsor  $\mathcal{P}_{\check{B}}$  with  $(\mathcal{P}_{\check{B}})_{\check{T}} \xrightarrow{\cong} \rho(\Omega)$  over  $\text{Spec}(\mathfrak{Z}_{\mathfrak{g},\text{crit},0})$  then (ii) upgrade  $\text{Aut}^+$ -equivalence on  $(\mathcal{P}_{\check{B}})_{\check{T}}$  to an  $\text{Aut}$ -equivalence. Now by Tannakian, a  $\check{G}$ -torsor over  $\text{Spec}(\mathfrak{Z}_{\mathfrak{g},\text{crit},0})$  is a monoidal functor  $\text{Rep}(\check{G}) \rightarrow \mathfrak{Z}_{\mathfrak{g},\text{crit},0}\text{-mod}$ . The functor we will take is

$$\mathcal{V} \mapsto \Gamma^{\text{IndCoh}}(\text{Gr}_G, \text{Sat}^{\text{nv}}(\mathcal{V}))^{G[[t]]}$$

Notice it is important to have naïve Satake lands in twisted  $\mathcal{D}$ -module  $\mathcal{D}\text{-mod}_{\text{crit}}(\text{Gr}_G)$  such that we could get the induced module at critical level. As a consequence of the construction, we get

$$\text{Sat}^{\text{nv}}(\mathcal{V}) \star \text{Vac}(G)_{\text{crit}} \xrightarrow{\cong} \mathfrak{t}^*(\mathcal{V}) \otimes_{\mathcal{O}_{\text{Op}_{\check{G}}}^{\text{reg}}} \text{Vac}(G)_{\text{crit}}$$

where  $\mathfrak{t} : \text{Op}_{\check{G}}^{\text{reg}} \rightarrow \text{LocSys}_{\check{G}}^{\text{reg}} = \text{pt}/\check{G}$ . For **Step 2**, we have affine Skryabin theorem (see [Ras21, Theorem 5.1.1]) that the Drinfeld-Sokolov functor  $\text{DS} : \hat{\mathfrak{g}}_{\kappa}\text{-mod} \rightarrow \mathbf{Vect}$  induces an equivalence:

$$\text{DS}^{\text{enh}} : \text{Whit}_*(\hat{\mathfrak{g}}_{\kappa}\text{-mod}) \longrightarrow (\text{DS}(\text{Vac}(G)_{\kappa})\text{-mod}^{\text{fact}})_{\text{ren}}$$

where on the RHS we need some renormalized factorization category and we have seen before how to define Whittaker category associated with category  $\mathcal{C}$  acted on by  $\mathcal{L}N$  in talk 2.1. Moreover, at critical level  $\kappa = \text{crit}$ , we have

$$\text{DS}(\text{Vac}(G)_{\text{crit}}) \cong \mathfrak{Z}_{\mathfrak{g}, \text{crit}, 0} \cong \mathcal{O}_{\text{Op}_G^{\text{reg}}}$$

where the first equivalence is affine Kostant theorem and the second one is Feigin-Frenkel. So we can construct:

$$\text{DS}^{\text{enh}} : \hat{\mathfrak{g}}_{\text{crit}}\text{-mod} \longrightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mer}})$$

where we use the fact 2.2 that  $\text{IndCoh}^*(\text{Op}_G^{\text{mer}}) \approx \mathcal{O}_{\text{Op}_G^{\text{reg}}}\text{-mod}^{\text{fact}}$ , and the desired functor in **Step 2** is

$$\text{KL}(G)_{\text{crit}} = (\hat{\mathfrak{g}}_{\text{crit}}\text{-mod})^{\mathcal{L}^+G} \longrightarrow \hat{\mathfrak{g}}_{\text{crit}}\text{-mod} \xrightarrow{\text{DS}_G^{\text{enh}}} \text{IndCoh}^*(\text{Op}_G^{\text{mer}})$$

## 2.6 Bonus B: proof of the FLE(Dennis Gaitsgory)

Now we are going to wrap up the previous talks and give a complete proof of FLE. We have seen:

(1)  $\text{KL}$  is a  $\text{Sph}$ -module category, let  $\star : \text{Sph} \otimes \text{KL} \rightarrow \text{KL}$  be the action and there is an isomorphism

$$\text{Sat}^{\text{nv}}(\mathcal{V}) \star \mathbb{V}_{\text{crit}} \cong \mathfrak{t}^*(\mathcal{V}) \otimes_{\mathfrak{Z}} \mathbb{V}_{\text{crit}}$$

(2)  $\mathbb{V}_{\text{crit}} \in \text{KL}(G)_{\text{crit}} \hookrightarrow \text{KM}(G)_{\text{crit}} = \hat{\mathfrak{g}}_{\text{crit}}\text{-mod}$ , composing with  $\text{DS}_G$  we obtain  $\mathcal{O}_{\text{Op}_G^{\text{reg}}} \cong \mathfrak{Z}_{\mathfrak{g}, \text{crit}, 0} \cong \text{DS}(\mathbb{V}_{\mathfrak{g}, \text{crit}})$ .  
(3)  $\text{DS}$  automatically enhances to a functor  $\text{KM}(G)_{\text{crit}} \rightarrow \text{DS}(\mathbb{V}_{\mathfrak{g}, \text{crit}})\text{-mod}^{\text{fact}}$ . We can conclude with the following diagram of categories:

$$\begin{array}{ccccc} \text{KL}(G)_{\text{crit}} & \xrightarrow{\quad} & \text{KM}(G)_{\text{crit}} & \xrightarrow{\quad \text{DS}^{\text{enh}} \quad} & \text{DS}(\mathbb{V}_{\mathfrak{g}, \text{crit}})\text{-mod}^{\text{fact}} \\ \text{FLE}_{\text{crit}} \downarrow \text{---} & & & & \uparrow \cong \\ \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}) & \xrightarrow{\iota_*} & \text{IndCoh}^*(\text{Op}_G^{\text{mer}}) & \xrightarrow{\approx} & \mathcal{O}_{\text{Op}_G^{\text{reg}}}\text{-mod}^{\text{fact}} \xrightarrow{\text{FF}} \mathfrak{Z}_{\mathfrak{g}, \text{crit}}\text{-mod}^{\text{fact}} \end{array}$$

For any category  $\mathcal{C}$  acted on by  $\mathcal{L}G$ , we denote by  $\text{Sph}(\mathcal{C}) := \mathcal{C}^{\mathcal{L}^+G}$ . For example if  $\mathcal{C} = \text{KM}$ , then  $\text{Sph}(\mathcal{C}) = \text{KL}$ , if  $\mathcal{C} = \mathcal{D}\text{-mod}(\mathcal{L}G)$ , then  $\text{Sph}(\mathcal{C}) \cong \mathcal{D}\text{-mod}(\mathcal{L}G/\mathcal{L}^+G)$ , which is acted on by  $\text{Sph}(G) = \mathcal{D}\text{-mod}(\mathcal{L}^+G \setminus \mathcal{L}G/\mathcal{L}^+G)$ . Also we write  $\mathcal{C}^{\text{sph-gen}} := \mathcal{D}\text{-mod}(\text{Gr}_G) \otimes_{\text{Sph}_G} \text{Sph}(\mathcal{C}) \hookrightarrow \mathcal{C}$ , the fully-faithful embedding has a right adjoint  $\mathcal{C} \rightarrow \mathcal{C}^{\text{sph-gen}}$  and both arrows are compatible with action of  $\mathcal{L}G$ .

Also, recall we have constructed  $\text{Whit}^!(\mathcal{C})$  and  $\text{Whit}_*(\mathcal{C})$  for  $\mathcal{C}$ . We have mentioned that if we take  $\mathcal{C} = \text{KM}$ , then  $\text{DS}_G^{\text{enh}} : \text{KM} \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mer}})$  factors through  $\text{Whit}_*(\mathcal{C})$ .

**Theorem 2.5** (Raskin [Ras21]). *We denote  $\overline{\text{DS}_G^{\text{enh}}} : \text{Whit}_*(\text{KM}) \rightarrow \text{IndCoh}^*(\text{Op}_G^{\text{mer}})$ , then it is an equivalence.*

What we also need is

$$\text{Whit}_*(G) \otimes_{\text{Sph}_G} \text{Sph}(\mathcal{C}) = \text{Whit}_*(G) \otimes_{\text{Sph}_G} \mathcal{C}^{\mathcal{L}^+G} \cong \text{Whit}_*(\mathcal{C}^{\text{sph-gen}}) \hookrightarrow \text{Whit}_*(\mathcal{C}) \quad (\star)$$

We can replace those Sph-categories with spectral version (due to derived Satake):

$$\begin{array}{ccc}
 \text{Rep}(\check{G}) & \xrightarrow{\text{Sat}^{\text{nv}}} & \text{Sph}_G \\
 \downarrow & \nwarrow \text{Sat} & \\
 \text{IndCoh}^* \left( \text{pt} / \check{G} \times_{\check{g} / \check{G}} \text{pt} / \check{G} \right) & =: & \text{Sph}_G^{\text{spec}}
 \end{array}$$

where we have  
 $\text{LocSys}_{\check{G}}^{\text{reg}} = \text{pt} / \check{G}$  and  $\text{LocSys}_{\check{G}}^{\text{mer}} = \check{g} / \check{G}$

And  $\text{Sph}_G^{\text{spec}}$  surjects to the tempered part  $\text{Sph}_{G, \text{temp}}^{\text{spec}} := \text{QCoh} \left( \text{pt} / \check{G} \times_{\check{g} / \check{G}} \text{pt} / \check{G} \right)$ . If  $\tilde{C}$  is acted on by  $\text{Sph}_G^{\text{spec}}$ , we can define  $\tilde{C}_{\text{temp}} := \text{Sph}_{G, \text{temp}}^{\text{spec}} \otimes_{\text{Sph}_G^{\text{spec}}} \tilde{C}$ . Moreover, on a disc, we have the following bimodule that defines a Morita equivalence:

$$\text{QCoh} \left( (\text{LocSys}_{\check{G}}^{\text{mer}})^{\wedge}_{\text{reg}} \right) \circ \text{QCoh} \left( \text{LocSys}_{\check{G}}^{\text{reg}} \right) \cong \text{Rep}(\check{G}) \circ \text{Sph}_{G, \text{temp}}^{\text{spec}}$$

where  $(\text{LocSys}_{\check{G}}^{\text{mer}})^{\wedge}_{\text{reg}}$  means the formal completion of  $\text{LocSys}_{\check{G}}^{\text{reg}}$  in  $\text{LocSys}_{\check{G}}^{\text{mer}}$ . It is a bimodule because  $\text{Sph}_{G, \text{temp}}^{\text{spec}}$  is exactly the endomorphism of  $\text{QCoh} \left( \text{LocSys}_{\check{G}}^{\text{reg}} \right)$  over  $\text{QCoh} \left( (\text{LocSys}_{\check{G}}^{\text{mer}})^{\wedge}_{\text{reg}} \right)$ .

Geometric Casselman-Shalika (Theorem 2.1) provides us an equivalence  $\text{Whit}^!(G) \cong \text{Rep}(\check{G})$  and  $\text{Whit}_*(G)$  is actually the dual category of  $\text{Whit}^!(G)$  that also isomorphic to  $\text{Rep}(\check{G})$ , we call this  $\text{FLE}_{\infty} : \text{Whit}_*(G) \cong \text{Rep}(\check{G})$ , we have  $\text{Sph}_G$  acting on the LHS of  $\text{FLE}_{\infty}$  and  $\text{Sph}_G^{\text{spec}}$  on the RHS, while  $\text{Sph} \cong \text{Sph}^{\text{spec}}$  by derived Satake. Thus we get from (★) that

$$\text{QCoh} \left( \text{LocSys}_{\check{G}}^{\text{reg}} \right) \otimes_{\text{Sph}_G^{\text{spec}}} \text{Sph}(C) \cong \text{Whit}_* (C^{\text{sph-gen}}) \quad (1)$$

A benefit of the above equivalence is that we can observe the action of  $\text{QCoh} \left( (\text{LocSys}_{\check{G}}^{\text{mer}})^{\wedge}_{\text{reg}} \right)$  on both sides. As a corollary, we get the following:

$$\text{Sph}(C)_{\text{temp}} \cong \text{QCoh} \left( \text{LocSys}_{\check{G}}^{\text{reg}} \right) \otimes_{\text{QCoh} \left( (\text{LocSys}_{\check{G}}^{\text{mer}})^{\wedge}_{\text{reg}} \right)} \text{Whit}_* (C^{\text{sph-gen}}) \quad (2)$$

We apply (1) to  $C = \text{KM}$ , we get

$$\begin{array}{ccccc}
 \text{QCoh} \left( \text{LocSys}_{\check{G}}^{\text{reg}} \right) \otimes_{\text{Sph}_G^{\text{spec}}} \text{KL} & \xrightarrow{\cong} & \text{Whit}_* (\text{KM}^{\text{sph-gen}}) & \hookrightarrow & \text{Whit}_* (\text{KM}) \\
 & \searrow \cong & \downarrow \cong & & \downarrow \cong \\
 & & \text{IndCoh}^* \left( \text{Op}_{\check{G}}^{\text{mer}} \right)_{\text{mon-free}} & \hookrightarrow & \text{IndCoh}^* \left( \text{Op}_{\check{G}}^{\text{mer}} \right)
 \end{array}$$

Theorem 2.5

where the left horizontal arrow comes from

**Lemma 2.2.**  $\text{Whit}_* (\text{KM}^{\text{sph-gen}})$  is isomorphic to  $\text{IndCoh}^* \left( \text{Op}_{\check{G}}^{\text{mer}} \right)_{\text{mon-free}}$ , which is category of ind-coherent sheaves on  $\text{Op}_{\check{G}}^{\text{mer}}$  that supported on monodromy-free part.

**Lemma 2.3.** *There exists a notion of  $\lambda$ -regular opers for  $\lambda \in \Lambda^+$  such that*

$$\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} = \coprod_{\check{\lambda} \in \check{\Lambda}^+} \mathrm{Op}_{\check{G}}^{\check{\lambda}\text{-reg}, \sim}$$

where  $\mathrm{Op}_{\check{G}}^{\check{\lambda}\text{-reg}}$  is an honest scheme, we take its formal completion in  $\mathrm{Op}_{\check{G}}^{\mathrm{mer}}$  then take out those part of finite codimension to get  $\mathrm{Op}_{\check{G}}^{\check{\lambda}\text{-reg}, \sim}$ . Also for any  $\mathbb{V}^{\check{\lambda}} \in \mathrm{KL}$ , we have  $\mathrm{DS}_G(\mathbb{V}^{\check{\lambda}}) \cong \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\check{\lambda}\text{-reg}}}$ .

**Corollary 2.1.** *As a corollary of (2), we obtain*

$$\mathrm{KL}_{\mathrm{temp}} \cong \mathrm{QCoh}\left(\mathrm{LocSys}_{\check{G}}^{\mathrm{reg}}\right) \otimes_{\mathrm{QCoh}\left((\mathrm{LocSys}_{\check{G}}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}}\right)} \mathrm{IndCoh}^*\left(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}\right)_{\mathrm{mon-free}}$$

Also we have:

$$\begin{array}{ccc} \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} & \longrightarrow & \left(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}\right)^{\wedge}_{\mathrm{mon-free}} \\ \downarrow & & \downarrow \\ \mathrm{LocSys}_{\check{G}}^{\mathrm{reg}} & \longrightarrow & \left(\mathrm{LocSys}_{\check{G}}^{\mathrm{mer}}\right)^{\wedge}_{\mathrm{reg}} \end{array}$$

which gives us

$$\mathrm{IndCoh}^*\left(\mathrm{Op}_{\check{G}}^{\mathrm{mon-free}}\right) \cong \mathrm{QCoh}\left(\mathrm{LocSys}_{\check{G}}^{\mathrm{reg}}\right) \otimes_{\mathrm{QCoh}\left((\mathrm{LocSys}_{\check{G}}^{\mathrm{mer}})^{\wedge}_{\mathrm{reg}}\right)} \mathrm{IndCoh}^*\left(\mathrm{Op}_{\check{G}}^{\mathrm{mer}}\right)_{\mathrm{mon-free}}$$

So we (almost) win, the last step is using the following fact to conclude Theorem 2.4.

**Proposition 2.1.** *We have  $\mathrm{KL} \cong \mathrm{KL}_{\mathrm{temp}}$ .*

### 3 Day 3 Loca(lization) Day

#### 3.1 KM localization on $\text{Bun}_G$ (Sam Raskin)

Let us begin with finite dimensional case: assume  $H$  affine algebraic group with subgroup  $K$ . We assume there are (smooth) stacks  $\mathcal{Y}$  and  $\tilde{\mathcal{Y}}$  that  $H$  acts on  $\tilde{\mathcal{Y}}$  so that  $\pi : \tilde{\mathcal{Y}} \rightarrow \mathcal{Y} \cong \tilde{\mathcal{Y}}/K$ . Then there is a localization functor

$$\text{Loc}_{\mathcal{Y}} : \mathfrak{h}\text{-mod}^K \longrightarrow \mathcal{D}\text{-mod}(\mathcal{Y})$$

where  $\mathfrak{h}\text{-mod}^K$  means  $K$ -integrated  $\mathfrak{h}$ -modules. To construct such functor, first we construct  $\text{Loc} : \mathfrak{h}\text{-mod} \rightarrow \mathcal{D}\text{-mod}(\tilde{\mathcal{Y}})$ . Since  $H$  acts on  $\tilde{\mathcal{Y}}$ , there is a map  $\mathfrak{h} \rightarrow$  vector fields on  $\tilde{\mathcal{Y}}$ . This induces  $U(\mathfrak{h}) \rightarrow \Gamma(\tilde{\mathcal{Y}}, \text{Diff}_{\tilde{\mathcal{Y}}}) = \text{End}(\text{Diff}_{\tilde{\mathcal{Y}}})$ . Given  $\mathcal{M} \in \mathfrak{h}\text{-mod}$ , then  $\text{Loc}(\mathcal{M}) := \text{Diff}_{\tilde{\mathcal{Y}}} \otimes_{U(\mathfrak{h})} \mathcal{M} \in \mathcal{D}\text{-mod}(\tilde{\mathcal{Y}})$ . We can also construct via pull-push along the correspondence (which is more straightforward):

$$\tilde{\mathcal{Y}}_{\text{dR}} \longleftarrow \tilde{\mathcal{Y}}/H^\wedge \longrightarrow \mathbf{B}H^\wedge$$

Secondly, we pass to  $K$ -invariant objects, we obtain

$$\text{Loc}_{\mathcal{Y}} : \mathfrak{h}\text{-mod}^K \longrightarrow \mathcal{D}\text{-mod}(\tilde{\mathcal{Y}})^K \cong \mathcal{D}\text{-mod}(\tilde{\mathcal{Y}}/K) = \mathcal{D}\text{-mod}(\mathcal{Y}) \quad \text{strong } K\text{-invariant}$$

Concretely, for  $\mathcal{M} \in \mathfrak{h}\text{-mod}^K$ , we can see that  $\text{Loc}(\mathcal{M}) \in \mathcal{D}\text{-mod}(\tilde{\mathcal{Y}})$  and the Harish-Chandra structure gives the descent datum. Or equivalently, as  $K_{\text{dR}}$  acts on above correspondence diagram, we can obtain the localization functor by quotienting  $K_{\text{dR}}$ .

**Example 3.1.** If we take  $K = H$ , then  $\mathfrak{h}\text{-mod}^H \cong \text{Rep}(H)$  and  $\text{Loc}_{\mathcal{Y}} : \mathcal{V} \mapsto \text{Diff}_{\mathcal{Y}} \otimes_{\mathcal{O}_{\mathcal{Y}}} \mathcal{V}_{\mathcal{Y}}$ , where  $\mathcal{V}_{\mathcal{Y}}$  is pullback of  $\mathcal{V}$  along  $\mathcal{Y} = \tilde{\mathcal{Y}}/H \rightarrow \text{pt}/H$ . A generalization of the above is for  $\mathcal{V} \in \text{Rep}(K)$ , then  $\text{ind}_{\mathfrak{k}=\text{Lie}(K)}^{\mathfrak{h}}(\mathcal{V}) \in \mathfrak{h}\text{-mod}^K$  and  $\text{Loc}$  will send it to  $\mathcal{D}$ -module induced from  $\mathcal{V}_{\mathcal{Y}}$ , again,  $\mathcal{V}_{\mathcal{Y}}$  is pullback from  $\mathcal{Y} \rightarrow \text{pt}/K$ . If we take  $\tilde{\mathcal{Y}} = H/H_0$  and  $\mathcal{Y} = K \backslash H/H_0$  for some subgroup  $H_0 \subseteq H$ , then the fiber of  $\text{Loc}(\mathcal{M})$  at  $1 \in K \backslash H/H_0$  is  $C.(\text{Lie}(H_0), \mathcal{M})$ , the derived coinvariants.

If  $\mathcal{Y}$  is quasi-compact, then  $\text{Loc}_{\mathcal{Y}}$  has a right adjoint  $\Gamma$  given by  $\Gamma(-) = \Gamma(\tilde{\mathcal{Y}}, \pi^!(-))$ .

Now our case is infinite dimensional analog. To set up, we choose a point  $x \in X$  (or a point  $\underline{x} \in \text{Ran}$ ). We let  $\tilde{\mathcal{Y}}$  to be  $\text{Bun}_G^{\text{level}_{\infty}, x} = \langle \mathcal{P}_G : G\text{-torsor on } X \backslash x, \text{ trivialized on } \check{D}_x \subseteq X \backslash x \rangle$ . There is an action of  $H = \mathcal{L}_x G$  on  $\text{Bun}_G^{\text{level}_{\infty}, x}$  and we know  $\mathcal{Y} = \text{Bun}_G \cong \text{Bun}_G^{\text{level}_{\infty}, x} / \mathcal{L}_x^+ G$ , where  $K$  is taken to be  $\mathcal{L}_x^+ G \subseteq \mathcal{L}_x G$ . We get:

$$\text{Loc}_{\kappa, x} : \text{KL}(G)_{\kappa, x} \longrightarrow \mathcal{D}\text{-mod}_{\kappa}(\text{Bun}_G)$$

We know that the induction  $\text{Rep}(\mathcal{L}_x^+ G) \rightarrow \text{KL}(G)_{\kappa}$  gives a large class of representations. In fact, we have

$$\begin{array}{ccc} \text{Rep}(\mathcal{L}_x^+ G) & \xrightarrow{\text{ind}_{\kappa}} & \text{KL}(G)_{\kappa} \\ \text{ev}^* \downarrow & & \downarrow \text{Loc}_{\kappa, x} \\ \text{QCoh}(\text{Bun}_G) & \xrightarrow{\text{ind}_{\kappa}} & \mathcal{D}\text{-mod}_{\kappa}(\text{Bun}_G) \end{array}$$

where the bottom horizontal arrow is the induction functor from quasi-coherent sheaves to  $\mathcal{D}$ -modules and  $\text{ev} : \text{Bun}_G = \text{Bun}_G^{\text{level}_{\infty,x}} / \mathcal{L}_x^+ G \rightarrow \mathcal{L}_x^+ G$  is the evaluation map. The above diagram commutes and the right vertical arrow is equivariant for action of  $\text{Sph}_\kappa(G)_x := \mathcal{D}\text{-mod}_\kappa(\mathcal{L}_x^+ G \backslash \mathcal{L}_x G / \mathcal{L}_x^+ G)$ .

**Example 3.2.** If we start with trivial representation, then the top horizontal arrow gives us  $\mathbb{V}_\kappa$ ,  $\text{Loc}_\kappa(\mathbb{V}_\kappa)$  is  $\text{Diff}_\kappa$ , the  $\kappa$ -twisted differential operators on  $\text{Bun}_G$ . On the other hand,  $\text{Diff}_\kappa = \text{ind}_\kappa(\mathcal{O}_{\text{Bun}_G})$ . More generally, we denote  $\mathbb{V}_{\kappa,x}^\lambda := \text{ind}_{\mathfrak{g}(\mathcal{O}_x)}^{\hat{\mathfrak{g}}_\kappa}(\mathcal{V}^\lambda)$ , then  $\text{Loc}_{\kappa,x}(\mathbb{V}_{\kappa,x}^\lambda)$  is twisted  $\mathcal{D}$ -module induced from  $\mathcal{E}_x^\lambda$  on  $\text{Bun}_G$ , where  $\mathcal{E}_x^\lambda$  is obtained by twisting  $\mathcal{V}^\lambda$  by universal sheaf  $\mathcal{P}_{G,x}^{\text{univ}}$ , which is pullback from  $\mathcal{P}_G^{\text{univ}}$  on  $X \times \text{Bun}_G$  by  $x \times \text{id} : \text{Bun}_G \rightarrow X \times \text{Bun}_G$ .

**Example 3.3.** What we are more interested in is the critical level, which is given in [BD] that for chosen  $x \in X$ , we have  $\text{Diff}_{\text{Bun}_G, \text{crit}} \cong \text{Loc}_{\text{crit},x}(\mathbb{V}_{\text{crit},x})$ . Recall we have  $\mathfrak{Z}_{\mathfrak{g}, \text{crit},x} \cong \text{Fun}(\text{Op}_{\check{G},x}^{\text{reg}})$  that acts on  $\mathbb{V}_{\text{crit},x}$ , so  $\mathfrak{Z}_{\mathfrak{g}, \text{crit},x}$  acts on  $\text{Diff}_{\text{Bun}_G, \text{crit}}$  by functoriality.

**Remark 3.1.** We note that the endomorphisms of  $\text{Diff}$  are in correspondence with global differential operators. It says that elements in  $\mathfrak{Z}_{\mathfrak{g},x}$  are global differential operators on  $\text{Bun}_G$ . Also the action of  $\mathfrak{Z}_{\mathfrak{g},x}$  factors through an action of  $\text{Fun}(\text{Op}_{\check{G}}(X))$  (functions on global opers over  $X$ ).

Further, by birth of opers yoga:  $\text{Sat}^{\text{nv}}(\mathcal{V}) \star \mathbb{V}_{\text{crit},x} \xrightarrow{\cong} \mathfrak{r}^*(\mathcal{V}) \otimes_{\mathfrak{Z}_{\text{crit},x}} \mathbb{V}_{\text{crit},x}$ , we can deduce that

$$\text{Sat}^{\text{nv}}(\mathcal{V}) \star \text{Diff}_{\text{crit}} \xrightarrow{\cong} \mathfrak{r}^*(\mathcal{V}) \otimes_{\mathfrak{Z}_{\text{crit},x}} \text{Diff}_{\text{crit}}$$

Aside: If one wants an "honest" Hecke eigensheaf, one should believe that  $\text{Fun}(\text{Op}_{\check{G}}(X))$  acts on  $\text{Diff}_{\text{crit}}$ , for  $\sigma \in \text{Op}_{\check{G}}$ , then quotient of  $\text{Diff}_{\text{crit}}$  by corresponding maximal ideal of  $\text{Fun}(\text{Op}_{\check{G}}(X))$  is eigensheaf, denoted by  $\mathcal{F}_\sigma$ . **FACT:**  $\mathcal{F}_\sigma$  is holonomic. If  $G$  is simply connected, then  $\mathcal{F}_\sigma$  is moreover regular holonomic.

### 3.2 Factorization homology and unital structures(Nick Rozenblyum)

For  $X$  smooth proper connected curve,  $\text{Ran}(X)(S) = \{\text{non-empty finite subset of } X(S^{\text{red}})\}$ ,  $\text{Ran}(X) \cong \text{colim}_{I \in (\text{fSet}^{\text{op}})} X_{\text{dR}}^I$  is a prestack and that  $\text{QCoh}(\text{Ran}(X))$  has some important features:

- (i) Let  $\pi : \text{Ran}(X) \rightarrow \text{pt}$ , then  $\pi^*$  has a left adjoint  $\pi_!$ . (coming from the properness of  $X$ )
- (ii)  $\text{Ran}(X)$  is contractible, which says  $\pi_! \pi^* \xrightarrow{\cong} \text{id}$ . (means homology is trivial)

There exists a sub-prestack  $(\text{Ran} \times \text{Ran})_{\text{disj}} \subseteq \text{Ran} \times \text{Ran}$  and  $\text{Ran}$  is a semigroup with group structure  $\bigcup : \text{Ran} \times \text{Ran} \rightarrow \text{Ran}$ . A factorization algebra  $\mathcal{A}$  is roughly the data of  $\mathcal{A}_{\text{Ran}} \in \mathcal{D}\text{-mod}(\text{Ran})$  such that  $\bigcup^!(\mathcal{A})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}} \cong (\mathcal{A} \boxtimes \mathcal{A})|_{(\text{Ran} \times \text{Ran})_{\text{disj}}}$  with some higher coherence.

**Definition 3.1.** Let  $\mathcal{A}$  be a factorization algebra, the **factorization** (aka. chiral) **homology** of  $\mathcal{A}$  is

$$\mathcal{C}_\bullet^{\text{fact}}(X, \mathcal{A}) = \pi_!(\mathcal{A}_{\text{Ran}})$$

**Example 3.4.** The category  $\text{FactAlg}(X)$  has a natural symmetric monoidal structure. We have notion of commutative factorization algebra, which is object in  $\text{CommAlg}(\text{FactAlg}(X))$ , which maps to  $\text{CommAlg}(\mathcal{D}\text{-mod}(X))$ .



Conversely, there is a functor  $\text{FA} : \text{CommAlg}(\mathcal{D}\text{-mod}(X)) \longrightarrow \text{CommAlg}(\text{FactAlg}(X))$  that associates commutative algebra on  $\mathcal{D}\text{-mod}(X)$  with commutative factorization algebra. Also, denote by  $\mathcal{A} = \text{FA}(A)$ , then  $C_{\bullet}^{\text{fact}}(X, \mathcal{A})$  is a commutative algebra, and if we take the spectrum  $\text{Spec}(C_{\bullet}^{\text{fact}}(X, \mathcal{A}))$  is the (derived) space of horizontal sections of  $\text{Spec}(A) \rightarrow X$ . There is an equivalence

$$\text{Maps}_{\text{CommAlg}}(C_{\bullet}^{\text{fact}}(X, \mathcal{A}), B) \cong \text{Maps}_{\text{CommAlg}(\mathcal{D}\text{-mod}(X))}(A, B \otimes \Omega_X)$$

**Example 3.5.** If we take  $A = \mathcal{O}_X \otimes \mathbf{k}[t]$  constant scheme  $X \times \mathbf{A}^1 \rightarrow X$ , then  $C_{\bullet}^{\text{fact}}(X, \mathcal{A}) = \text{Sym}(H_{\text{dR}}^*(X))^{\vee}$ .

**Example 3.6.** We can consider the Kac-Moody case (at level 0). Let  $\text{Vac}(G)$  be the factorization algebra corresponds to the vacuum Kac-Moody module, then we have

$$C_{\bullet}^{\text{fact}}(X, \text{Vac}(G)) \cong C_{\bullet}^{\text{Lie}}(H^*(X) \otimes \mathfrak{g})$$

there is a natural Lie bracket on  $H^*(X) \otimes \mathfrak{g}$  and  $C_{\bullet}^{\text{Lie}}$  is the Chevalley-Eilenberg complex. Exercise: In the abelian case, we can deduce the above equality from Example 3.4.

Reminiscent: this should be how fibers of Localization map look like, an observation is: pick a  $G$ -torsor  $\mathcal{P} \in \text{Bun}_G$ , then  $\text{Diff}_{\mathcal{P}}$ , the stalk of sheaf of differential operators on  $\text{Bun}_G$  is isomorphic to  $C_{\bullet}^{\text{Lie}}(H^*(X, \mathfrak{g}_{\mathcal{P}}))$ .

Next, we are going to discuss the *units* of factorization algebras. What is a unital factorization algebra? *Version 0:* There is a 'unit' algebra  $\omega_{\text{Ran}(X)}$ , which is a factorization algebra, we can naively ask  $\omega_{\text{Ran}(X)} \rightarrow \mathcal{A}$  map of factorization algebras. But a unit is more!

Rough idea is that for every point  $x \in X$ , we want to have a map  $\mathbf{k} \rightarrow \mathcal{A}_x$  compatible with factorization. For example, take  $x \neq y \in X$ , there should be a map  $\mathcal{A}_x \rightarrow \mathcal{A}_{\{x,y\}} \cong \mathcal{A}_x \otimes \mathcal{A}_y$ . This suggests there is something to do with inclusion of subsets and gives the idea that, the unital structure should remember the inclusion relation on subsets. A geometric construction need to consider categorical prestacks(=prestacks valued in categories), i.e. functors  $(\text{Sch}^{\text{aff}})^{\text{op}} \rightarrow \text{Cat}$ . Main example is the **unital Ran space**:

$$\text{Ran}^{\text{unlt}}(X)(S) = \langle \text{poset of finite subsets of } X(S^{\text{red}}) \rangle$$

For  $\mathcal{X}$  categorical prestacks, there are notions of  $\text{QCoh}(\mathcal{X})$ ,  $\text{IndCoh}(\mathcal{X})$  and  $\mathcal{D}\text{-mod}(\mathcal{X})$ . For example  $\mathcal{F} \in \text{QCoh}(\mathcal{X})$  is the data of

(i)  $\forall x : S \rightarrow X, \mathcal{F}_{S,x} \in \text{QCoh}(S)$ ; (ii)  $\alpha : x \rightarrow x' \in \mathcal{X}(S), \mathcal{F}_{S,x} \rightarrow \mathcal{F}_{S,x'}$  and (iii) compatibility with composition.

**Definition 3.2.** A **unital factorization algebra**  $\mathcal{A}$  is the data of

- (i)  $\mathcal{A}_{\text{Ran}} \in \mathcal{D}\text{-mod}(\text{Ran}^{\text{unlt}})$ ;
- (ii) Isomorphism  $\bigcup^! (\mathcal{A}_{\text{Ran}}) |_{(\text{Ran}^{\text{unlt}} \times \text{Ran}^{\text{unlt}})_{\text{disj}}} \cong (\mathcal{A} \boxtimes \mathcal{A}) |_{(\text{Ran}^{\text{unlt}} \times \text{Ran}^{\text{unlt}})_{\text{disj}}}$ .
- (iii)  $\mathcal{A}_{\emptyset} = \mathbf{k}$ . (notice we allow empty set for  $\text{Ran}^{\text{unlt}}$ ).

Consider some  $\mathbf{k}$ -points of  $\text{Ran}^{\text{unlt}}(X)$ . Let  $J \subseteq X(\mathbf{k})$  be a finite subset, then the factorization structure gives  $\mathcal{A}_J \cong \bigotimes_{j \in J} \mathcal{A}_j$ , and the unital structure asks if  $J \subseteq J'$ , there is a map  $\mathcal{A}_J \rightarrow \mathcal{A}_{J'}$ . For example if we take  $J = \emptyset, J' = \{x\}$ , we get  $\mathbf{k} \rightarrow \mathcal{A}_x$  and if  $J = \{x\} \subseteq J' = \{x, y\}$ , we get  $\mathcal{A}_x \rightarrow \mathcal{A}_{\{x,y\}}$  as we want.

To discuss factorization homology in unital cases, note that  $\text{Ran}^{\text{untl}}$  can be expressed in terms of  $X^I$  so that it is a (lax) colimit of projections, which are proper. As a consequence,  $(\pi^{\text{untl}})^! : \mathbf{Vect} \rightarrow \mathcal{D}\text{-mod}(\text{Ran}^{\text{untl}})$  has a left adjoint functor  $\pi_!^{\text{untl}}$ .

**Definition 3.3.** Let  $\mathcal{A}$  be a unital factorization algebra, we define similarly the factorization homology to be

$$C_{\bullet}^{\text{fact}}(X, \mathcal{A}) = \pi_!^{\text{untl}}(\mathcal{A}_{\text{Ran}})$$

We denote by  $\text{FactAlg}^{\text{untl}}(X)$  the category of unital factorization algebras. There is a natural map  $t : \text{Ran} \rightarrow \text{Ran}^{\text{untl}}$  that gives restriction  $t^! : \text{FactAlg}^{\text{untl}}(X) \rightarrow \text{FactAlg}(X)$ . Also as a corollary of contractibility of  $\text{Ran}(X)$ , for any  $\mathcal{F} \in \mathcal{D}\text{-mod}(\text{Ran}^{\text{untl}})$ , we have  $\pi_! t^!(\mathcal{F}) = \pi_!^{\text{untl}}(\mathcal{F})$ .

On the categorical level, we have notion of unital factorization categories. Concretely, if  $\mathcal{Y}$  is a categorical prestack, underlying  $\mathcal{Y}$ , there are prestacks  $\mathcal{Y}^{\text{gpd}}$  and  $\mathbf{Maps}([1], \mathcal{Y})^{\text{gpd}}$ , where  $\mathcal{Y}^{\text{gpd}}(S) = \mathcal{Y}(S)^{\text{gpd}}$  is the prestack of objects and  $\mathbf{Maps}([1], \mathcal{Y})^{\text{gpd}}$  is the prestack of 1-morphisms that classifies data of  $\langle x, y \in \mathcal{Y}(S); f : x \rightarrow y \rangle$ .

There are two maps  $\mathbf{Maps}([1], \mathcal{Y})^{\text{gpd}} \xrightarrow[s]{t} \mathcal{Y}^{\text{gpd}}$ . And we can think of sheaves on  $\mathcal{Y}$  as data of a sheaf  $\mathcal{F}$  on  $\mathcal{Y}^{\text{gpd}}$  with  $s^! \mathcal{F} \rightarrow t^! \mathcal{F}$  satisfies compatibilities.

We apply this to  $\text{Ran}(X)^{\text{untl}}$ , observe that we have

$$(\text{Ran}(X)^{\text{untl}})^{\text{gpd}} = \text{Ran}(X) \quad \text{and} \quad \mathbf{Maps}([1], \text{Ran}(X)^{\text{untl}})^{\text{gpd}}(S) = \langle \underline{x}, \underline{x}' \subseteq X(S^{\text{red}}); \underline{x} \subseteq \underline{x}' \rangle$$

We rename  $\mathbf{Maps}([1], \text{Ran}(X)^{\text{untl}})^{\text{gpd}}$  as  $\text{Ran}_{\subseteq}$ , the maps from  $\text{Ran}_{\subseteq}$  to  $\text{Ran}(X)^{\text{untl}} = \text{Ran}(X)$  are projections

$$\text{Ran} \xleftarrow{p_{\text{small}}} \text{Ran}_{\subseteq} \xrightarrow{p_{\text{big}}} \text{Ran} \quad \text{where } p_{\text{small}}(\underline{x} \subseteq \underline{x}') = \underline{x}; p_{\text{big}}(\underline{x} \subseteq \underline{x}') = \underline{x}'.$$

Upshot: For factorization category  $\mathcal{C}$ , the data of unital structure on  $\mathcal{C}$  is given by  $p_{\text{small}}^*(\underline{\mathcal{C}}) \rightarrow p_{\text{big}}^*(\underline{\mathcal{C}})$ . In fact, the functor  $p_{\text{small}}^*$  has a right adjoint  $(p_{\text{small}})_*$ , so we can write the structure as a map  $\underline{\mathcal{C}} \mapsto (p_{\text{small}})_* p_{\text{big}}^*(\underline{\mathcal{C}})$ . More explicitly, suppose  $\underline{x} \subseteq \underline{x}'$ , what we get from the structure map is a functor

$$\mathbf{ins.unit}_{\underline{x} \subseteq \underline{x}'} : \mathcal{C}_{\underline{x}} \rightarrow \mathcal{C}_{\underline{x}'}$$

**Example 3.7.** (1) The Kazhdan-Lusztig category  $\text{KL}$  is a unital factorization category.

(2) Let  $\mathcal{A}$  be a factorization algebra, then there exists a (weak) factorization category  $\mathcal{A}\text{-mod}^{\text{fact}}$ , in this case,  $\mathbf{ins.unit}_{\emptyset \subseteq \{x\}}$  is given by  $\mathbf{k} \rightarrow \mathcal{A}_x$ .

Let  $\underline{\mathcal{C}}^{\text{loc}}$  be a sheaf of category on  $\text{Ran}^{\text{untl}}$  and  $\mathcal{C}^{\text{glob}}$  be a category.

**Definition 3.4.** A **unital local-to-global** functor is a map of sheaves  $F : \underline{\mathcal{C}}^{\text{loc}} \rightarrow \underline{\mathcal{D}\text{-mod}}(\text{Ran}^{\text{untl}}) \otimes \mathcal{C}^{\text{glob}}$  with unital structure  $F_{\underline{x}} \xrightarrow{\cong} F_{\underline{x}'} \circ \mathbf{ins.unit}_{\underline{x} \subseteq \underline{x}'}$

where  $F_{\underline{x}} : \mathcal{C}_{\underline{x}}^{\text{loc}} \rightarrow \mathcal{C}^{\text{glob}}$  and the unital structure is a natural equivalence

$$\begin{array}{ccc} \mathcal{C}_{\underline{x}}^{\text{loc}} & \xrightarrow{\text{ins.unit}_{\underline{x} \subseteq \underline{x}'}} & \mathcal{C}_{\underline{x}'}^{\text{loc}} \\ & \searrow F_{\underline{x}} \quad \swarrow F_{\underline{x}'} & \\ & \mathcal{C}^{\text{glob}} & \end{array}$$

We can weaken the condition to "lax" unital structure, namely we only require natural transformation for all  $\underline{x} \subseteq \underline{x}'$ . We have an inclusion of categories:

$$\text{Fun}^{\text{loc-glob,until}}(\underline{\mathcal{C}}, \mathcal{C}^{\text{glob}}) \hookrightarrow \text{Fun}^{\text{loc-glob,lax-until}}(\underline{\mathcal{C}}, \mathcal{C}^{\text{glob}})$$

**Claim:** the above inclusion has a left adjoint given by  $F \mapsto F^{\int_{\text{ins.unit}}}$ , where  $F^{\int_{\text{ins.unit}}}$  is constructed as following:

For  $Z \rightarrow \text{Ran}$ , we denote  $Z^{\subseteq} := Z \times_{\text{Ran}}^{p_{\text{small}}} \text{Ran}_{\subseteq}$ , then we construct:

$$F^{\int_{\text{ins.unit}}} : \mathcal{C}_Z^{\text{loc}} \longrightarrow \mathcal{C}_{Z^{\subseteq}}^{\text{loc}} \xrightarrow{F} \mathcal{C}^{\text{glob}} \otimes \mathcal{D}\text{-mod}(Z^{\subseteq}) \xrightarrow{(p_{\text{small}})!} \mathcal{C}^{\text{glob}} \otimes \mathcal{D}\text{-mod}(Z)$$

**Example 3.8.** The key example is:  $\mathcal{C}^{\text{loc}} = \mathcal{A}\text{-mod}^{\text{fact}}$ , where  $\mathcal{A}$  factorization algebra. We take  $\mathcal{C}^{\text{glob}}$  to be  $\mathbf{Vect}$  and a unital factorization functor  $F = \text{oblv} : \mathcal{A}\text{-mod}^{\text{fact}} \rightarrow \mathbf{Vect}$ . We claim  $F^{\int_{\text{ins.unit}}} \cong \mathcal{C}_{\bullet}^{\text{fact}}(X; \mathcal{A}, -)$ . The point is for  $Z \rightarrow \text{Ran}$ ,  $F_Z^{\int_{\text{ins.unit}}}(M) = (p_{\text{small}})!(M_{Z^{\subseteq}})$ .

### 3.3 Making branes interact: Localization vs Whittaker(Lin Chen)

Now we are ready to get into the most important diagram:

$$\begin{array}{ccc} \text{KL}(G)_{\text{crit,Ran}} & \xrightarrow{\text{FLE}_{\text{crit}}} & \text{IndCoh}^*(\text{Op}_G^{\text{mon-free}}) \\ \downarrow \text{Loc}_{\text{crit}} & & \downarrow \text{Poinc}^{\text{spec}} \\ \mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\text{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \\ \downarrow \text{coeff}_* & & \downarrow \Gamma^{\text{spec}} \\ \text{Whit}^!(G)_{\text{crit,Ran}} & \xrightarrow{\text{CS}} & \text{Rep}(\check{G})_{\text{Ran}} \end{array}$$

Fundamental Diagram 1

Most of the components in Fundamental diagram 1 have been introduced before, except the functors  $\text{Loc}$  and  $\text{Poinc}^{\text{spec}}$ , which will be introduced soon in this and next talk 3.4. Our primary goal is to describe the left vertical, i.e.  $\text{coeff}_* \circ \text{Loc}$ , via local data (+ Ran magic), then to describe  $\text{oblv} \circ \text{Loc}$  and  $\text{CT}_* \circ \text{Loc}$ .

The first step is to describe  $\text{coeff}^{\text{vac}} \circ \text{Loc}$  (recall Definition 2.2). The second step is to look at Hecke action and use Ran magic.

Recall for  $x \in X$ , we have  $\text{Bun}_G \cong \text{Bun}_G^{\text{level}_\infty, x} / \mathcal{L}_x^+ G$  and acted on by  $\mathcal{L}_x G$ . The analog to finite dimensional localization gives us  $\mathcal{L}_x \mathfrak{g} \cong \mathfrak{g}(K_x) \rightarrow T(\text{Bun}_G^{\text{level}_\infty, x})$ , which gives us a functor  $\mathcal{L}_x \mathfrak{g}\text{-mod} \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G^{\text{level}_\infty, x})$  that maps  $\mathcal{M} \mapsto \text{Diff} \otimes_{U(\mathcal{L}_x \mathfrak{g})} \mathcal{M}$ . Take (strong)  $\mathcal{L}_x^+ G$ -invariant, we obtain a functor  $\mathcal{L}_x \mathfrak{g}\text{-mod}^{\mathcal{L}_x^+ G} \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$ . In fact, this works for any  $\underline{x} \in \text{Ran}(X)$ . So we obtain a lax unital local-to-global functor

$$\underline{\text{Loc}} : \underline{\text{KL}}(G) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \underline{\text{Vect}} \quad \text{in } \text{ShvCat}(\text{Ran})$$

In fact, it is even in  $\text{ShvCat}(\text{Ran}^{\text{until}})$ , we can consider the following diagram (Exercise: verify it commutes)

$$\begin{array}{ccc} \text{KL}(G)_{\underline{x}} & \xrightarrow{\text{ins.vac}_{\underline{x}, \underline{x}'}} & \text{KL}(G)_{\underline{x}'} \\ \text{Loc}_{\underline{x}} \downarrow & & \downarrow \text{Loc}_{\underline{x}'} \\ \mathcal{D}\text{-mod}(\text{Bun}_G) & \xlongequal{\quad} & \mathcal{D}\text{-mod}(\text{Bun}_G) \end{array} \quad (\underline{x} \subseteq \underline{x}') \in \text{Ran}^{\text{until}}$$

The Ran version of localization functor in Fundamental diagram 1 is given by

$$\text{Loc} : \text{KL}(G)_{\text{Ran}} \xrightarrow{\text{Loc}_{\text{Ran}}} \mathcal{D}\text{-mod}(\text{Bun}_G \times \text{Ran}) \xrightarrow{\int_{\text{Ran}}} \mathcal{D}\text{-mod}(\text{Bun}_G)$$

where the first map is the global section of  $\underline{\text{Loc}} \in \text{ShvCat}(\text{Ran}^{\text{until}})$  and we write  $\int_{\text{Ran}}$  for taking the direct image along  $\text{Ran} \rightarrow \text{pt}$ .

Recall  $\text{Whit}^!(G)_x = \mathcal{D}\text{-mod}(\text{Gr}_{G,x})^{(\mathcal{L}N, \chi)}$  and there is forgetful map  $\pi : \text{Gr}_{G,x} \rightarrow \text{Bun}_G$  that forget the trivialization data. As we have seen before in section 2.1, there is a pair of adjunction:

$$\text{Poinc}_{!,x} : \text{Whit}^!(G)_x \begin{array}{c} \xrightarrow{\pi_!} \\ \xleftarrow{\text{Av}_* \circ \pi^!} \end{array} \mathcal{D}\text{-mod}(\text{Bun}_G) : \text{coeff}_{*,x}$$

The above adjunction can be upgraded into sheaves of categories on Ran, we get

$$\underline{\text{Poinc}}_! : \underline{\text{Whit}}^!(G) \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{array} \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \underline{\text{Vect}} : \underline{\text{coeff}}_*$$

Let us denote by  $\underline{\text{unit}}$  the map  $\underline{\text{Vect}} \rightarrow \underline{\text{Whit}}^!(G)$ , then it is actually given by  $p_!(\exp_{\text{Gr}_N})$ , for  $p : \text{Gr}_N \rightarrow \text{Gr}_G$  and  $\exp_{\text{Gr}_N}$  is pullback from  $\exp$  on  $G_a$  with respect to character  $\chi$ . One can verify this in family and check the compatibility. Roughly speaking, the unit for  $\mathcal{D}\text{-mod}(\text{Gr}_G)$  is given by  $\delta$ -function on unit section, but to be compatible with Whittaker condition, we need to make a twist by  $\exp$ .

Exercise: (1)  $\underline{\text{unit}}$  admits a right adjoint  $\underline{\text{unit}}^R : \underline{\text{Whit}}^!(G) \xrightarrow{e^!} \underline{\text{Vect}}$ . (2) There is commutative diagram:

$$\begin{array}{ccc} \underline{\text{Whit}}^!(G) & \xrightarrow{\text{CS}} & \underline{\text{Rep}}(\check{G}) \\ & \searrow e^! & \swarrow \text{inv} \\ & \underline{\text{Vect}} & \end{array}$$

Taking global section of  $\underline{\text{coeff}}_*$  gives us:

$$\begin{array}{ccccc}
 & & \text{coeff}_* & & \\
 & \nearrow & & \searrow & \\
 D\text{-mod}(\text{Bun}_G) & \xrightarrow{\quad} & D\text{-mod}(\text{Bun}_G \times \text{Ran}) & \xrightarrow{\text{coeff}_{\text{Ran}}} & \text{Whit}^!(G)_{\text{Ran}} \\
 & \searrow & & & \downarrow e'_{\text{Ran}} \\
 & & & & D\text{-mod}(\text{Ran}) \\
 & & \text{coeff}^{\text{Vac}} & & \downarrow \\
 & & & & \mathbf{Vect}
 \end{array}$$

As in section 2.1. One can verify that  $\text{coeff}^{\text{Vac}}$  is given by  $q_* \left( p^!(-) \overset{!}{\otimes} \chi^!(\text{exp}) \right)$ , where  $\text{Bun}_G \xleftarrow{p} \text{Bun}_N^{\Omega_X} \xrightarrow{q} \text{pt}$ .

Exercise: (1) One can show  $\underline{\text{Poinc}}$  is unital, while  $\underline{\text{coeff}}_*$  is right lax unital. (2)  $\text{coeff}_{*,\emptyset} : D\text{-mod}(\text{Bun}_G) \rightarrow \mathbf{Vect}$  coincides with  $\text{coeff}^{\text{Vac}}$ .

**Theorem 3.1.** *To know the behavior of  $\text{coeff}_* \circ \text{Loc}$ , the following diagram commutes:*

$$\begin{array}{ccccc}
 \text{KL}(G)_{\text{Ran}} & \xrightarrow{\quad \text{Loc} \quad} & D\text{-mod}(\text{Bun}_G) & & \\
 \text{ins.vac} \downarrow & & \downarrow \text{coeff}^{\text{Vac}} & & \\
 \text{KL}(G)_{\text{Ran}^\subseteq} & \xrightarrow{\text{DS}} & D\text{-mod}(\text{Ran}^\subseteq) & \xrightarrow{\int_{\text{Ran}^\subseteq}} & \mathbf{Vect}
 \end{array}$$

where for  $\underline{C} \in \text{ShvCat}(\text{Ran})$ ,  $C_{\text{Ran}^\subseteq} := \Gamma \left( \text{Ran}^\subseteq, p_{\text{big}}^*(\underline{C}) \right)$ .

Recall  $\text{DS} : \text{KL}(G) \rightarrow \mathbf{Vect}$  is given by  $\mathcal{M} \mapsto \text{DS}(\mathcal{M}) = C^{\frac{\infty}{2}}(\mathcal{L}n, \mathcal{L}^+N, \mathcal{M} \otimes \mathbf{k}_\chi)$ , which (by construction of semi-infinite cohomology) can be written as a colimit

$$\text{colim}_i C^* \left( n_i, (\mathcal{M} \otimes \mathbf{k}_\chi) \otimes \det(n_i/n_0) [\dim(N_i/N_0)] \right), \text{ where } N_i = \text{Ad}_{t^{-ip}}(\mathcal{L}^+N) \text{ and } \mathcal{L}N = \bigcup_i N_i$$

We can upgrade DS to factorization functor  $\underline{\text{DS}} : \text{KL}(G) \rightarrow \mathbf{Vect}$ , which is a lax unital functor. In fact, it is only lax unital since we know  $\text{DS}(\mathbb{V}_{\mathfrak{g}}) = \mathfrak{Z}_{\mathfrak{g}} = Z(\text{Vac}(G))$ , which sends unit to non-unit in  $\mathbf{Vect}$ .

**Remark 3.2.** For any  $F : \underline{C} \rightarrow \underline{D}$  lax unital functor, we know  $F(1_C)$  is a factorization algebra in  $\underline{D}$ . Then  $F$  always factors through  $\underline{F(1_C)\text{-mod}^{\text{fact}}}$ , we call the factorization functor  $C \rightarrow \underline{F(1_C)\text{-mod}^{\text{fact}}}$  by  $\underline{F}^{\text{enh}}$ .

In our case of Drinfeld-Sokolov functor, the enhanced factorization functor  $\underline{\text{DS}}^{\text{enh}} : \text{KL}(G) \rightarrow \underline{\mathfrak{Z}_{\mathfrak{g}}\text{-mod}^{\text{fact}}}$ , is nothing but the enhanced Drinfeld-Sokolov. Also, the factorization functor  $\underline{\mathfrak{Z}_{\mathfrak{g}}\text{-mod}^{\text{fact}}} \rightarrow \mathbf{Vect}$  is in fact  $\underline{C_{\bullet}^{\text{fact}}(X, \mathfrak{Z}, -)}$ . Taking global section, we can decompose  $\text{DS} : \text{KL}(G)_{\text{Ran}} \rightarrow \mathbf{Vect}$  as

$$\text{KL}(G)_{\text{Ran}} \xrightarrow{\text{DS}^{\text{enh}}} \underline{\mathfrak{Z}_{\mathfrak{g}}\text{-mod}^{\text{fact}}_{\text{Ran}}} \xrightarrow{C_{\bullet}^{\text{fact}}(X, \mathfrak{Z}, -)} D\text{-mod}(\text{Ran}) \xrightarrow{\int_{\text{Ran}}} \mathbf{Vect}$$

Now we have two directions going from  $\underline{\text{KL}}(G) \rightarrow \underline{\text{Vect}}$ :

$$\begin{array}{ccc}
 \underline{\text{KL}}(G) & \xrightarrow{\underline{\text{Loc}}} & \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \underline{\text{Vect}} \\
 \underline{\text{DS}} \downarrow & \nearrow & \downarrow \text{coeff}^{\text{Vac}} \otimes \text{Id} \\
 \underline{\text{Vect}} & \xlongequal{\quad} & \underline{\text{Vect}}
 \end{array}$$

However, the diagram is not commutative, since both  $\text{coeff}^{\text{Vac}} \otimes \text{Id}$  and  $\underline{\text{Loc}}$  are unital, so their composition is again unital, while  $\underline{\text{DS}}$  is lax unital. In fact, we can test at unit in  $\underline{\text{KL}}(G)$ , which is  $\mathbb{V}_g$ . We have  $\underline{\text{Loc}}(\text{Vac}(G)) = \text{Diff} \boxtimes \underline{\mathbf{1}}$  and  $\text{coeff}^{\text{Vac}} \otimes \text{Id}(\text{Diff} \boxtimes \underline{\mathbf{1}})$  is constant over  $\text{Ran}$ . However,  $\underline{\text{DS}}(\text{Vac}(G)) = \mathfrak{Z}_g$ , which is non-constant. One can still expect there is some interaction between those two functors. One reason is we can regard  $\underline{\text{DS}}$  or the semi-infinite cohomology as certain (local) Hamiltonian reduction and  $\text{coeff}^{\text{Vac}}$  as a global Hamiltonian reduction (as it can be given by pulling back to  $\text{Bun}_N^{\Omega_X}$ ), so there should be some local-to-global comparison for Hamiltonian reductions. In general, recall if we have  $F : \underline{\mathcal{C}} \rightarrow \mathcal{D} \otimes \underline{\text{Vect}}$  lax unital local-to-global functor, we can make it unital by 'inserting units'  $F \mapsto F^{\text{ins.unit}}$ , let us denote  $F^{\text{ins.unit}}$  by  $F^{\text{fix}}$  for simplicity.

**Theorem 3.2.** *There is a natrual equivalence*

$$\underline{\text{DS}}^{\text{fix}} \xrightarrow{\cong} \text{coeff}^{\text{Vac}} \circ \underline{\text{Loc}}$$

One can verify that Theorem 3.2 and Theorem 3.1 are equivalent.

### Loc vs oblv

Next, we consider the composition of  $\text{oblv} \circ \underline{\text{Loc}}$ . We first recall in Section 3.1 a commutative diagram:

$$\begin{array}{ccc}
 \text{Rep}(\mathcal{L}^+G) & \xrightarrow{\underline{\text{Loc}}^{\text{QCoh}}} & \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{Vect}} \\
 \text{ind} \downarrow & & \downarrow \text{ind} \otimes \text{Id} \\
 \underline{\text{KL}}(G) & \xrightarrow{\underline{\text{Loc}}} & \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \underline{\text{Vect}}
 \end{array}$$

Sanity check for  $\text{triv} \in \text{Rep}(\mathcal{L}^+G)$ , in one direction, we have  $\underline{\text{Loc}}^{\text{QCoh}}(\text{triv}) = \mathcal{O}_{\text{Bun}_G} \boxtimes \underline{\mathbf{1}}$  and apply  $\text{ind} \otimes \text{id}$  we get  $\text{Diff} \boxtimes \underline{\mathbf{1}}$ . In the other direction, we get  $\text{ind}(\text{triv}) = \text{Vac}(G)$  and then apply  $\underline{\text{Loc}}$  we get  $\text{Diff} \boxtimes \underline{\mathbf{1}}$ . In fact, both compositions of functors are unital. We then pass to right adjoint, we get the following diagram

$$\begin{array}{ccc}
 \text{Rep}(\mathcal{L}^+G) & \xrightarrow{\underline{\text{Loc}}^{\text{QCoh}}} & \text{QCoh}(\text{Bun}_G) \otimes \underline{\text{Vect}} \\
 \text{oblv} \uparrow & \searrow & \uparrow \text{oblv} \otimes \text{Id} \\
 \underline{\text{KL}}(G) & \xrightarrow{\underline{\text{Loc}}} & \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \underline{\text{Vect}}
 \end{array}$$

unfortunately it is not commutative, due to the same reason above. We have  $(\text{oblv} \otimes \text{Id}) \circ \underline{\text{Loc}}$  is unital, while  $\underline{\text{Loc}}^{\text{QCoh}} \circ \text{oblv}$  is lax-unital. Fortunately, we have similar result as Thereom 3.2.

**Theorem 3.3.** *There is a natural equivalence*

$$(\underline{\text{Loc}}^{\text{QCoh}} \circ \underline{\text{oblv}})^{\text{fix}} \xrightarrow{\cong} (\underline{\text{oblv}} \otimes \text{Id}) \circ \underline{\text{Loc}}$$

### Loc vs pullback and pushforward

Let us consider first the general picture:

$$\begin{array}{ccc} H_1 & \xrightarrow{\quad} & Y_1 \\ \downarrow & & \downarrow p \\ H_2 & \xrightarrow{\quad} & Y_2 \end{array}$$

that affine group scheme  $H_i$  acts on stack  $Y_i, i = 1, 2$ .

Then we have localization functors  $\text{Loc}_i : \mathfrak{h}_i\text{-mod} \rightarrow \mathcal{D}\text{-mod}(Y_i)$  and the following natural transformation:

$$\begin{array}{ccc} \mathfrak{h}_1\text{-mod} & \xrightarrow{\text{Loc}_1} & \mathcal{D}\text{-mod}(Y_1) \\ \text{res} \uparrow & \searrow & \uparrow p^! \\ \mathfrak{h}_2\text{-mod} & \xrightarrow{\text{Loc}_2} & \mathcal{D}\text{-mod}(Y_2) \end{array}$$

Similarly, we have natural transformation for loop groups version:

$$\begin{array}{ccc} \underline{\text{KL}}(H_1) & \xrightarrow{\underline{\text{Loc}}_{H_1}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_1}) \otimes \underline{\text{Vect}} \\ \text{res} \uparrow & \searrow & \uparrow p^! \\ \underline{\text{KL}}(H_2) & \xrightarrow{\underline{\text{Loc}}_{H_2}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_2}) \otimes \underline{\text{Vect}} \end{array}$$

Notice that  $\text{res}$  is not unital, as it does not send  $\text{Vac}(H_2)$  to  $\text{Vac}(H_1)$ . But similar to Theorem 3.2 and 3.3

**Theorem 3.4.** *We have  $(\underline{\text{Loc}}_{H_1} \circ \text{res})^{\text{fix}} \xrightarrow{\cong} p^! \circ \underline{\text{Loc}}_{H_2}$*

*Proof.* The Theorem 3.4 follows from the Theorem 3.3. Since  $\underline{\text{oblv}} : \mathcal{D}\text{-mod}(\text{Bun}_{H_1}) \rightarrow \text{QCoh}(\text{Bun}_{H_1})$  is conservative, only need to show that

$$(\underline{\text{oblv}}_1 \otimes \text{Id}) \circ (\underline{\text{Loc}}_{H_1} \circ \text{res})^{\text{fix}} \xrightarrow{\cong} (\underline{\text{oblv}}_1 \otimes \text{Id}) \circ p^! \circ \underline{\text{Loc}}_{H_2} \quad ((\star))$$

We can look at the following diagram, where the right square is commutative:

$$\begin{array}{ccccc} \underline{\text{KL}}(H_1) & \xrightarrow{\underline{\text{Loc}}_{H_1}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_1}) \otimes \underline{\text{Vect}} & \xrightarrow{\underline{\text{oblv}}_1 \otimes \text{Id}} & \text{QCoh}(\text{Bun}_{H_1}) \otimes \underline{\text{Vect}} \\ \text{res} \uparrow & \searrow & \uparrow p^! & & \uparrow p^{!, \text{QCoh}} \\ \underline{\text{KL}}(H_2) & \xrightarrow{\underline{\text{Loc}}_{H_2}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_2}) \otimes \underline{\text{Vect}} & \xrightarrow{\underline{\text{oblv}}_2 \otimes \text{Id}} & \text{QCoh}(\text{Bun}_{H_2}) \otimes \underline{\text{Vect}} \end{array}$$

The RHS of  $(\star)$  is equivalent to

$$\begin{aligned} (\underline{\text{oblv}}_1 \otimes \text{Id}) \circ p^! \circ \underline{\text{Loc}}_{H_2} &\cong p^{!, \text{QCoh}} \circ (\underline{\text{oblv}}_2 \otimes \text{Id}) \circ \underline{\text{Loc}}_{H_2} \\ &\cong p^{!, \text{QCoh}} \circ (\underline{\text{Loc}}_{H_2}^{\text{QCoh}} \circ \underline{\text{oblv}}_2)^{\text{fix}} \end{aligned}$$

where by Theorem 3.3 we have (for  $i = 1, 2$ ) the following natural equivalence

$$\begin{array}{ccc} \underline{\text{Rep}}(\mathcal{L}^+ H_i) & \xrightarrow{\underline{\text{Loc}}_{H_i}^{\text{QCoh}}} & \text{QCoh}(\text{Bun}_{H_i}) \otimes \underline{\text{Vect}} \\ \text{oblv}_i \uparrow & \searrow & \uparrow \text{oblv} \otimes \text{Id} \\ \underline{\text{KL}}(H_i) & \xrightarrow{\underline{\text{Loc}}_{H_i}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_i}) \otimes \underline{\text{Vect}} \end{array}$$

Now we can consider the following diagram

$$\begin{array}{ccccc} \underline{\text{KL}}(H_1) & \xrightarrow{\text{oblv}_1} & \underline{\text{Rep}}(\mathcal{L}^+ H_1) & \xrightarrow{\underline{\text{Loc}}_{H_1}^{\text{QCoh}}} & \text{QCoh}(\text{Bun}_{H_1}) \otimes \underline{\text{Vect}} \\ \text{res} \uparrow & & \uparrow \text{res}' & & \uparrow p^!, \text{QCoh} \\ \underline{\text{KL}}(H_2) & \xrightarrow{\text{oblv}_2} & \underline{\text{Rep}}(\mathcal{L}^+ H_2) & \xrightarrow{\underline{\text{Loc}}_{H_2}^{\text{QCoh}}} & \text{QCoh}(\text{Bun}_{H_2}) \otimes \underline{\text{Vect}} \end{array}$$

The RHS of  $(*)$  can be further written as

$$\begin{aligned} (p^!, \text{QCoh} \circ \underline{\text{Loc}}_{H_2}^{\text{QCoh}} \circ \text{oblv}_2)^{\text{fix}} &\cong (\underline{\text{Loc}}_{H_1}^{\text{QCoh}} \circ \text{oblv}_1 \circ \text{res})^{\text{fix}} \cong ((\text{oblv}_1 \otimes \text{Id}) \circ \underline{\text{Loc}}_{H_1} \circ \text{res})^{\text{fix}} \\ &\cong ((\text{oblv}_1 \otimes \text{Id}) \circ \underline{\text{Loc}}_{H_1}) \circ (\text{res})^{\text{fix}} \cong (\text{oblv}_1 \otimes \text{Id}) \circ (\underline{\text{Loc}}_{H_1} \circ \text{res})^{\text{fix}} \\ &= \text{LHS of } (*) \end{aligned}$$

As we want. ■

Now we want to reverse the arrows by considering the dual categories and dual functors. We can admit that all the category in Theorem 3.4 are self-dual (actually not true for  $\text{Bun}_{H_i}$ ), we have

$$\begin{array}{ccc} \underline{\text{KL}}(H_1) & \xleftarrow{\Gamma_{H_1}^{\text{level}\infty}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_1}) \otimes \underline{\text{Vect}} \\ \text{res}^\vee \downarrow & \searrow & \downarrow p_* \\ \underline{\text{KL}}(H_2) & \xleftarrow{\Gamma_{H_2}^{\text{level}\infty}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_2}) \otimes \underline{\text{Vect}} \end{array}$$

where the global section functor  $\Gamma$  is the dual functor to  $\text{Loc}$ . One should be careful here that it is true only when  $\text{Bun}_H$  is quasi-compact (which rarely happens), in this case  $\mathcal{D}\text{-mod}(\text{Bun}_H)$  is self-dual. But we can pass to quasi-compact opens that cover  $\text{Bun}_H$ . The natural transformation is in the direction shown in above diagram. To get  $p_* \circ \underline{\text{Loc}}_{H_1}$ , we pass to left adjoint (at least for quasi-compact cases)  $\text{Loc} \dashv \Gamma$ , we get a natural transformation in the following direction.

$$\begin{array}{ccc} \underline{\text{KL}}(H_1) & \xrightarrow{\underline{\text{Loc}}_{H_1}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_1}) \otimes \underline{\text{Vect}} \\ \text{res}^\vee \downarrow & \nearrow & \downarrow p_* \\ \underline{\text{KL}}(H_2) & \xrightarrow{\underline{\text{Loc}}_{H_2}} & \mathcal{D}\text{-mod}(\text{Bun}_{H_2}) \otimes \underline{\text{Vect}} \end{array}$$

deduced from Theorem 3.4  
by taking duality and Beck-Chevalley



If  $H_1 \twoheadrightarrow H_2$  with kernel being some unipotent group  $N$ , then the dual restriction functor is nothing but the BRST reduction functor  $\mathcal{C}_{\bullet}^{\infty}(\mathcal{L}\mathfrak{n}, \mathcal{L}^+N, -)$ .

**Theorem 3.5.** *When  $H_1 \twoheadrightarrow H_2$  with kernel unipotent  $N$ , we have  $(\text{Loc}_{H_2} \circ \text{BRST})^{\text{fix}} \xrightarrow{\cong} p_* \circ \text{Loc}_{H_1}$ .*

**Remark 3.3.** Under the assumption as in Theorem 3.5, we know the fiber of  $p : \text{Bun}_{H_1} \rightarrow \text{Bun}_{H_2}$  is quasi-compact, which implies the well-definedness of  $p_*$ .

*Proof of Theorem 3.5.* For simplicity, we assume  $H_2$  to be trivial and  $H_1 = N$ . It is enough to verify for generators in  $\text{KL}(N)$ . We first restrict at some point  $x \in \text{Ran}$  and want to prove the commutativity of the following diagram:

$$\begin{array}{ccccc} \text{KL}(N)_x & \xrightarrow{\text{Loc}} & \mathcal{D}\text{-mod}(\text{Bun}_N) & \xrightarrow{\Gamma_{\text{dR}}} & \mathbf{Vect} \\ \text{ins.vac} \downarrow & & & \nearrow \int_{\text{Ran}_x} & \\ \text{KL}(N)_{\text{Ran}_x} & \xrightarrow{\text{BRST}} & \mathcal{D}\text{-mod}(\text{Ran}_x) & & \end{array}$$

It is enough to prove the commutativity for generators. Let us consider  $\text{Vac}(N)_x$ ,  $\text{ins.vac}(\text{Vac}(N)) = \text{Vac}(N)_{\text{Ran}_x}$ . Its image in  $\mathcal{D}\text{-mod}(\text{Ran}_x)$  under BRST is  $(\mathcal{O}_{\mathbf{BL}^+N})_{\text{Ran}_x}$ . A way to think about this: the semi-infinite cohomology behaves like homology for loop part and cohomology for arc part, our  $\text{Vac}(G)$  is induced from trivial representation on arc part. Thus  $\text{BRST}(\text{Vac}(G)) \cong \text{End}_{\text{Rep}(\mathcal{L}^+N)}(\text{triv}, \text{triv}) \cong \mathcal{O}_{\mathbf{BL}^+N}$ . Let us denote by  $\Psi_N := \mathcal{O}_{\mathbf{BL}^+N}$ , its image under  $\int_{\text{Ran}_x}$  is the factorization homology  $\mathcal{C}_{\bullet}^{\text{fact}}(X, \Psi_N; \Psi_N)$ .

For the other direction, we have seen that  $\text{Loc}(\text{Vac}(G))$  is Diff, sheaf of differential operators on  $\text{Bun}_N$ . Taking de Rham cohomology  $\Gamma_{\text{dR}}$ , we get  $\mathcal{O}_{\text{Bun}_N}$ . So it reduces to show  $\mathcal{O}_{\text{Bun}_N} \cong \mathcal{C}_{\bullet}^{\text{fact}}(X, \Psi_N; \Psi_N)$ . We note  $\text{Bun}_N = \mathbf{Maps}(X, \mathbf{BN}) = \mathbf{Maps}_X(X, \mathbf{BN} \times X)$ , while  $\mathbf{BL}^+N = \mathbf{Maps}_X(\mathbb{D}, \mathbf{BN} \times X)$  for disc  $\mathbb{D}$ . Recall for  $\mathcal{Y} \rightarrow X$  affine  $\mathcal{D}$ -scheme, then  $\mathcal{O}_{\text{Sect}_{\mathcal{Y}}(X, \mathcal{Y})} = \mathcal{C}_{\bullet}^{\text{fact}}(X, \mathcal{O}(\mathcal{L}_{\mathcal{Y}}^+ \mathcal{Y}))$ . Now in our case,  $\mathcal{Y}$  is taken as the Weil restriction of  $\mathbf{BN} \times X$  along  $X \rightarrow X_{\text{dR}}$ , which is not affine  $\mathcal{D}$ -scheme over  $X$ . However,  $\mathbf{BN}$  is as 'good' as affine, in the sense that  $\text{QCoh}(\mathbf{BN}) \cong \mathcal{O}_{\mathbf{BN}}\text{-mod}$ . ■

### 3.4 Spectral Poincaré functors(Kevin Lin)

Recall we denote  $\text{Op}_{\check{G}}^{\text{mer}}$  the  $\check{G}$ -opers on  $\mathring{D}_X$ . For example, if  $\check{G} = \text{GL}_3$ ,  $\text{Op}_{\check{G}}^{\text{mer}} = \left\{ \frac{d}{dt} + \begin{pmatrix} a_1 & a_2 & a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \middle| a_i \in \mathbf{k}((t)) \right\}$ .

In general, we can write  $\text{Op}_{\check{G}}^{\text{mer}} = \text{colim } V_{\leq n}$ , where each  $V_n = \text{lim } V_{\leq n} / V_{\leq -m}$  is a limit of finite dimensional schemes. We want to define a functor  $\text{Poinc}_{\bullet}^{\text{spec}} : \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}} \rightarrow \text{IndCoh}(\text{LocSys}_{\check{G}}))$ , where we recall  $\text{Op}_{\check{G}}^{\text{mon-free}} = \text{Op}_{\check{G}}^{\text{mer}} \times_{\text{LocSys}_{\check{G}}^{\text{mer}}} \text{LocSys}_{\check{G}}^{\text{reg}}$  is the space of monodromy free opers.

Let  $\text{Op}_{\check{G}}^{\text{mer, glob}}$  be the stack of opers on  $X \setminus x$  and  $\text{LocSys}_{\check{G}}^{\text{mer, glob}}$  be the stack of local systems on  $X \setminus x$ , we have

$$\text{Op}_{\check{G}}^{\text{mer}} \longleftarrow \text{Op}_{\check{G}}^{\text{mer, glob}} \longrightarrow \text{LocSys}_{\check{G}}^{\text{mer, glob}}$$

base change along  $\text{LocSys}_{\check{G}}^{\text{reg}}(D_x) \rightarrow \text{LocSys}_{\check{G}}^{\text{mer}}(D_x)$ , we get

$$\text{Op}_{\check{G}}^{\text{mon-free}} \xleftarrow{\text{ev}^{\text{mon-free}}} \text{Op}_{\check{G}}^{\text{mon-free,glob}} \xrightarrow{\tau} \text{LocSys}_{\check{G}}$$

Our first attempt to define the spectral Poincaré functor is taking  $\text{Poinc}_*^{\text{spec}} := \tau_* \circ (\text{ev}^{\text{mon-free}})^*$ . Because of the base change along  $\text{LocSys}_{\check{G}}^{\text{reg}} \rightarrow \text{LocSys}_{\check{G}}^{\text{mer}}$ , we know  $\text{Poinc}_*^{\text{spec}}$  is linear over  $\text{IndCoh}(\text{LocSys}_{\check{G}}^{\text{reg}} \times_{\text{LocSys}_{\check{G}}^{\text{mer}}} \text{LocSys}_{\check{G}}^{\text{reg}})$ .

Moreover, one can show  $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}})$  is tempered, means it equals to  $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}})^{\text{temp}}$ . As a result,  $\text{Poinc}_*^{\text{spec}}$  lands in  $\text{IndCoh}(\text{LocSys}_{\check{G}})^{\text{temp}} \cong \text{QCoh}(\text{LocSys}_{\check{G}})$ .

However, we need to warn that even if  $f : X \rightarrow Y$  is a map of finite type schemes,  $f_* : \text{IndCoh}(X) \rightarrow \text{IndCoh}(Y)$  might not admit a left adjoint  $f^*$ . In general, we need  $f$  to have finite tor amplitude, for example if  $f$  quasi-smooth, then  $f^*$  exists, for example  $\text{pt} \rightarrow \text{pt} \times_{\mathbb{A}^1} \text{pt}$  has no  $\text{IndCoh}^*$ -pullback. So to define  $\text{Poinc}_*^{\text{spec}}$ , we need the existence of  $(\text{ev}^{\text{mer}})^*$ . According to the presentation  $\text{Op}_{\check{G}}^{\text{mer}} = \text{colim } V_{\leq n}$ , we know  $\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mer}}) = \text{colim}_{*-push} \text{IndCoh}^*(V_{\leq n})$  and for each  $n$ ,  $\text{IndCoh}^*(V_{\leq n}) = \text{colim}_{*-pull} \text{IndCoh}^*(V_{\leq n}/V_{\leq -m})$ . Similar to the presentation of  $\text{Op}_{\check{G}}^{\text{mer}}$ , when  $\check{G} = \text{GL}_3$ , we have global opers

$$\text{Op}_{\check{G}}^{\text{mer,glob}} = \left\{ \frac{d}{dt} + \begin{pmatrix} a_1 & a_2 & a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \right\}$$

here  $\alpha_i$ 's are no longer in  $\mathbf{k}((t))$ , but in the global section  $\Gamma(X \setminus x, \Omega_X^{\otimes i})$ . In general, we can write  $\text{Op}_{\check{G}}^{\text{mer,glob}}$  as  $\text{colim } L_{\leq n}$ , with each  $L_n$  finite dimensional and we have

$$\begin{array}{ccccc} L_{\leq n} & \longrightarrow & V_{\leq n} & \longrightarrow & V_{\leq n}/V_{\leq -m} \\ \downarrow & & \downarrow & & \\ \text{Op}_{\check{G}}^{\text{mer,glob}} & \longrightarrow & \text{Op}_{\check{G}}^{\text{mer}} & & \end{array}$$

To define the pullback  $(\text{ev}^{\text{mon-free}})^*$ , it suffices to have the existence of  $(\text{ev}^{\text{mer}})^*$ , which can be deduced from the fact that the  $*$ -pullback along  $L_{\leq n} \rightarrow V_{\leq n}/V_{\leq -m}$  exists.

**Theorem 3.6.** *We have commutative diagram:*

$$\begin{array}{ccccc} \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) & \xrightarrow{\text{Poinc}_*^{\text{spec}}} & \text{IndCoh}^*(\text{LocSys}_{\check{G}}) & & \\ \downarrow (t^+)^* & & \downarrow \Gamma & & \\ \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{reg}}) & \longrightarrow & \mathcal{O}_{\text{Op}_{\check{G}}^{\text{reg}}} \text{-mod}_x^{\text{fact}} & \xrightarrow{\text{C}_{\bullet}^{\text{fact}}(-)} & \mathbf{Vect} \end{array}$$

*Proof.* We prove for an example: let us plug in  $\iota_*^+ \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}}$ . We have

$$\begin{array}{ccccc}
 \mathrm{Op}_G^{\mathrm{reg}} & \xleftarrow{\mathrm{ev}^{\mathrm{reg}}} & \mathrm{Op}_G^{\mathrm{glob}} & \xrightarrow{\tilde{\tau}} & \mathrm{LocSys}_{\check{G}} \\
 \downarrow \iota^+ & & \downarrow \iota & & \\
 \mathrm{Op}_G^{\mathrm{mon-free}} & \xleftarrow{\mathrm{ev}^{\mathrm{mon-free}}} & \mathrm{Op}_G^{\mathrm{mon-free,glob}} & \xrightarrow{\tau} & \mathrm{LocSys}_{\check{G}}
 \end{array}$$

So  $\mathrm{Poinc}_*^{\mathrm{spec}} \left( \iota_*^+ \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} \right) = \tilde{\tau}_* (\mathrm{ev}^{\mathrm{reg}})^* \left( \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} \right)$ . So the upper circuit in the Theorem is functions on global opers  $\mathrm{Op}_G^{\mathrm{glob}}$  and lower circuit of the diagram in Theorem is  $\mathrm{C}_{\bullet}^{\mathrm{fact}} \left( X, \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} \right)$ . We have seen also  $\mathrm{Op}_G^{\mathrm{glob}}$  is an affine scheme with functions  $\mathrm{C}_{\bullet}^{\mathrm{fact}} \left( X, \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} \right)$ . However, the object  $\iota_*^+ \left( \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} \right)$  does not generate whole category. How to get around this problem? Roughly, if  $\mathcal{Y} \rightarrow X$  affine  $\mathcal{D}$ -scheme, it is possible to find  $\mathcal{Y} \rightarrow \mathcal{Y}_i$  modifying  $\mathcal{Y}$  at  $x$ , such that  $\mathcal{L}_{\nabla} \mathcal{Y} = \mathrm{colim} \mathcal{L}_{\nabla}^+ \mathcal{Y}_i$ .  $\blacksquare$

**Remark 3.4.** The Theorem 3.6 can be regard as an consequence of the following general paradigm:

$$\begin{array}{ccc}
 \mathrm{IndCoh}^* (\mathcal{L}_{\nabla} \mathcal{Y})_x & \xrightarrow{\quad} & \mathcal{O}_{\mathrm{Op}_G^{\mathrm{reg}}} \text{-mod}_x^{\mathrm{fact}} \\
 \downarrow \text{*pullback} & & \downarrow \mathrm{C}_{\bullet}^{\mathrm{fact}}(-) \\
 \mathrm{IndCoh}^* (\mathrm{Sect}_{\nabla}(X, \mathcal{Y})) & \xrightarrow{\Gamma} & \mathbf{Vect}
 \end{array}$$

Last, let us mention how to define the functor  $\Gamma^{\mathrm{spec}}$  in Fundamental diagram (1). We have

$$(\mathrm{BG})_{\mathrm{Ran}} \xleftarrow{\mathrm{ev}} \mathrm{LocSys}_{\check{G}} \times \mathrm{Ran} \xrightarrow{p} \mathrm{LocSys}_{\check{G}}$$

Then  $\Gamma^{\mathrm{spec}}$  is defined as  $\mathrm{ev}_* \circ p^!$ , which is right adjoint to  $p_! \circ \mathrm{ev}^* =: \mathrm{Loc}^{\mathrm{spec}}$ .

## 4 Day 4 Eis(senstein) Day

### 4.1 Why Whittaker coefficients are cool(Sam Raskin)

Let us consider  $G = \mathrm{PGL}_2$ . We start with analogy with number theory: Given local system  $\sigma$ , which could be understood as certain Galois representation  $\mathrm{Gal}_{\mathbb{Q}} \rightarrow \tilde{G} = \mathrm{SL}_2$ , we take it to be an irreducible (odd) representation. We are able to define  $a_p = \mathrm{tr}(\sigma(\mathrm{Fr}_p))$ ,  $\mathrm{Fr}_p \in \mathrm{Gal}_{\mathbb{Q}}$ . Write  $q = e^{2\pi i \tau}$ ,  $f(\tau) = \sum a_n q^n$ , where  $a_0 = 0$  (constant term = 0, called "cuspidal"),  $a_1 = 1$ ,  $a_n \cdot a_m = a_{nm}$  if  $\gcd(n, m) = 1$  and recursion formula  $a_{p^{n+1}} + a_{p^{n-1}} = a_p \cdot a_{p^n}$ . In this setup, the Langlands conjecture says  $f(\tau)$  that defined on upper half plan  $\mathrm{Im}(\tau) > 0$ , is a modular form of level 1 and essentially every modular form has this type.

In geometry,  $\sigma \in \mathrm{LocSys}_{\tilde{G}} = \mathrm{LocSys}_{\mathrm{SL}_2}$ , ideally irreducible. What (Betti) geometric Langlands says there exists a Hecke eigensheaf  $\mathcal{F}_{\sigma} \in \mathcal{D}\text{-mod}(\mathrm{Bun}_G)$ . Motto is: we want to characterize  $\mathcal{F}_{\sigma}$  by its "q"-expansion. Analogy of  $\{\text{modular forms}\} \rightarrow \{\text{functions on } \mathbb{R}^{>0}\} : f \mapsto a_0(f)$  is  $\mathrm{CT}_* : \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \rightarrow \mathcal{D}\text{-mod}(\mathrm{Bun}_M)$ . The analogy of  $a_0(f) = 0$  is that for  $\sigma$  irreducible,  $\mathrm{CT}_*(\mathcal{F}_{\sigma}) = 0$  for any proper  $M$ ,  $\mathcal{F}_{\sigma}$  is called cuspidal in this case. The geometric analog of map  $f \mapsto a_1(f)$  is the functor vacuum Whittaker coefficient  $\mathrm{coeff}^{\mathrm{Vac}} : \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \rightarrow \mathbf{Vect}$ . The condition that  $a_1(f) = 1$  is translated to  $\mathrm{coeff}^{\mathrm{Vac}}(\mathcal{F}_{\sigma}) = \mathbf{k}$ .

Those analogies come from comparisons between harmonic analysis over  $\mathbb{R}$ ; adelic stuffs and geomtry For

| Harmonic analysis/ $\mathbb{R}$                                        | Adèles stuffs                                                                  | Geometry                                                    |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------|
| $\mathbb{Z} \subseteq \mathbb{R}$                                      | $\mathbb{Q} \subseteq \mathbb{A}_{\mathbb{Q}}$                                 | $\mathbf{k}(X) \subseteq \mathbb{A}_X$                      |
| $K \backslash G(\mathbb{R}) / G(\mathbb{Z})$                           | $K \backslash G(\mathbb{A}_{\mathbb{Q}}) / G(\mathbb{Q})$                      | $\mathrm{Bun}_G$                                            |
| $\mathbb{R} / \mathbb{Z}$                                              | integral adèles $\backslash \mathbb{A}_{\mathbb{Q}} / \mathbb{Q}$              | $\mathrm{Bun}_{G_a}^{\Omega_X} = \mathrm{Bun}_N^{\Omega_X}$ |
| $a_1(f) = \int_{\mathbb{R}/\mathbb{Z}} f(\tau) e^{-2\pi i \tau} d\tau$ | $a_1(f) = \int_{\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}} f(\tau) \psi(-\tau) d\tau$ | $\int_{\mathrm{Bun}_N^{\Omega_X}} \mathcal{F} \cdot \exp$   |

Table 2: Analogs of Whittaker coefficients

higher coefficients  $a_n, n \geq 1, n \Leftrightarrow p_1^{r_1} \dots p_s^{r_s}$  that corresponds to a effective divisor  $D = \sum_i r_i [p_i]$  on  $\mathrm{Spec} \mathbb{Z}$ . This indicates that the higher Whittaker coefficients for  $\mathrm{PGL}_2$  are indexed by effective divisor on  $X$ . Let  $D$  be an effective divisor, we have

$$\mathrm{Bun}_{\mathrm{PGL}_2} \longleftarrow \mathrm{Bun}_N^{\Omega_X(-D)} = \{0 \rightarrow \Omega_X(-D) \rightarrow \mathcal{E} \rightarrow \mathcal{O} \rightarrow 0\} \longrightarrow H^1(X, \Omega_X(-D)) \longrightarrow H^1(X, \Omega_X) = \mathbb{A}^1$$

We denote the composition  $\mathrm{Bun}_N^{\Omega_X(-D)} \rightarrow H^1(X, \Omega_X(-D)) \rightarrow \mathbb{A}^1$  by  $\psi_D$ , then  $\mathrm{coeff}_D$  is given by  $\psi_D^!(\exp)$ . To conclude, the procedure getting coefficients  $a_n$  for  $f(\tau)$  has an analog that for any  $D = \sum_i n_i x_i$  and  $\sigma \in \mathrm{LocSys}_{\tilde{G}}$ ,

$$\mathrm{coeff}_D(\mathcal{F}_{\sigma}) = \bigotimes_i \mathrm{Sym}^{n_i}(\sigma_{x_i})$$

For  $d \geq 0$ , can easily define  $\mathrm{coeff}_d : \mathcal{D}\text{-mod}(\mathrm{Bun}_{\mathrm{PGL}_2}) \rightarrow \mathcal{D}\text{-mod}(\mathrm{Sym}^d X)$  such that the composition of  $\mathrm{coeff}_d$  with  $!$ -fiber:  $\mathcal{D}\text{-mod}(\mathrm{Sym}^d X) \rightarrow \mathbf{Vect}$  at  $D$ ,  $\deg D = d$  is  $\mathrm{coeff}_D$ .

For general  $G$ , Whittaker coefficients are indexed by divisors  $D = \sum \lambda_i x_i$  on  $X$ , valued in  $\Lambda^+$  dominant coweights of  $G$ . In this case, the above constructions for  $\mathrm{PGL}_2$  become:

$$\begin{aligned} \mathrm{Bun}_G &\leftarrow \mathrm{Bun}_N^{\Omega_X(-D)} \rightarrow \prod_i H^1(X, \Omega_X(\alpha_i(-D))) \rightarrow \prod_i H^1(X, \Omega_X) \xrightarrow{\text{sum}} \mathbb{A}^1 \\ \mathrm{coeff}_D(\mathcal{F}_\sigma) &= \bigotimes_i \mathcal{V}_{\sigma|_{x_i}}^{\lambda_i} \quad \text{for } D = \sum_i \lambda_i x_i \end{aligned}$$

We get  $\mathcal{D}\text{-mod}(\mathrm{Bun}_{\mathrm{PGL}_2}) \rightarrow \prod_{d \geq 0} \mathcal{D}\text{-mod}(\mathrm{Sym}^d X)$ , which cannot be possibly fully-faithful. One can regard  $\mathcal{D}\text{-mod}(\mathrm{Bun}_{\mathrm{PGL}_2})$  as  $\{\text{modular forms}\}$  and another side as  $\{q\text{-expansions}\}$ . To fix, we want to define

$$\mathrm{coeff} : \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \rightarrow \mathrm{Whit}_*(\mathrm{Gr}_{G,\mathrm{Ran}}) \cong \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$$

which is more subtle to define. In fact, what happened secretly on Day 1 is that

$$\begin{array}{ccc} \mathcal{D}\text{-mod}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_{G,\mathrm{coarse}}} & \mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}) \\ & \searrow \mathrm{coeff} & \downarrow \\ & & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \end{array}$$

so we can see (a posteriori) that  $\mathrm{coeff} : \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \rightarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$  is essentially fully-faithful.

**Theorem 4.1.** [FR25] *In fact, we have inclusions  $\mathcal{D}\text{-mod}(\mathrm{Bun}_G)_{\mathrm{cusp}} \subseteq \mathcal{D}\text{-mod}(\mathrm{Bun}_G)_{\mathrm{temp}} \subseteq \mathrm{Rep}(\check{G})_{\mathrm{Ran}}$ .*

Now let us consider the upper part of Fundamental diagram (1), which is

$$\begin{array}{ccc} \mathrm{KL}(G)_{\mathrm{crit},\mathrm{Ran}} & \xrightarrow{\mathrm{FLE}_{\mathrm{crit}}} & \mathrm{IndCoh}^* \left( \mathrm{Op}_{\check{G}}^{\mathrm{mon-free}} \right) \\ \mathrm{Loc}_{\mathrm{crit}} \downarrow & & \downarrow \mathrm{Poinc}^{\mathrm{spec}} \\ \mathcal{D}\text{-mod}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}}) \end{array}$$

Fundamental Diagram 2

where  $\mathrm{Loc}_{\mathrm{crit}}(\mathbf{V}_{\mathrm{crit}}) = \mathrm{Diff}_{\mathrm{crit}} \in \mathcal{D}\text{-mod}(\mathrm{Bun}_G)$  and  $\mathrm{Fun}(\mathrm{Op}_{\check{G}}^{\mathrm{glob}}) \curvearrowright \mathrm{Diff}_{\mathrm{crit}}$ . Recall for  $\sigma \in \mathrm{Op}_{\check{G}} \subseteq \mathrm{LocSys}_{\check{G}}$ , we define  $\mathcal{F}_\sigma$  as  $\mathrm{Diff}_{\mathrm{crit}} / \mathfrak{m}_\sigma$ , where  $\mathfrak{m}_\sigma$  is the corresponding maximal ideal in  $\mathrm{Fun}(\mathrm{Op}_{\check{G}}^{\mathrm{glob}})$ . Let us assume the above diagram commutes, then  $\mathbb{L}_G(\mathrm{Diff}_{\mathrm{crit}}) = \mathrm{Poinc}_*^{\mathrm{spec}} \circ \mathrm{FLE}_{\mathrm{crit}}(\mathbf{V}_{\mathrm{crit}}) = \mathrm{Poinc}_*^{\mathrm{spec}}(\mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{reg}}}) = \mathcal{O}_{\mathrm{Op}_{\check{G}}^{\mathrm{glob}}}$ . This shows  $\mathbb{L}_G(\mathcal{F}_\sigma)$  is the skyscraper sheaf at  $\sigma$ , in other words,  $\mathcal{F}_\sigma$  has the right Whittaker coefficient.

## 4.2 Langlands functor and critical localization(Nick Rozenblyum)

This section is devoted to prove the commutativity of the Fundamental diagram (1). Recall we have

$$\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}) \leftarrow \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \curvearrowright \mathcal{D}\text{-mod}(\mathrm{Bun}_G) \xrightarrow{\mathrm{coeff}^{\mathrm{Vac}}} \mathbf{Vect}$$

From this we obtain the coarse Langlands functor  $\mathbb{L}_{G,\text{coarse}} : \mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LocSys}_{\check{G}})$  and the Langlands functor  $\mathbb{L}_G$  is a renormalized version of  $\mathbb{L}_{G,\text{coarse}}$ . Moreover,  $\text{coeff}^{\text{Vac}}$  factors as  $\mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \text{Whit}'(G)_{\text{crit,Ran}} \rightarrow \mathbf{Vect}$ . As a result, we obtain a commutative diagram extract from the Fundamental diagram:

$$\begin{array}{ccc}
\mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \\
\text{coeff}_* \downarrow & & \downarrow \Gamma^{\text{spec}} \\
\text{Whit}'(G)_{\text{crit,Ran}} & \xleftarrow{\text{CS}} & \text{Rep}(\check{G})_{\text{Ran}} \\
\downarrow & & \downarrow \\
\mathbf{Vect} & \xlongequal{\quad\quad\quad} & \mathbf{Vect}
\end{array}$$

The theorem we really want to prove is:

**Theorem 4.2.** *The upper circuit (i.e. Fundamental diagram (2)) of Fundamental diagram (canonically) commutes:*

$$\begin{array}{ccc}
\text{KL}(G)_{\text{crit,Ran}} & \xrightarrow{\text{FLE}_{\text{crit}}} & \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \\
\text{Loc}_{\text{crit}} \downarrow & & \downarrow \text{Poinc}^{\text{spec}} \\
\mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})
\end{array}$$

To show Theorem 4.2, it is enough to show the outer diagram in Fundamental diagram (1) commutes. The point is,  $\text{Sph}_G$  acts everywhere in the diagram and the Langlands functor  $\mathbb{L}_G$  commutes with  $\text{Sph}_G$ -actions. So we shall consider the tempered categories:  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{temp}} \hookrightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$  and  $\text{IndCoh}(\text{LocSys}_{\check{G}})_{\text{temp}} \cong \text{QCoh}(\text{LocSys}_{\check{G}})$ . And for local categories, we have following facts

$$(i) \text{KL} \cong \text{KL}_{\text{temp}} \quad (ii) \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \cong (\text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}))_{\text{temp}}$$

we can consider the following diagram:

$$\begin{array}{ccc}
\text{KL}(G)_{\text{crit,Ran}} & \xrightarrow{\text{FLE}_{\text{crit}}} & \text{IndCoh}^*(\text{Op}_{\check{G}}^{\text{mon-free}}) \\
\downarrow \text{Loc}_{\text{crit}} & & \downarrow \text{Poinc}^{\text{spec}} \\
\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{temp}} & \xrightarrow{\mathbb{L}_{G,\text{coarse}}|_{\text{temp}}} & \text{QCoh}(\text{LocSys}_{\check{G}}) \\
\downarrow & & \downarrow \\
\mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \\
\downarrow \text{coeff}_* & & \downarrow \Gamma^{\text{spec}} \\
\text{Whit}'(G)_{\text{crit,Ran}} & \xleftarrow{\text{CS}} & \text{Rep}(\check{G})_{\text{Ran}} \\
\downarrow & & \downarrow \\
\mathbf{Vect} & \xlongequal{\quad\quad\quad} & \mathbf{Vect}
\end{array}$$

And we have seen that  $\text{coeff}^{\text{Vac}} \circ \text{Loc}$  and  $\Gamma^{\text{spec}} \circ \text{Poinc}_*^{\text{spec}}$  are (respectively) given by

$$\begin{aligned} & \text{KL}_{\text{crit}, \text{ran}} \xrightarrow{\text{DS}^{\text{enh}}} \mathfrak{Z}_{\text{g}, \text{crit}}\text{-mod}^{\text{fact}} \xrightarrow{\text{C}_*^{\text{fact}}(X, \mathfrak{Z}, -)} \underline{\mathbf{Vect}} \xrightarrow{f_{\text{Ran}}} \mathbf{Vect} \\ \text{and } \text{IndCoh}^* \left( \text{Op}_{\hat{G}}^{\text{mon-free}} \right)_{\text{Ran}} & \xrightarrow{\iota_*^+} \text{IndCoh}^* \left( \text{Op}_{\hat{G}}^{\text{mer}} \right)_{\text{Ran}} \longrightarrow \mathfrak{Z}_{\text{g}, \text{crit}}\text{-mod}^{\text{fact}} \xrightarrow{\text{C}_*^{\text{fact}}(X, \mathfrak{Z}, -)} \underline{\mathbf{Vect}} \xrightarrow{f_{\text{Ran}}} \mathbf{Vect} \end{aligned}$$

Now we can conclude with the statement of  $\text{FLE}_{\text{crit}}$ , namely, the functor  $\text{FLE}_{\text{crit}}$  is constructed such that  $\iota_*^+ \circ \text{FLE}_{\text{crit}}$  agrees with the composition

$$\text{KL}(G)_{\text{crit}} \hookrightarrow \text{KM}(G)_{\text{crit}} \rightarrow \text{Whit}_*(\text{KM}) \xrightarrow{\text{DS}^{\text{enh}}} \text{IndCoh}^* \left( \text{Op}_{\hat{G}}^{\text{mer}} \right)$$

here we recall  $\text{KM}(G)_{\text{crit}} = \hat{\mathfrak{g}}_{\text{crit}}\text{-mod}$  and the above agreement has been showed in section 2.5.

### 4.3 Eisenstein series and constant term(Justin Campbell)

We take parabolic subgroup  $P = M \cdot N_P \subseteq G$ , where  $M$  is the Levi and  $N_P$  the unipotent radical. We have:

$$\text{Bun}_G \xleftarrow{p} \text{Bun}_P \xrightarrow{q} \text{Bun}_M$$

we can define the functors of constant term and Eisenstein series (on automorphic side)

$$\begin{aligned} \text{CT}_* &:= q_* p^! : \mathcal{D}\text{-mod}(\text{Bun}_G) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_M); \\ \text{Eis}_! &:= p_! q^* : \mathcal{D}\text{-mod}(\text{Bun}_M) \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G). \end{aligned}$$

**Proposition 4.1** (Ref: [BG02] and [DG16]). *The functor  $\text{CT}_*$  admits a left adjoint which is  $\text{Eis}_!$ .*

**Remark 4.1.** Since for GLC, we work over critical level, one can modify  $\text{Eis}_!$  and  $\text{CT}_*$  so that they become

$$\begin{array}{ccc} & \xrightarrow{\text{Eis}_!} & \\ \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_M) & \xleftrightarrow{\perp} & \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_G) \\ & \xleftarrow{\text{CT}_*} & \end{array}$$

We can define the Eisenstein generated part and cuspidal part in  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ :

$$\begin{aligned} \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{Eis}} &\subseteq \mathcal{D}\text{-mod}(\text{Bun}_G) \quad \text{full subcategory generated by images of } \text{Eis}_{P,!} \text{ for all } P \subsetneq G \\ \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} &:= \left\{ \mathcal{F} \in \mathcal{D}\text{-mod}(\text{Bun}_G) \mid \text{CT}_{P,*}(\mathcal{F}) = 0, \forall P \subsetneq G \right\} = (\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{Eis}})^\perp \end{aligned}$$

**Theorem 4.3** (Drinfeld-Gaitsgory [DG15]). *There exists  $j : U \hookrightarrow \text{Bun}_G$  open substack such that:*

- (i)  $U$  has quasi-compact intersection with connected components of  $\text{Bun}_G$ .
- (ii) For any  $\mathcal{F} \in \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}}$ ,  $j_! j^! \mathcal{F} \xrightarrow{\cong} \mathcal{F} \xrightarrow{\cong} j_* j^* \mathcal{F}$  (cleaness).

On the spectral side, we have:

$$\text{LocSys}_{\check{G}} \xleftarrow{p^{\text{spec}}} \text{LocSys}_{\check{P}} \xrightarrow{q^{\text{spec}}} \text{LocSys}_{\check{M}}$$

As mentioned before, to define Eisenstein series on spectral side, we need  $(q^{\text{spec}})^*$  to exist, in fact  $q^{\text{spec}}$  is quasi-smooth. Also,  $p^{\text{spec}}$  is proper, so  $(p^{\text{spec}})_*$  exists, as a left adjoint of  $(p^{\text{spec}})^!$ .

**Proposition 4.2** (Arinkin-Gaitsgory [AG15]). *The spectral Eisenstein series  $\text{Eis}^{\text{spec}} := (p^{\text{spec}})_*(q^{\text{spec}})^*$  and the spectral constant term  $\text{CT}^{\text{spec}} := (q^{\text{spec}})_*(p^{\text{spec}})^!$  preserve nilpotent singular support. This gives us adjoint pair:*

$$\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{M}}) \begin{array}{c} \xrightarrow{\text{Eis}^{\text{spec}}} \\ \perp \\ \xleftarrow{\text{CT}^{\text{spec}}} \end{array} \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})$$

The key point is if  $\alpha \in \mathfrak{g}^*$  such that its image in  $\mathfrak{p}^*$  lands in  $\mathfrak{m}^*$ , then  $\alpha$  is nilpotent.

**Theorem 4.4** (Arinkin-Gaitsgory). (1)  $\text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}}) \hookrightarrow \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})$  is an equivalence.  
(2)  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})$  is generated by  $\text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}})$  and  $\text{Eis}^{\text{spec}}(\text{QCoh}(\text{LocSys}_{\check{M}}))$ ,  $\forall P \subsetneq G$ .

### (spectral) constant term via factorization homology

We want to prove the compatibility between Langlands functor and CT, Eis's in the next section. We need some local-to-global arguments. The local analog of spectral constant functor is the Jacquet functor:

$$J^{\text{spec},!} := (q_{\text{loc}}^{\text{spec}})_*(p_{\text{loc}}^{\text{spec}})^! : \text{Rep}(\check{G}) = \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{reg}}) \longrightarrow \text{Rep}(\check{M}) = \text{QCoh}(\text{LocSys}_{\check{M}}^{\text{reg}})$$

In fact, as we mentioned before,  $J^{\text{spec},!}$  is a lax unital functor, we have natural transform:

$$\begin{array}{ccc} \text{Rep}(\check{G}) & \xrightarrow{J^{\text{spec},!}} & \text{Rep}(\check{M}) \\ \text{Loc}_{\check{G}}^{\text{spec}} \downarrow & \swarrow & \downarrow \text{Loc}_{\check{M}}^{\text{spec}} \\ \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \otimes \underline{\mathbf{Vect}} & \xrightarrow{\text{CT}^{\text{spec}} \otimes \text{Id}} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{M}}) \otimes \underline{\mathbf{Vect}} \end{array}$$

**Proposition 4.3.** *There exists a natural transformation  $\eta$  of lax unital local-to-global functors such that*

$$\left( \text{Loc}_{\check{M}}^{\text{spec}} \circ J^{\text{spec},!} \right)^{\int_{\text{ins.unit}}} \xrightarrow{\cong} (\text{CT}^{\text{spec}} \otimes \text{Id}) \circ \text{Loc}_{\check{G}}^{\text{spec}}$$

*Proof.* By definition, we have

$$\begin{array}{ccc} \text{Rep}(\check{G}) & \xrightarrow{\text{Loc}_{\check{G}}^{\text{spec}}} & \text{QCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \otimes \underline{\mathbf{Vect}} \\ \text{Loc}_{\check{G}}^{\text{spec}} \downarrow & \swarrow & \\ \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \otimes \underline{\mathbf{Vect}} & & \end{array}$$



And in [AG15], it is shown that

$$\begin{array}{ccc}
 \mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}) & \xrightarrow{\mathrm{CT}_{\mathrm{QCoh}}^{\mathrm{spec}}} & \mathrm{QCoh}(\mathrm{LocSys}_{\check{M}}) \\
 \downarrow & \swarrow & \downarrow \\
 \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}}) & \xrightarrow{\mathrm{CT}^{\mathrm{spec}}} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{M}})
 \end{array}$$

Also it comes (tautologically) that

$$\begin{array}{ccc}
 \mathrm{Rep}(\check{G}) & \xrightarrow{\mathrm{res}_{\check{G}}^{\check{P}}} & \mathrm{Rep}(\check{P}) \\
 \mathrm{Loc}_{\check{G}}^{\mathrm{spec}} \downarrow & \swarrow & \downarrow \mathrm{Loc}_{\check{P}}^{\mathrm{spec}} \\
 \mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}) \otimes \underline{\mathbf{Vect}} & \xrightarrow{(p^{\mathrm{spec}})^* \otimes \mathrm{Id}} & \mathrm{QCoh}(\mathrm{LocSys}_{\check{P}}) \otimes \underline{\mathbf{Vect}}
 \end{array}$$

Draw the same picture and replace  $\check{P} \rightarrow \check{G}$  by  $\check{P} \rightarrow \check{M}$ , then pass to right adjoints of horizontal functors:

$$\begin{array}{ccc}
 \mathrm{Rep}(\check{P}) & \xrightarrow{\mathrm{inv}_{\check{P}}^{\check{M}}} & \mathrm{Rep}(\check{M}) \\
 \mathrm{Loc}_{\check{P}}^{\mathrm{spec}} \downarrow & \swarrow & \downarrow \mathrm{Loc}_{\check{M}}^{\mathrm{spec}} \\
 \mathrm{QCoh}(\mathrm{LocSys}_{\check{P}}) \otimes \underline{\mathbf{Vect}} & \xrightarrow{(q^{\mathrm{spec}})_* \otimes \mathrm{Id}} & \mathrm{QCoh}(\mathrm{LocSys}_{\check{M}}) \otimes \underline{\mathbf{Vect}}
 \end{array}$$

We want to show that  $\left( \mathrm{Loc}_{\check{M}}^{\mathrm{spec}} \circ \mathrm{inv}_{\check{P}}^{\check{M}} \right)^{f_{\mathrm{ins.unit}}} \cong ((q^{\mathrm{spec}})_* \otimes \mathrm{Id}) \circ \mathrm{Loc}_{\check{P}}^{\mathrm{spec}}$ . It enough to show after base-change along  $\sigma : \mathrm{Spec}(\mathbf{k}) \rightarrow \mathrm{LocSys}_{\check{M}}$ . For simplicity, let us take  $\sigma = \sigma^{\mathrm{triv}}$ . The above diagram reduces to

$$\begin{array}{ccc}
 \mathrm{Rep}(N_{\check{P}}) & \xrightarrow{\mathrm{inv}_{N_{\check{P}}}^{\check{M}}} & \underline{\mathbf{Vect}} \\
 \mathrm{Loc}_{\check{P}}^{\mathrm{spec}} \downarrow & & \parallel \\
 \mathrm{QCoh}(\mathrm{LocSys}_{N_{\check{P}}}) \otimes \underline{\mathbf{Vect}} & \xrightarrow{\Gamma(\mathrm{LocSys}_{N_{\check{P}}}, -)} & \underline{\mathbf{Vect}}
 \end{array}$$

And we can conclude the desired property with the fact that  $\mathrm{C}_{\bullet}^{\mathrm{fact}}(X, \mathrm{inv}_{N_{\check{P}}}(\mathbf{k})) \cong \Gamma(\mathrm{LocSys}_{N_{\check{P}}}, \mathcal{O}_{\mathrm{LocSys}_{N_{\check{P}}}})$ . ■

#### 4.4 Langlands functor vs Eis/CT(Dennis Gaitsgory)

The goal is to show the commutativity of

$$\begin{array}{ccc}
 \mathcal{D}\text{-mod}(\mathrm{Bun}_M) & \xrightarrow{\mathrm{L}_M} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{M}}) \\
 \mathrm{Eis}_! \downarrow & \uparrow \mathrm{CT}_* & \mathrm{Eis}^{\mathrm{spec}} \downarrow \quad \uparrow \mathrm{CT}^{\mathrm{spec}} \\
 \mathcal{D}\text{-mod}(\mathrm{Bun}_G) & \xrightarrow{\mathrm{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}})
 \end{array}$$

Thats to say we want  $\text{Eis}^{\text{spec}} \circ \mathbb{L}_M \cong \mathbb{L}_G \circ \text{Eis}_!$  and  $\mathbb{L}_M \circ \text{CT}_* \cong \text{CT}^{\text{spec}} \circ \mathbb{L}_G$ . Let us first consider the Eisenstein part.

$$\begin{array}{ccccc}
\text{Whit}^!(\text{Gr}_M)_{\text{Ran}} & \xleftarrow{\text{CS}_M} & & \xrightarrow{\quad} & \text{Rep}(\check{M})_{\text{Ran}} \\
\uparrow \text{coeff}_M & \searrow & & & \searrow \text{ind}^{(\text{fix})^\vee} \\
& & \text{Whit}^!(\text{Gr}_G)_{\text{Ran}} & \xleftarrow{\text{CS}_G} & \text{Rep}(\check{G})_{\text{Ran}} \\
& & \uparrow & \downarrow \Gamma_M^{\text{spec}} & \uparrow \Gamma_{\check{G}}^{\text{spec}} \\
D\text{-mod}_{\text{crit}}(\text{Bun}_M) & \xrightarrow{\quad} & \text{QCoh}(\text{LocSys}_{\check{M}}) & & \\
& \searrow \text{Eis}_! & \uparrow \text{coeff}_G & \searrow \text{Eis}_{\text{QCoh}}^{\text{spec}} & \\
& & D\text{-mod}_{\text{crit}}(\text{Bun}_G) & \xrightarrow{\quad} & \text{QCoh}(\text{LocSys}_{\check{G}}) \\
& & \uparrow \mathbb{L}_{M,\text{coarse}} & & \uparrow \mathbb{L}_{G,\text{coarse}}
\end{array}$$

What we want is the commutativity of the bottom lid, what we have is the commutativity of the front, back and right lids. So it remains to construct  $\text{coJ} : \text{Whit}^!(M) \rightarrow \text{Whit}^!(G)$  so that it can make:

(a) the upper lid commute; (b) the left lid commute.

Let us construct first  $J : \text{Whit}^!(G) \rightarrow \text{Whit}^!(M)$ , which is supposed to be compatible with CS and ind. Consider

$$\begin{array}{ccc}
\text{Whit}^!(\text{Gr}_G) \otimes D\text{-mod}(\text{Gr}_G)^{\mathcal{L}N_{P^-} \cdot \mathcal{L}^+ M} & \xrightarrow{\quad} & \text{Whit}^!(\text{Gr}_M) \\
\cong & & \downarrow \\
\text{Whit}^!(\text{Gr}_G) \otimes D\text{-mod}(\text{Gr}_G \times \text{Gr}_M)^{\mathcal{L}P^-} & & \\
\downarrow & & \downarrow \\
D\text{-mod}(\text{Gr}_G) \otimes D\text{-mod}(\text{Gr}_G) \otimes D\text{-mod}(\text{Gr}_M) & \xrightarrow{\langle \cdot, \cdot \rangle \otimes \text{Id}} & D\text{-mod}(\text{Gr}_M)
\end{array}$$

The above diagram claims that if we consider the composition of  $\langle \cdot, \cdot \rangle \otimes \text{Id}$  with inclusion  $\text{Whit}^!(\text{Gr}_G) \otimes D\text{-mod}(\text{Gr}_G \times \text{Gr}_M)^{\mathcal{L}P^-} \hookrightarrow D\text{-mod}(\text{Gr}_G) \otimes D\text{-mod}(\text{Gr}_G) \otimes D\text{-mod}(\text{Gr}_M)$ , then the essential image in  $D\text{-mod}(\text{Gr}_M)$  actually lands in  $\text{Whit}^!(\text{Gr}_M)$ . Let us denote by  $S^-$  the  $\frac{\infty}{2}$ -orbit  $\mathcal{L}N_{P^-} \cdot 1$  inside of  $\text{Gr}_G$

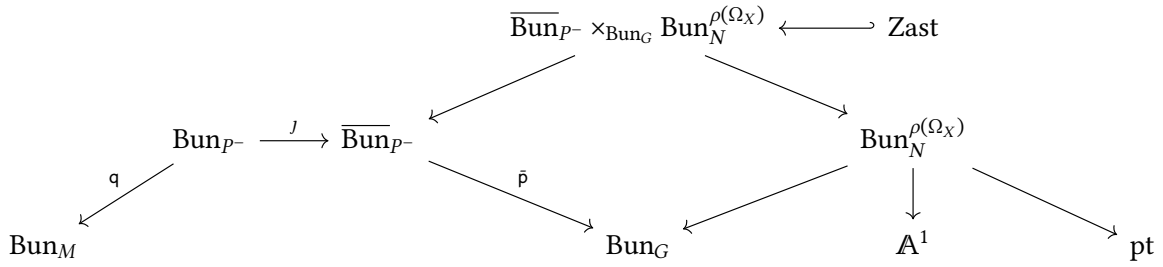
$$\Delta^{-, \frac{\infty}{2}} := i_!(\omega_{S^-}) \quad \text{where } i : S^- \hookrightarrow \text{Gr}_G$$

We define  $J : \text{Whit}^!(\text{Gr}_G) \rightarrow \text{Whit}^!(\text{Gr}_M)$  by inserting  $\Delta^{-, \frac{\infty}{2}} \in D\text{-mod}(\text{Gr}_G)^{\mathcal{L}N_{P^-} \cdot \mathcal{L}^+ M}$  along the above diagram.

**Theorem 4.5 (Sith).** *We have commutative diagram:*

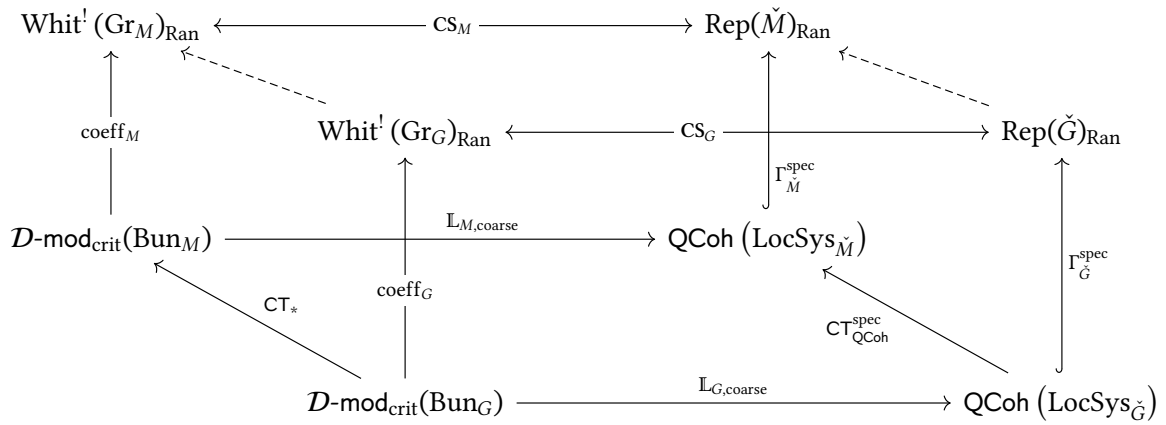
$$\begin{array}{ccc}
\text{Whit}^!(\text{Gr}_G) & \xleftarrow{\text{CS}_G} & \text{Rep}(\check{G}) \\
\downarrow J & & \downarrow j^{\text{spec}} \\
\text{Whit}^!(\text{Gr}_M) & \xleftarrow{\text{CS}_M} & \text{Rep}(\check{M})
\end{array}$$

The (global) geometry of the left lid goes like the following:

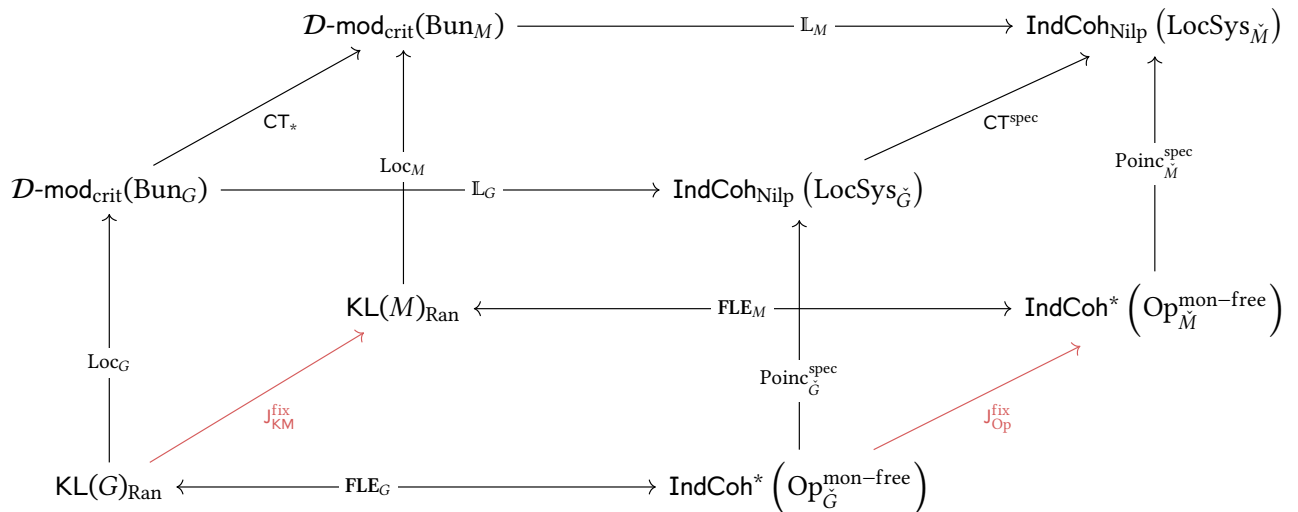


where  $\overline{\text{Bun}}_{P^-}$  is compactification of  $\text{Bun}_{P^-}$  that makes  $\bar{p}$  a proper map. The Zastava space  $\text{Zast}$  is open substack in the fiber product  $\overline{\text{Bun}}_{P^-} \times_{\text{Bun}_G} \text{Bun}_N^{\rho(\Omega_X)}$  that classifies the data where  $P^-$ -reduction and  $N$ -reduction are transversal over generic point of  $X$ . The geometry of Zastava space depends on  $\text{Gr}_M$ .

Next, we goes into the reverse direction for CT:



However, the problem in this case is, there is no local functor that can make either right or left lid commute. We will draw a different cube, using the another part in the Fundamental diagram (1).



Now we want to show the commutativity of the upper lid, we have shown in Theorem 4.2 the commutativity of the front and back lids. It remains to construct two arrows (marked as red in the above diagram) so that they make the left, right and the bottom lids commute. There is also problem in practical that we show the commutativity by precomposing the upper lid with localization functors. However, the functors  $\text{Loc}_G$  and  $\text{Loc}_M$  are not exactly quotient functors, they only surject to category of  $\mathcal{D}$ -modules on (in fact any) quasi-compact open subset of  $\text{Bun}_G$  (and  $\text{Bun}_M$  respectively).

Briefly, the functors  $J_{KM}$  and  $J_{Op}$  to fill the diagram are defined as following:

$$J_{KM} = \text{BRST} = C_{\bullet}^{\infty}(\mathcal{L}\mathfrak{n}_P, \mathcal{L}^+ N_P; -)$$

$$J_{Op} = \text{pull-push along } \text{Op}_{\check{G}}^{\text{mon-free}} \xleftarrow{p^{\text{Miu}}} \text{MOp}_{\check{P}^-}^{\text{mon-free}} \xrightarrow{q^{\text{Miu}}} \text{Op}_{\check{M}}^{\text{mon-free}}$$

where  $\text{MOp}_{\check{P}^-}$  is the stack of *Miura Ops*, see [GLC-III, Section 4.2] for more details.

**Theorem 4.6.** *The bottom lid commutes.*

**Corollary 4.1.** *The Langlands functor  $\mathbb{L}_G$  admits a left adjoint  $(\mathbb{L}_G)^{\perp}$ .*

*Proof.* Want to show that  $(\mathbb{L}_G)^{\perp}$  is defined on a generating set of objects.

**Lemma 4.1.** *Assume we have*

$$\begin{array}{ccc} C_1 & \xrightarrow{F_1} & D_1 \\ \uparrow i_C & & \uparrow i_D \\ C & \xrightarrow{F} & D \end{array}$$

*and that  $F_1, i_C, i_D$  admit a left adjoint, then there exists a left adjoint*

$F^{\perp}$  *defined on essential image of  $i_D^{\perp}$  such that the diagram*

$$\begin{array}{ccc} C_1 & \xleftarrow{F_1^{\perp}} & D_1 \\ \downarrow i_C^{\perp} & & \downarrow i_D^{\perp} \\ C & \xleftarrow{F^{\perp}} & D \end{array}$$

*commutes.*

We apply the Lemma to two circumstances: the first one is

$$\begin{array}{ccc} \text{Whit}^!(\text{Gr}_G)_{\text{Ran}} & \xleftarrow{\quad} \text{CS}_G \xrightarrow{\quad} & \text{Rep}(\check{G})_{\text{Ran}} \\ \text{Poinc}_! \downarrow & \uparrow \text{coeff}_G & \uparrow \Gamma_{\check{G}}^{\text{spec}} \\ & & \text{Loc}^{\text{spec}} \downarrow \\ \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_G) & \xrightarrow{\quad \mathbb{L}_G \quad} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \end{array}$$

As a result, we obtain that  $\mathbb{L}_G^{\perp}$  is defined on  $\text{QCoh}(\text{LocSys}_{\check{G}}) \subseteq \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})$ . Another circumstance is

$$\begin{array}{ccc} \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_M) & \xrightarrow{\quad \mathbb{L}_M \quad} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{M}}) \\ \text{Eis}_! \downarrow & \uparrow \text{CT}_* & \uparrow \text{CT}^{\text{spec}} \\ & & \text{Eis}^{\text{spec}} \downarrow \\ \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_G) & \xrightarrow{\quad \mathbb{L}_G \quad} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \end{array}$$

We obtain (by induction assumption) that  $\mathbb{L}_G^L$  is defined on essential image of  $\text{Eis}^{\text{spec}}$  and the following diagram commutes:

$$\begin{array}{ccc} \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_M) & \xleftarrow{\mathbb{L}_M^L} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{M}}) \\ \text{Eis}_! \downarrow & & \downarrow \text{Eis}^{\text{spec}} \\ \mathcal{D}\text{-mod}_{\text{crit}}(\text{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) \end{array}$$

Recall the Theorem 4.4 says that  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})$  is generated by  $\text{QCoh}(\text{LocSys}_{\check{G}})$  and essential image of  $\text{Eis}^{\text{spec}}$ , so we can conclude.  $\blacksquare$

Similar to  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{Eis}}$ , we can define  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{Eis}}$  to be generated by  $\text{Eis}^{\text{spec}}$  for all proper parabolics in  $\check{G}$ . And  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{cusp}}$  the part that is perpendicular to  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{Eis}}$ . Moreover,  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{Eis}}$  is the same as  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{red}}$ : the category of ind-coherent sheaves with nilpotent support on reducible local systems and  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{cusp}} \cong \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}^{\text{irred}})$ . By irreducibility, we know all nilpotent singular supports reduce to zero section, thus  $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{cusp}} \cong \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}})$ . By such decomposition, we have Langlands functor on Eisenstein part and cuspidal/irreducible part respectively:

$$\begin{array}{ccccc} \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{Eis}} & \xleftrightarrow{\quad} & \mathcal{D}\text{-mod}(\text{Bun}_G) & \xleftrightarrow[e]{e^L} & \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \\ \mathbb{L}_{G,\text{Eis}}^L \uparrow \downarrow \mathbb{L}_{G,\text{Eis}} & & \mathbb{L}_G^L \uparrow \downarrow \mathbb{L}_G & & \mathbb{L}_{G,\text{cusp}}^L \uparrow \downarrow \mathbb{L}_{G,\text{cusp}} \\ \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{red}} & \xleftrightarrow{\quad} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) & \xleftrightarrow[e^{\text{spec}}]{e^{\text{spec},L}} & \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}}) \end{array}$$

**Theorem 4.7.** *The adjoint pair of functors  $(\mathbb{L}_{G,\text{Eis}}^L, \mathbb{L}_{G,\text{Eis}})$  are mutually inverse equivalence.*

**Remark 4.2.** We want an isomorphism  $\text{Eis}^{\text{spec}} \xrightarrow{\cong} \mathbb{L}_G^L \circ \mathbb{L}_G \circ \text{Eis}^{\text{spec}}$ . Note that the RHS is isomorphic to  $\mathbb{L}_G \circ \text{Eis}_! \circ \mathbb{L}_M^L$ , which is  $\text{Eis}^{\text{spec}} \circ \mathbb{L}_M \circ \mathbb{L}_M^L$  and by induction hypothesis, this is isomorphic to  $\text{Eis}^{\text{spec}}$ , which appears on LHS. However, this is not the end of the proof, since we want a map that is an isomorphism for two sides.

On the cuspidal/irreducible part, let us consider the following commutative diagram:

$$\begin{array}{ccccc} \text{Whit}^!(\text{Gr}_G)_{\text{Ran}} & \xleftarrow{\quad} & \text{CS}_G & \xrightarrow{\quad} & \text{Rep}(\check{G})_{\text{Ran}} \\ & \uparrow \text{coeff}_G & & & \uparrow \Gamma_{\check{G}}^{\text{spec}} \\ \mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{\quad \mathbb{L}_G \quad} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) & & \\ & \uparrow & \uparrow & & \\ \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} & \xrightarrow{\quad \mathbb{L}_{G,\text{cusp}} \quad} & \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{red}}) & & \end{array}$$

where the right outer arrow is fully-faithful since  $\Gamma_{\check{G}}^{\text{spec}}$  is fully-faithful. Also we have

**Theorem 4.8.** *The functor  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \rightarrow \text{Whit}^!(\text{Gr}_G)_{\text{Ran}}$  is fully-faithful.*

As a corollary, we know  $\mathbb{L}_{G,\text{cusp}}$  is fully-faithful, as the diagram commutes and CS is an equivalence.

#### 4.5 Vanishing theorem and Hecke-eigen property(Lin Chen)

Recall in Theorem 1.6, we claim that the action  $\text{Rep}(\check{G})_{\text{Ran}}$  on  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  factors through  $\text{QCoh}(\text{LocSys}_{\check{G}})$ , which permit us to define the coarse Langlands functor. Now we explain how to prove this Theorem via Loc.

**Theorem 4.9.** *We define  $\text{KL}_{\text{Ran}}^{\text{Op}} := \left( \text{KL}(G) \otimes_{\text{QCoh}(\text{Op}_{\check{G}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{\check{G}}^{\text{mon-free,glob}}) \right)_{\text{Ran}}$ . Then the localization map  $\text{Loc} : \text{KL}(G)_{\text{Ran}} \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$  factors as*

$$\begin{array}{ccc} \text{KL}(G)_{\text{Ran}} & \xrightarrow{\text{Loc}} & \mathcal{D}\text{-mod}(\text{Bun}_G) \\ & \searrow & \nearrow \text{Loc}^{\text{Op}} \\ & \text{KL}(G)_{\text{Ran}}^{\text{Op}} & \end{array}$$

where  $\text{Loc}^{\text{Op}}$  is  $\text{Rep}(\check{G})_{\text{Ran}}$ -linear and the local-to-global functor  $\text{Loc}^{\text{Op}}$  is unital.

If we restrict to a point  $x \in X$  or more generally  $\underline{x} \in \text{Ran}$ , by FLE, we know  $\text{QCoh}(\text{Op}_{\check{G},x}^{\text{mon-free}}) \cong \text{KL}(G)_x$ , thus  $\text{KL}_x^{\text{Op}} \cong \text{QCoh}(\text{Op}_{\check{G},x}^{\text{mon-free,glob}})$ , where we recall  $\text{Op}_{\check{G},x}^{\text{mon-free,glob}} = \text{Op}_{\check{G}}(X \setminus x) \times_{\text{LocSys}_{\check{G}}(X \setminus x)} \text{LocSys}_{\check{G}}(X)$  that carries a projection to the second factor  $\text{LocSys}_{\check{G}}$ . Thus, as a corollary of Theorem 4.9, we obtain an action of  $\text{QCoh}(\text{LocSys}_{\check{G}})$  on  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ , which is  $\text{Rep}(\check{G})_{\text{Ran}}$ -linear, i.e. we obtain the vanishing Theorem 1.6.

In fact,  $\text{Loc}$  is 'almost' a quotient, meaning for any  $j : U \hookrightarrow \text{Bun}_G$  quasi-compact open,  $j^! \circ \text{Loc} : \text{KL}(G)_{\text{Ran}} \rightarrow \text{Bun}_G$  is a quotient. This tells us that  $\text{Loc}^{\text{Op}}$  almost generates  $\mathcal{D}\text{-mod}(\text{Bun}_G)$ . Hence, only need  $\text{Rep}(\check{G})_{\text{Ran}} \curvearrowright \text{KL}(G)_{\text{Ran}}^{\text{Op}}$  factors through  $\text{QCoh}(\text{LocSys}_{\check{G}})$ , but it follows from the definition of  $\text{Rep}(\check{G})$ -action on  $\text{KL}(G)_{\text{Ran}}^{\text{Op}}$ , which is action on second factor  $\text{Op}_{\check{G}}^{\text{mon-free,glob}} \rightarrow \text{LocSys}_{\check{G}}$ .

Here are some history in [BD]: As is claimed in Theorem 4.9,  $\text{Loc}^{\text{Op}} : \text{KL}(G)^{\text{Op}} \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes \mathbf{Vect}$  is unital, which means it is in  $\text{ShvCat}(\text{Ran}^{\text{untl}})$ . so we can take the stalk at  $\emptyset \in \text{Ran}^{\text{untl}}$  and this gives us:

$$\mathbf{Vect} \longrightarrow \text{QCoh}(\text{Op}_{\check{G}}(X)) \xrightarrow{\text{Loc}_{\emptyset}^{\text{Op}}} \mathcal{D}\text{-mod}(\text{Bun}_G)$$

This functor is first constructed by Beilinson-Drinfeld in [BD], which sends  $\mathbf{k} \mapsto \mathcal{O}_{\text{Op}_{\check{G}}} \mapsto \text{Diff}$ . As we have seen before, for  $\sigma \in \text{Op}_{\check{G}}$ , the image of  $\delta_{\sigma}$  under  $\text{Loc}_{\emptyset}^{\text{Op}}$  is  $\mathcal{F}_{\sigma}$ . By  $\text{Rep}(\check{G})$ -linearity, we obtain an action  $\text{Rep}(\check{G}) \curvearrowright \mathcal{F}_{\sigma}$  and  $\mathcal{F}_{\sigma}$  satisfies the Hecke-eigen property for  $\text{oblv}(\sigma) \in \text{LocSys}_{\check{G}}(X)$ .

We know that  $\text{Op}_{\check{G}}^{\text{reg}} = \mathcal{L}_{\check{V}}^+(\text{Op}_{\check{G}}) \stackrel{\text{FF}}{\cong} \text{Spec } \mathfrak{Z}_{\mathfrak{g}}$  and  $\text{Op}_{\check{G}} = \text{Spec } \mathbf{C}_{\bullet}^{\text{fact}}(X, \mathfrak{Z}_{\mathfrak{g}})$ . Also,  $\text{Loc}$  sends  $\text{Vac}(G)_x$  to  $\text{Diff}$  for any  $\underline{x} \in \text{Ran}$ , and  $\mathfrak{Z}_{\mathfrak{g},\underline{x}}$  acts on  $\text{Diff}$ . The local-to-global functor gives us

$$\begin{array}{ccccccc} \mathbf{Vect} = \text{KL}_{\emptyset} & \xrightarrow{\text{ins.vac}} & \text{KL}_{\text{Ran}} & \xrightarrow{\text{Loc}} & \mathcal{D}\text{-mod}(\text{Bun}_G \times \text{Ran}) & \xrightarrow{f_{\text{Ran}}} & \mathcal{D}\text{-mod}(\text{Bun}_G) \\ \psi & & \psi & & \psi & & \psi \\ \mathbf{k} & \longmapsto & \text{Vac}(G)_{\text{Ran}} & \longmapsto & \text{Loc}(\text{Vac}(G)) \otimes \mathcal{O}_{\text{Ran}} & \longmapsto & \text{Diff} \end{array}$$

As we can see in the above map,  $\mathfrak{Z}_{\mathfrak{g}, \text{Ran}}$  acts on  $\text{Vac}(G)_{\text{Ran}} \in \text{KL}_{\text{Ran}}$  and  $\text{Loc}(\text{Vac}(G)) \otimes \mathcal{O}_{\text{Ran}}$ . In the last step,  $\mathfrak{Z}_{\mathfrak{g}, \text{Ran}} \mapsto \mathcal{C}_{\bullet}^{\text{fact}}(X, \mathfrak{Z}_{\mathfrak{g}})$  that acts on  $\text{Diff}$ , in other words,  $\text{Fun}(\text{Op}_{\check{G}}) \circ \text{Diff}$ . As a result, we get

$$\text{QCoh}(\text{Op}_{\check{G}}(X)) \longrightarrow \mathcal{D}\text{-mod}(\text{Bun}_G); \quad \mathcal{M} \mapsto \text{Diff} \otimes_{\text{Fun}(\text{Op}_{\check{G}}(X))} \mathcal{M}$$

For  $\underline{x} \in \text{Ran}^{\text{unfl}}$ , we want to construct  $\text{KL}(G)_{\underline{x}} \otimes_{\text{QCoh}(\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free, glob}}) \longrightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$ . Now the map  $\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free, glob}} \rightarrow \text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}}$  is affine, we can pretend  $\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}} = \text{Spec } R$ , where  $R \in \mathfrak{Z}\text{-mod}_{\underline{x}}^{\text{fact}}$ , because  $\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}}$  is (spectrum of) a factorization module for  $\text{Op}_{\check{G}}^{\text{reg}} = \text{Spec } \mathfrak{Z}$ . Then we claim that

$$\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free, glob}} = \text{Spec } \mathcal{C}_{\bullet}^{\text{fact}}(X, \mathfrak{Z}_{\mathfrak{g}}; R)$$

which we will denote by  $\text{Spec } R^{\text{glob}}$ . Over  $\text{Ran}_{\underline{x}}$ , we have

$$\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free} \rightsquigarrow \text{reg}} = \left\{ \underline{y} \in \text{Ran}_{\underline{x}} \left| \begin{array}{l} \text{opers mon-free at } \underline{x} \\ \text{opers regular at } \underline{y} \setminus \underline{x} \end{array} \right. \right\} \longrightarrow \text{Ran}_{\underline{x}}$$

Pass to **FLE**, we get following version of Beilinson-Drinfeld (with poles):

$$\begin{array}{ccccccc} \text{KL}(G)_{\underline{x}} & \xrightarrow{\text{ins.vac}} & \text{KL}_{\text{Ran}} & \xrightarrow{\text{Loc}_{\text{Ran}_{\underline{x}}}} & \mathcal{D}\text{-mod}(\text{Bun}_G \times \text{Ran}) & \xrightarrow{\int_{\text{Ran}_{\underline{x}}}} & \mathcal{D}\text{-mod}(\text{Bun}_G) \\ \psi & & \psi & & \psi & & \psi \\ \mathcal{M} & \longmapsto & (\mathcal{M} \rightsquigarrow \text{Vac}(G)_{\text{Ran}}) & \longmapsto & (\text{Loc}_{\underline{x}}(\mathcal{M}) \rightsquigarrow \text{Diff}) & \longmapsto & \text{Loc}_{\underline{x}}(\mathcal{M}) \\ \circ & & \circ & & \circ & & \circ \\ R & \longmapsto & (R \rightsquigarrow \mathfrak{Z}_{\mathfrak{g}}) & \longmapsto & (R \rightsquigarrow \mathfrak{Z}_{\mathfrak{g}}) & \longmapsto & \mathcal{C}_{\bullet}^{\text{fact}}(X, \mathfrak{Z}_{\mathfrak{g}}; R) \end{array}$$

In particular, we get a functor  $\text{KL}(G)_{\underline{x}} \longrightarrow R^{\text{glob}}\text{-mod}(\mathcal{D}\text{-mod}(\text{Bun}_G)) \cong R^{\text{glob}}\text{-mod} \otimes \mathcal{D}\text{-mod}(\text{Bun}_G)$ , which is  $(R\text{-mod})$ -linear. This gives us the functor  $\text{KL}(G)_{\underline{x}} \otimes_{R\text{-mod}} R^{\text{glob}}\text{-mod} \longrightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$ , which is the functor

$$\text{KL}(G)_{\underline{x}} \otimes_{\text{QCoh}(\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free}})} \text{QCoh}(\text{Op}_{\check{G}, \underline{x}}^{\text{mon-free, glob}}) \longrightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$$

by definition.

## 5 Day 5 Doomsday

### 5.1 Bonus C: $\mathbb{L}_{G,\text{cusp}}$ is conservative (Joakim Færgeman)

Recall by induction, we have  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{Eis}} \cong \text{IndCoh}(\text{LocSys}_{\check{G}})_{\text{Eis}}$ . It leaves to show the equivalence

$$\mathbb{L}_{G,\text{cusp}} : \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \xrightarrow{\cong} \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}})$$

The goal of this section is to show first that  $\mathbb{L}_{G,\text{cusp}}$  is conservative. In fact, we are going to prove a stronger result that  $\mathbb{L}_{G,\text{temp}}$  is conservative (recall:  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \subseteq \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{temp}}$ ).

**Remark 5.1.** (1) To prove GLC from here, it remains to show that  $\mathbb{L}_{G,\text{temp}}^{\mathbb{L}}$  is fully-faithful.

(2) the proof of conservativity is micro-local in nature.

**Claim:** To show  $\mathbb{L}_{G,\text{temp}}$  is conservative, it suffices to show its restriction to  $\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)_{\text{temp}}$  is conservative. In fact, given  $0 \neq \mathcal{F} \in \mathcal{D}\text{-mod}(\text{Bun}_G)$ ,  $\exists \sigma : \text{Spec } \mathbf{k} \rightarrow \text{LocSys}_{\check{G}}$  such that  $\delta_\sigma \otimes \mathcal{F} \neq 0$  and  $\sigma : \text{Spec } \mathbf{k} \rightarrow \text{LocSys}_{\check{G}}^{\text{restr}} \rightarrow \text{LocSys}_{\check{G}}$  factors through restricted local systems. Thus

$$\delta_\sigma \otimes \mathcal{F} \in \mathcal{D}\text{-mod}(\text{Bun}_G) \otimes_{\text{QCoh}(\text{LocSys}_{\check{G}})} \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{restr}}) \xrightarrow[\text{Theorem 1.12}]{\cong} \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)$$

Also  $\mathbb{L}_{G,\text{temp}}(\delta_\sigma \otimes \mathcal{F}) = \delta_\sigma \otimes \mathbb{L}_{G,\text{temp}}(\mathcal{F})$ . So if  $\mathbb{L}_{G,\text{temp}}(\delta_\sigma \otimes \mathcal{F}) \neq 0$ , then  $\mathbb{L}_{G,\text{temp}}(\mathcal{F}) \neq 0$ . So it reduces to describe the kernel

$$\text{Ker}(\mathbb{L}_{G,\text{temp}} : \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G) \rightarrow \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{restr}}))$$

For fixed non-degenerate character  $\psi \in \mathbf{n}^*$ , consider  $\mathfrak{g}^*/N = \mathfrak{g}^*/G \times_{\text{pt}/G} \text{pt}/N$ . Define Kostant section

$$\begin{array}{ccc} \text{Kos}_\psi & \longrightarrow & \psi/N \\ \downarrow & & \downarrow \\ \mathfrak{g}^*/N & \longrightarrow & \mathbf{n}^*/N \end{array}$$

Identifying  $\mathfrak{g}^* \cong \mathfrak{g}$ ,  $\mathbf{n}^* \cong \mathbf{n}^-$  and  $\psi = f$ , we get  $\text{Kos}_\psi \cong f + \mathfrak{b}/N \cong f + \mathfrak{g}^e$  and the map  $f + \mathfrak{b}/N \rightarrow \mathfrak{g}_{\text{reg}}/G$  is a bijection on  $\mathbf{k}$ -points. We denote  $\mathfrak{g}_{\text{irreg}} = \mathfrak{g} \setminus \mathfrak{g}_{\text{reg}}$  and  $\mathcal{N}_{\text{irreg}} = \mathcal{N} \cap \mathfrak{g}_{\text{irreg}}$ . Correspondingly, we can define global irregular nilpotent cone  $\text{Nilp}_{\text{irreg}} \subseteq T^* \text{Bun}_G$  to be the pairs

$$(\mathcal{P}_G, \varphi \in \Gamma(X, \mathfrak{g}_{\mathcal{P}_G}^* \otimes \Omega_X^1)) \quad \text{such that } \varphi \text{ is nilpotent irregular, i.e. locally factors through } \mathcal{N}_{\text{irreg}}$$

**Theorem A.**  $\text{Ker}(\mathbb{L}_{G,\text{coarse}}|_{\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)}) = \mathcal{D}\text{-mod}_{\text{Nilp}_{\text{irreg}}}(\text{Bun}_G)$ .

**Theorem B.**  $\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)_{\text{temp}} \cap \mathcal{D}\text{-mod}_{\text{Nilp}_{\text{irreg}}}(\text{Bun}_G) = \emptyset$ .

**Example 5.1.** If we take  $G = \text{SL}_2$ , then  $\text{Nilp}_{\text{irreg}} = \{0\}$  and

$$\mathcal{D}\text{-mod}_{\text{Nilp}_{\text{irreg}}}(\text{Bun}_G) = \left\{ \mathcal{F} \in \mathcal{D}\text{-mod}(\text{Bun}_G) \mid H^i(\mathcal{F}) = \bigcup (\text{extension of vector bundles with } \nabla) \right\}$$



To see Theorem A, we compose  $\mathbb{L}_{G,\text{coarse}}$  with  $\Gamma : \text{QCoh}(\text{LocSys}_{\check{G}}) \rightarrow \mathbf{Vect}$ , which gives us

$$\text{coeff}^{\text{Vac}} : \mathcal{D}\text{-mod}(\text{Bun}_G) \longrightarrow \mathbf{Vect}$$

We define a global version of Kostant section: character  $\psi$  induces  $\psi : \text{Bun}_N^{\Omega_X} \rightarrow \mathbb{A}^1$ , taking differential, we get

$$d\psi : \text{Bun}_N^{\Omega} \rightarrow T^* \text{Bun}_N^{\Omega}$$

and define the global Kostant section as  $\text{Kos}_{\psi}^{\text{glob}} := \text{Bun}_N^{\Omega} \times_{T^* \text{Bun}_N^{\Omega}} (T^* \text{Bun}_G \times_{\text{Bun}_G} \text{Bun}_N^{\Omega})$ . The Key property of  $\text{Kos}_{\psi}^{\text{glob}}$  is  $\text{Kos}_{\psi}^{\text{glob}} \times_{T^* \text{Bun}_G} \text{Nilp} = \{f^{\text{glob}}\}$ , this is an analog that  $\text{Kos}_{\psi} \times_{\mathfrak{g}^*/G} \mathcal{N}/G = \{f\}$ . Define  $\text{Nilp}^{\text{Kos}} \subseteq \text{Kos}_{\psi}^{\text{glob}}$  the irreducible component containing  $f^{\text{glob}}$ .

**Example 5.2.** For example if we take  $G = \text{SL}_2$ , then

$$\begin{aligned} \text{Nilp} &= \bigcup_{d \geq 1-g} \overline{\text{Nilp}^d} \cup \{0\}, \quad \text{where,} \\ \text{Nilp}^d &= \{0 \rightarrow \mathcal{L} \rightarrow \mathcal{E} \rightarrow \mathcal{L}^{\vee}, 0 \neq \varphi : \mathcal{L}^{\vee} \rightarrow \mathcal{L} \otimes \Omega_X^1\} \end{aligned}$$

where the Higgs field is given by  $\mathcal{E} \rightarrow \mathcal{L}^{\vee} \rightarrow \mathcal{L} \otimes \Omega_X^1 \rightarrow \mathcal{L} \otimes \Omega_X^1$ . In this case,

$$f^{\text{glob}} = \left( \Omega_X^{-\frac{1}{2}} \oplus \Omega_X^{\frac{1}{2}}; \varphi : \Omega_X^{-\frac{1}{2}} \oplus \Omega_X^{\frac{1}{2}} \xrightarrow{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}} (\Omega_X^{-\frac{1}{2}} \oplus \Omega_X^{\frac{1}{2}}) \otimes \Omega_X^1 = \Omega_X^{\frac{1}{2}} \oplus \Omega_X^{\frac{3}{2}} \right)$$

As a result, we see that  $\text{Nilp}^{\text{Kos}} = \text{Nilp}^{1-g}$ .

**Theorem 1A.**  $\text{coeff}^{\text{Vac}}|_{\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)}$  satisfies: (1) It is  $t$ -exact; (2) It commutes with Verdier duality; (3) If  $\mathcal{F} \in \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)^{\heartsuit}$ , then  $\dim(\text{coeff}^{\text{Vac}}(\mathcal{F})) = \text{mult}(\text{Nilp}^{\text{Kos}}, \text{CC}(\mathcal{F}))$ .

**Theorem 2A.** Pick  $x \in X$ , if  $\mathcal{F} \in \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)$  such that  $\text{SS}(\mathcal{F}) \not\subseteq \text{Nilp}_{\text{irreg}}$ , then there exists  $\check{\lambda} \in \check{\Lambda}^+$  so that

$$\text{SS}(\mathcal{H}_{\mathcal{V}_x^{\check{\lambda}}} \star \mathcal{F}) \supseteq \text{Nilp}^{\text{Kos}}$$

Exercise: Formulate and prove the analog of Theorem 2A for  $\mathcal{D}\text{-mod}_{\text{Nilp}}(\mathfrak{g}/G)$  using Fourier transform.

**Claim:** Theorem 1A + Theorem 2A implies Theorem A, namely  $\text{Ker}(\mathbb{L}_{G,\text{coarse}}|_{\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)}) = \mathcal{D}\text{-mod}_{\text{Nilp}_{\text{irreg}}}(\text{Bun}_G)$ :

*Proof of the Claim.* To prove  $\subseteq$ , take  $\mathcal{F} \in \mathcal{D}\text{-mod}_{\text{Nilp}_{\text{irreg}}}(\text{Bun}_G)$ , then Theorem 1A + 2A implies  $\text{coeff}^{\text{Vac}}(\mathcal{H}_{\mathcal{V}_x^{\check{\lambda}}} \star \mathcal{F}) = 0$ , which means  $\Gamma(\text{LocSys}_{\check{G}}^{\text{restr}}, \mathbb{L}_{G,\text{coarse}}(\mathcal{F}) \otimes E_x^{\check{\lambda}}) = 0$ . It implies  $\mathbb{L}_{G,\text{coarse}}(\mathcal{F}) = 0$ , since  $\text{LocSys}_{\check{G}}^{\text{restr}} \times_{\text{pt}/\check{G}} \text{pt}$  is a formal affine scheme.

For  $\supseteq$ , given  $\mathcal{F} \in \text{Ker}(\mathbb{L}_{G,\text{coarse}})$ , we know

$$\Gamma\left(\text{LocSys}_{\check{G}}^{\text{restr}}, \mathbb{L}_{G,\text{coarse}}(\mathcal{F}) \otimes E_x^{\check{\lambda}}\right) = \Gamma\left(\text{LocSys}_{\check{G}}, \text{coeff}^{\text{vac}}\left(\mathcal{H}_{\mathcal{V}_x^{\check{\lambda}}} \star \mathcal{F}\right)\right) = 0$$

for any  $\check{\lambda} \in \check{\Lambda}$ , then by Theorem 2A, we know  $\text{SS}(\mathcal{F}) \subseteq \text{Nilp}_{\text{irreg}}$ . ■

Now Theorem A and Theorem B together imply that

$$\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{temp}} \cap \mathcal{D}\text{-mod}_{\text{Nilp}_{\text{irreg}}}(\text{Bun}_G) = 0 \implies \text{Ker}\left(\mathbb{L}_{G,\text{coarse}} \Big|_{\mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)}\right) \cap \mathcal{D}\text{-mod}_{\text{Nilp}}(\text{Bun}_G)_{\text{temp}} = 0$$

This gives the conservativity we want.

## 5.2 Ambidexterity-I(Dennis Gaitsgory)

We have proved in the previous section that  $\mathbb{L}_{G,\text{cusp}}$  is conservative, the next goal in this and the following sections is to show that the left adjoint  $\mathbb{L}_{G,\text{cusp}}^{\text{L}}$  is fully-faithful. Recall the following diagram:

$$\begin{array}{ccccc} \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{Eis}} & \xrightleftharpoons{\quad} & \mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightleftharpoons[e]{e^{\text{L}}} & \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \\ \mathbb{L}_{G,\text{Eis}}^{\text{L}} \uparrow \downarrow \mathbb{L}_{G,\text{Eis}} & & \mathbb{L}_G^{\text{L}} \uparrow \downarrow \mathbb{L}_G & & \mathbb{L}_{G,\text{cusp}}^{\text{L}} \uparrow \downarrow \mathbb{L}_{G,\text{cusp}} \\ \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}})_{\text{red}} & \xrightleftharpoons{\quad} & \text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{\check{G}}) & \xrightleftharpoons[e^{\text{spec}}]{e^{\text{spec,L}}} & \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}}) \end{array}$$

As we have proved  $\mathbb{L}_{G,\text{temp}}$  is conservative, thus by Barr-Beck-Lurie, we have monad:

$$\mathbb{L}_{G,\text{temp}} \circ \mathbb{L}_{G,\text{temp}}^{\text{L}} \cong \mathcal{A}_G \otimes - : \text{QCoh}(\text{LocSys}_{\check{G}}) \longrightarrow \text{QCoh}(\text{LocSys}_{\check{G}}), \quad \mathcal{A}_G \in \text{AssoAlg}(\text{QCoh}(\text{LocSys}_{\check{G}}))$$

The functor  $e^{\text{spec}}$  (resp.  $e^{\text{spec,L}}$ ) is given by  $(j^{\text{spec}})_*$  (resp.  $(j^{\text{spec}})^*$ ), we take  $\mathcal{A}_{G,\text{cusp}} := (j^{\text{spec}})^* \mathcal{A}_G$ , then

$$\mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^{\text{L}} \cong \mathcal{A}_{G,\text{cusp}} \otimes - \in \text{End}(\text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}})) \quad \mathcal{A}_{G,\text{cusp}} \in \text{AssoAlg}(\text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}}))$$

To show the fully-faithfulness of  $\mathbb{L}_{G,\text{temp}}$  (resp.  $\mathbb{L}_{G,\text{cusp}}$ ), it is enough to show that  $\mathcal{O}_{\text{LocSys}_{\check{G}}} \cong \mathcal{A}_G$  (resp.  $\mathcal{O}_{\text{LocSys}_{\check{G}}^{\text{irred}}} \cong \mathcal{A}_{G,\text{cusp}}$ ). We know already  $((i^{\text{spec}})^{\wedge})^! \mathcal{O}_{\text{LocSys}_{\check{G}}} \cong ((i^{\text{spec}})^{\wedge})^! \mathcal{A}_G$ , where  $(i^{\text{spec}})^{\wedge}$  is pullback to reducible locus.

**Theorem 5.5** (Ambidexterity).  $\mathbb{L}_{G,\text{cusp}}$  is ambidexterous, namely  $\mathbb{L}_{G,\text{cusp}}^{\text{L}} \cong \mathbb{L}_{G,\text{cusp}}^{\text{R}}$ .

**Corollary 5.1.**  $\mathcal{A}_{G,\text{cusp}}$  is perfect in  $\text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}})$  and self-dual.

*Proof.* By definition we know  $\mathcal{A}_{G,\text{cusp}} \otimes - = \mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^{\text{L}}$ , we have

$$(\mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^{\text{L}})^{\text{R}} = (\mathbb{L}_{G,\text{cusp}}^{\text{L}})^{\text{R}} \circ \mathbb{L}_{G,\text{cusp}}^{\text{R}} = \mathbb{L}_{G,\text{cusp}} \circ \mathbb{L}_{G,\text{cusp}}^{\text{L}} \quad \blacksquare$$

## Duality of DG categories

Let  $\mathcal{C}$  be a compactly generated category, then its dual  $\mathcal{C}^\vee = \text{Func}_{\text{cont}}(\mathcal{C}, \mathbf{Vect})$ . Another way to see the dual is  $\mathcal{C} = \text{Ind}(\mathcal{C}^c)$ , then  $\mathcal{C}^\vee = \text{Ind}((\mathcal{C}^c)^{\text{op}})$ . There exist pairing and unit:

$$\mathcal{C} \otimes \mathcal{C}^\vee \xrightarrow{\text{ev}} \mathbf{Vect}; \quad \text{unit}_{\mathcal{C}} \in \mathcal{C} \otimes \mathcal{C}^\vee$$

If  $F : \mathcal{C} \rightarrow \mathcal{D}$  functor between  $\mathcal{C}$  and another compactly generated  $\mathcal{D}$ , we will have dual functor  $F^\vee : \mathcal{D}^\vee \rightarrow \mathcal{C}^\vee$ . Also, if  $F : \mathcal{C} \rightarrow \mathcal{D}$  preserves compactness (for example when  $F$  admits a right adjoint  $F^R$ ), then we write  $F : \mathcal{C}^c \rightarrow \mathcal{D}^c$ , this gives  $F^{\text{conj}} : (\mathcal{C}^c)^{\text{op}} \rightarrow (\mathcal{D}^c)^{\text{op}}$ . Pass to ind-completion, we get  $F^{\text{conj}} = (F^R)^\vee : \mathcal{C}^\vee \rightarrow \mathcal{D}^\vee$ .

**Lemma 5.1.** *Under the above assumption/construction,  $(F^{\text{conj}}, F^\vee)$  forms an adjoint pair.*

In fact, denote  $\mathbb{D}_{\mathcal{C}} : \mathcal{C}^c \rightarrow (\mathcal{C}^\vee)^c$  the Verdier duality on  $\mathcal{C}^c$ , then  $F^{\text{conj}} = \mathbb{D}_{\mathcal{D}} \circ F \circ \mathbb{D}_{\mathcal{C}}$ .

**Example 5.3.** For example if  $\mathcal{Y}$  is a quasi-compact algebraic stack, then  $\mathcal{C} = \mathcal{D}\text{-mod}(\mathcal{Y})$  is compactly generated and self-dual. The pairing is given by  $\mathcal{D}\text{-mod}(\mathcal{Y}) \otimes \mathcal{D}\text{-mod}(\mathcal{Y}) \xrightarrow{\langle \cdot, \cdot \rangle} \mathbf{Vect}; (F_1, F_2) \mapsto C_{\text{dR}}^\bullet \left( \mathcal{Y}, F_1 \overset{!}{\otimes} F_2 \right)$ . And the unit is given by  $\Delta_*(\omega_{\mathcal{Y}}) \in \mathcal{D}\text{-mod}(\mathcal{Y} \times \mathcal{Y}) \cong \mathcal{D}\text{-mod}(\mathcal{Y}) \otimes \mathcal{D}\text{-mod}(\mathcal{Y})$ . We have  $(\mathcal{D}\text{-mod}(\mathcal{Y})^c)^{\text{op}} = \mathcal{D}\text{-mod}(\mathcal{Y})^c$ . This is the Verdier duality  $\mathbb{D}_{\mathcal{Y}}$  on  $\mathcal{Y}$ .

Assume  $f : \mathcal{Y}_1 \rightarrow \mathcal{Y}_2$  map between quasi-compact stacks, can consider  $f_* : \mathcal{D}\text{-mod}(\mathcal{Y}_1) \rightarrow \mathcal{D}\text{-mod}(\mathcal{Y}_2)$ , whose dual functor  $(f_*)^\vee : \mathcal{D}\text{-mod}(\mathcal{Y}_2) \rightarrow \mathcal{D}\text{-mod}(\mathcal{Y}_1)$  is nothing but  $f^!$ . To have  $(f_*)^{\text{conj}}$ , we want  $f_*$  to preserve compact objects, namely  $f$  need to be proper. In this case  $(f_*)^{\text{conj}}$  is  $f_! = f_*$ . Now suppose  $f$  is smooth, in this case  $f^!$  preserves compactness, then  $(f^!)^{\text{conj}} = f^* = f^!$ .

Why we care about duality? For example if  $U \subseteq \text{Bun}_G$  quasi-compact open, then  $\mathcal{D}\text{-mod}(U)$  is self-dual (NB:  $\text{Bun}_G$  is not quasi-compact,  $\mathcal{D}\text{-mod}(\text{Bun}_G)$  is dualizable, in a weird/miraculous way). So we can take  $U \xrightarrow{j} \text{Bun}_G$  large enough such that for  $\mathcal{F} \in \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}}$ ,  $\mathcal{F} \cong j_* j^*(\mathcal{F})$ .

Assume  $\mathcal{C}_1 \xleftarrow[e]{e^\perp} \mathcal{C}$ , this induces  $\mathcal{C}_1^\vee \xleftarrow[(e^\perp)^\vee]{e^\vee} \mathcal{C}^\vee$ . In our case, the embedding  $e : \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \hookrightarrow$

$\mathcal{D}\text{-mod}(\text{Bun}_G)$  decomposes as

$$\begin{array}{ccccc} \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} & \xleftarrow{(e_U)^\perp} & \mathcal{D}\text{-mod}(U) & \xrightarrow{j_*} & \mathcal{D}\text{-mod}(\text{Bun}_G) \\ & \xrightarrow{e_U} & & & \\ & \searrow e & & & \end{array}$$

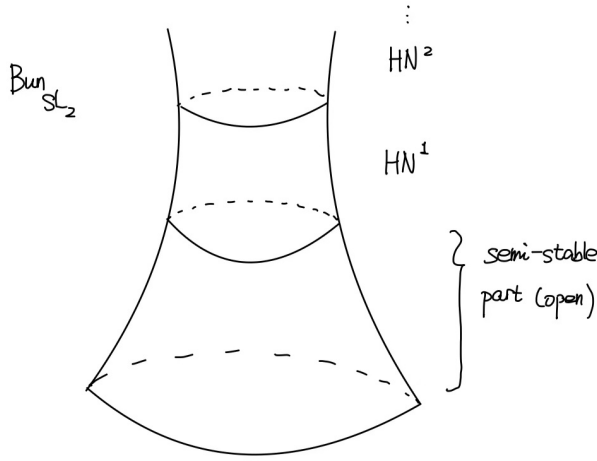
We know  $\mathcal{D}\text{-mod}(U)$  is self-dual, by above paradigm, we have  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}}^\vee \xleftarrow[(e_U^\perp)^\vee]{(e_U)^\vee} \mathcal{D}\text{-mod}(U)$ .

**Lemma 5.2.** *We have*

$$\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}}^\vee \cong \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \subseteq \mathcal{D}\text{-mod}(U)$$

where  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \subseteq \mathcal{D}\text{-mod}(U)$  can be realized as  $(j^* \circ \text{Eis})^\perp$  and  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}}^\vee \subseteq \mathcal{D}\text{-mod}(U)$  can be realized as  $(\mathbb{D}_U \circ (j^* \circ \text{Eis}) \circ \mathbb{D}_{\text{Bun}_G})^\perp$ .

**Example 5.4.** Let  $G = \mathrm{SL}_2$ , we can draw a Caton of  $\mathrm{Bun}_{\mathrm{SL}_2}$ :



$\mathrm{HN}^d = \{0 \rightarrow \mathcal{L} \rightarrow \mathcal{E} \rightarrow \mathcal{L}^\vee \rightarrow 0 \mid \deg \mathcal{L} = d\}$ .

Exercise 1:  $\mathrm{HN}^d \rightarrow \mathrm{Bun}_{\mathrm{G}_m}$  is a unipotent gerbe for  $d \gg 0$ , in which case,  $H^i(\mathcal{L}^2) = 0$ .

Exercise 2:  $\mathrm{CT}_*^d = !\text{-restriction to } \mathrm{HN}^d \text{ for large } d$ .

Figure 3: Carton of  $\mathrm{Bun}_{\mathrm{SL}_2}$

**Claim:** After composing with  $j^*$ ,  $\mathrm{Eis}_!$  and  $\mathrm{Eis}_*$  are finite extensions of copies of each other, where the set of finite extensions comes the strata of Drinfeld compactification.

We have  $\mathcal{D}\text{-mod}(\mathrm{Bun}_G)_{\mathrm{cusp}} \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}} \mathrm{QCoh}\left(\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}\right)$ , its dual functor is  $\mathrm{QCoh}\left(\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}\right) \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}^\vee} \mathcal{D}\text{-mod}(\mathrm{Bun}_G)_{\mathrm{cusp}}$ .

**Theorem 5.6.** (i)  $\mathbb{L}_{G,\mathrm{cusp}}^\vee \cong \mathbb{L}_{G,\mathrm{cusp}}^{\mathrm{L}}$ ; (ii)  $\mathbb{L}_{G,\mathrm{cusp}}^\vee \cong \mathbb{L}_{G,\mathrm{cusp}}^{\mathrm{R}}$ .

The above theorem implies the ambidexterity we want.

*Proof of Theorem 5.6(i).* It equivalent to show  $\mathbb{L}_{G,\mathrm{cusp}}^{\mathrm{L}} \cong (\mathbb{L}_{G,\mathrm{cusp}}^{\mathrm{L}})^{\mathrm{conj}}$ . Recall

$$\begin{array}{ccc}
 \mathrm{Whit}^!(\mathrm{Gr}_G)_{\mathrm{Ran}} & \xleftarrow{\mathrm{CS}_G} & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} \\
 \uparrow \mathrm{coeff}_G & & \uparrow \Gamma_{\check{G}}^{\mathrm{spec}} \\
 \mathcal{D}\text{-mod}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}}) \\
 \uparrow e & & \uparrow (j^{\mathrm{spec}})^* \\
 \mathcal{D}\text{-mod}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{\mathbb{L}_{G,\mathrm{cusp}}} & \mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}^{\mathrm{red}})
 \end{array}$$

Taking the dual functors gives

$$\begin{array}{ccc}
 & \mathrm{Vect} & \\
 & \downarrow & \\
 & \mathrm{Rep}(\check{G})_{\mathrm{Ran}} & \\
 \swarrow (\mathrm{coeff}^{\mathrm{Vac}} \circ e)^\vee & & \searrow (\Gamma(\mathrm{LocSys}_{\check{G}}, -) \circ j_*^{\mathrm{spec}})^\vee \\
 \mathcal{D}\text{-mod}(\mathrm{Bun}_G)_{\mathrm{cusp}} & \xrightarrow{(\mathbb{L}_{G,\mathrm{cusp}})^\vee} & \mathrm{QCoh}(\mathrm{LocSys}_{\check{G}}^{\mathrm{red}})
 \end{array}$$

Taking the left adjoint functors gives

$$\begin{array}{ccc}
 & \text{Vect} & \\
 & \downarrow & \\
 & \text{Rep}(\check{G})_{\text{Ran}} & \\
 \swarrow & & \searrow \\
 (\text{coeff}^{\text{Vac}} \circ e)^{\text{L}} & & (\Gamma(\text{LocSys}_{\check{G}}, -) \circ j_*^{\text{spec}})^{\text{L}} \\
 \downarrow & & \downarrow \\
 \text{D-mod}(\text{Bun}_G)_{\text{cusp}} & \xrightarrow{\mathbb{L}_{G, \text{cusp}}^{\text{L}}} & \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{red}})
 \end{array}$$

. We want

to show outer arrows in above two diagrams of dual/left adjoint coincide.

(i)  $(\Gamma^{\text{spec}} \circ j_*^{\text{spec}})^{\text{L}} = (j_*^{\text{spec}})^{\text{L}} \circ \Gamma(\text{LocSys}_{\check{G}}, -)^{\text{L}} = (j^{\text{spec}})^* \circ \mathcal{O}_{\text{LocSys}_{\check{G}}} = \mathcal{O}_{\text{LocSys}_{\check{G}}^{\text{irred}}}$ . On the other hand  $(\Gamma^{\text{spec}} \circ j_*^{\text{spec}})^{\vee} = (j_*^{\text{spec}})^{\vee} \circ \Gamma(\text{LocSys}_{\check{G}}, -)^{\vee} = (j^{\text{spec}})^* \circ \mathcal{O}_{\text{LocSys}_{\check{G}}} = \mathcal{O}_{\text{LocSys}_{\check{G}}^{\text{irred}}}$ .

(ii)  $(\text{coeff}^{\text{Vac}} \circ e)^{\text{L}} = e^{\text{L}} \circ (\text{coeff}^{\text{Vac}})^{\text{L}} = e^{\text{L}} \circ (\text{Poinc}_!^{\text{Vac}})$  while  $((\text{coeff}^{\text{Vac}} \circ e)^{\text{L}})^{\text{conj}} \cong (\text{coeff}^{\text{Vac}} \circ e)^{\text{L}} = e^{\text{L}}(\text{Poinc}_!^{\text{Vac}})$ . We want to compare  $e^{\text{L}}(\text{Poinc}_!^{\text{Vac}})$  and  $e^{\text{L}}(\mathbb{D}(\text{Poinc}_!^{\text{Vac}}))$ . Note that  $\text{Poinc}_*^{\text{Vac}} := \mathbb{D}(\text{Poinc}_!^{\text{Vac}}) \neq \text{Poinc}_!^{\text{Vac}}$ . We claim, however, after composing  $e^{\text{L}}$  we have

$$e^{\text{L}}(\text{Poinc}_!^{\text{Vac}}) = e^{\text{L}}(\text{Poinc}_*^{\text{Vac}})$$

And the Claim is enough to show the Theorem 5.6 (i), as we can compare:

$$\begin{array}{ccc}
 & \text{Vect} & \\
 \swarrow & & \searrow \\
 e_U^{\text{L}} \circ (j_U^* \circ \text{Poinc}_!^{\text{Vac}}) & & \mathcal{O}_{\text{LocSys}_{\check{G}}^{\text{irred}}} \\
 \downarrow & & \downarrow \\
 \text{D-mod}(\text{Bun}_G)_{\text{cusp}} & \xleftarrow{\mathbb{L}_{G, \text{cusp}}^{\text{L}}} & \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{red}})
 \end{array}$$

$$\text{where } e_U^{\text{L}} \circ (j_U^* \circ \text{Poinc}_!^{\text{Vac}}) = e^{\text{L}}(\text{Poinc}_!^{\text{Vac}})$$

$$\begin{array}{ccc}
 & \text{Vect} & \\
 \swarrow & & \searrow \\
 (e_U^{\text{L}})^{\text{conj}} \circ \mathbb{D}_U \circ (j_U^* \circ \text{Poinc}_!^{\text{Vac}}) & & \mathcal{O}_{\text{LocSys}_{\check{G}}^{\text{irred}}} \\
 \downarrow & & \downarrow \\
 \text{D-mod}(\text{Bun}_G)_{\text{cusp}} & \xleftarrow{(\mathbb{L}_{G, \text{cusp}}^{\text{L}})^{\text{conj}}} & \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{red}})
 \end{array}$$

$$(e_U^{\text{L}})^{\text{conj}} \circ \mathbb{D}_U \circ (j_U^* \circ \text{Poinc}_!^{\text{Vac}}) = e^{\text{L}}(\text{Poinc}_*^{\text{Vac}})$$

■

### 5.3 Ambidexterity-II(Lin Chen)

We have seen the previous section half of the ambidexterity. Next, we divide into three goals:

**Goal 1:** We show  $\text{KL}(G)_{\text{Ran}} \xrightarrow{\text{Loc}} \text{D-mod}(\text{Bun}_G) \xrightarrow{j^*} \text{D-mod}(U)$  is a Verdier quotient, as is mentioned before.

**Goal 2:** We want to show the second half of Theorem 5.6,  $\mathbb{L}_{G, \text{cusp}}^{\vee} \cong \mathbb{L}_{G, \text{cusp}}^{\text{R}}$ .

**Goal 3:** We want to describe the stalks of  $\mathcal{A}_{G, \text{cusp}}$  at  $\sigma \in \text{LocSys}_{\check{G}}^{\text{irred}}$ , we recall  $\mathcal{A}_{G, \text{cusp}} \otimes - \stackrel{\text{def}}{=} \mathbb{L}_{G, \text{cusp}} \circ \mathbb{L}_{G, \text{cusp}}^{\text{L}}$ . In fact, we will see that  $\mathcal{A}_{G, \text{cusp}}|_{\sigma}$  is some singular homology  $\mathcal{C}_{\bullet}(-)$ .

**Goal 1:** Recall the general paradigm  $C \xrightleftharpoons[F^R]{F} D ; C^\vee \xrightleftharpoons[F^\vee]{(F^\vee)^\perp = (F^R)^\vee} D^\vee$ , where  $(F^\vee)^\perp = (F^R)^\vee$  is called  $F^{\text{conj}}$  in the previous section. For  $\text{unit}_1 \in C \otimes C^\vee$  and  $\text{unit}_2 \in D \otimes D^\vee$ , we can construct  $F \otimes F^{\text{conj}}(\text{unit}_1) \rightarrow \text{unit}_2$ . Moreover, we can show the following are equivalent (proof leaves as an exercise):

$$F \text{ is a (Verdier) quotient} \Leftrightarrow F^{\text{conj}} \text{ is a (Verdier) quotient} \Leftrightarrow F \otimes F^{\text{conj}}(\text{unit}_1) \xrightarrow{\cong} \text{unit}_2 \Leftrightarrow F \circ F^R \cong \text{Id}$$

**Proposition 5.1.**  $j^* \circ \text{Loc} : \text{KL}(G)_{\text{Ran}} \rightarrow \mathcal{D}\text{-mod}(U)$  is a quotient for any quasi-compact open  $U \subseteq \text{Bun}_G$ .

*Proof.* We first look at the following diagram:

$$\begin{array}{ccccc} \text{Rep}(\mathcal{L}^+G)_{\text{Ran}} & \xrightarrow{\text{Loc}^{\text{QCoh}}} & \text{QCoh}(\text{Bun}_G) & \xrightarrow{j^*} & \text{QCoh}(U) \\ \downarrow \text{ind} & & \downarrow \text{ind} & & \downarrow \text{ind} \\ \text{KL}(G)_{\text{Ran}} & \xrightarrow{\text{Loc}} & \mathcal{D}\text{-mod}(\text{Bun}_G) & \xrightarrow{j^*} & \mathcal{D}\text{-mod}(U) \end{array}$$

We can see  $j^* \circ \text{Loc}^{\text{QCoh}}$  is the given by first  $*$ -pullback then  $\int_{\text{Ran}}$  along the diagram  $\mathbf{B}\mathcal{L}^+G \leftarrow U \times \text{Ran} \rightarrow \text{Ran}$ ,  $*$ -pullback admits a (continuous) right adjoint as  $U$  is quasi-compact, thus it preserves compact objects, also  $\int_{\text{Ran}}$  admits continuous right adjoint since  $\text{Ran}$  is a union of proper schemes. So we see that  $j^* \circ \text{Loc}^{\text{QCoh}}$  preserves compact objects. Combining the fact that  $\text{ind}$  preserves compactness, we get  $j^* \circ \text{Loc}$  preserves compact objects. So we can apply the general paradigm. Notice that  $\text{KL} \cong \text{KL}^\vee$  and  $\mathcal{D}\text{-mod}(U) \cong \mathcal{D}\text{-mod}(U)^\vee$ . We claim that the functor  $j^* \circ \text{Loc}$  is self-conjugate. To wit, we consider the localization functor  $\text{Loc}^{l/r} : \mathfrak{h}\text{-mod}^K \rightarrow \mathcal{D}\text{-mod}^{l/r}(\tilde{\mathcal{Y}}/K)$ , where the superscript  $l/r$  is to distinguish left and right  $\mathcal{D}$ -modules. Then we have conjugation  $\text{Loc}^l \xleftrightarrow{\text{conj}} \text{Loc}^r$ .

We can use the paradigm and the criterion of being a quotient, we only need to show

$$(j^* \otimes j^*)(\text{Loc} \otimes \text{Loc})(\text{unit}_{\text{KL}}) \xrightarrow{\cong} \Delta_{*,\text{dR}}(\omega_U)$$

which is equivalent to show  $(\text{Loc} \otimes \text{Loc})(\text{unit}_{\text{KL}}) \xrightarrow{\cong} \Delta_{*,\text{dR}}(\omega_{\text{Bun}_G})$ .

**Remark 5.2.** (i)  $\text{unit}_{\text{KL}}$  is a chiral algebra of differential operators (CADO), where

$$\mathcal{D}\text{-mod}(\mathcal{L}G) \xrightarrow{\Gamma_{\text{IndCoh}}} \mathcal{L}\mathfrak{g}\text{-mod} \otimes \mathcal{L}\mathfrak{g}\text{-mod}; \quad \delta_{\mathcal{L}^+G} \mapsto \text{CADO} = \text{unit}_{\text{KL}} \in \text{KL} \otimes \text{KL}$$

that sends  $(\mathcal{L}^+G \times \mathcal{L}^+G)$ -invariant modules to  $(\mathcal{L}^+G \times \mathcal{L}^+G)$ -integrable modules.

(ii)  $j^* \circ \text{Loc}^{\text{QCoh}} : \text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{QCoh}(U)$ , then

$$\text{Loc}^{\text{QCoh}} \otimes \text{Loc}^{\text{QCoh}}(\text{unit}_{\text{Rep}(\mathcal{L}^+G)}) \cong \Delta_* \mathcal{O}_{\text{Bun}_G} \iff j^* \circ \text{Loc}^{\text{QCoh}} \text{ is a quotient}$$

that the LHS in the above holds would imply  $(\text{Loc} \otimes \text{Loc})(\text{unit}_{\text{KL}}) \xrightarrow{\cong} \Delta_{*,\text{dR}}(\omega_{\text{Bun}_G})$ . In fact, we can use

$$(\text{ind} \otimes \text{Id})(\text{unit}_{\text{Rep}(\mathcal{L}^+G)}) \cong (\text{Id} \otimes \text{oblv})(\text{unit}_{\text{KL}})$$

(iii) For the localization on spectral side,  $\text{Loc}^{\text{spec}} : \text{Rep}(\check{G})_{\text{Ran}} \rightarrow \text{QCoh}(\text{LocSys}_{\check{G}})$ , can show

$$\text{Loc}^{\text{spec}} \otimes \text{Loc}^{\text{spec}} \left( \text{unit}_{\text{Rep}(\check{G})} \right) \cong \Delta_* \mathcal{O}_{\text{LocSys}_{\check{G}}} \iff \text{Loc}^{\text{spec}} \text{ is a quotient}$$

(iv) We have use many times the equality  $\text{Fun}(\text{Sect}_{\mathcal{V}}(X, Z)) = C_{\bullet}^{\text{fact}}(X, \text{Fun}(\mathcal{L}_{\mathcal{V}}^+(Z)))$ , where  $X \rightarrow Z$  affine and  $Z \rightarrow \text{pt}$ , there is actually a relative version when  $Z \rightarrow \mathcal{Y}$  is affine. We could apply this to  $\check{B}\check{G} \rightarrow \check{B}\check{G} \times \check{B}\check{G}$  to obtain (iii) and apply to  $W(\mathbf{B}G) \rightarrow W(\mathbf{B}G \times \mathbf{B}G)$  for (ii), where  $W$  denotes the Weil restriction along  $X \rightarrow X_{\text{dR}}$ .  
(v) We have shown that  $\text{Loc}$  is (almost) a quotient, and one may wonder if  $\text{Poinc}_! : \text{Whit}^!(G)_{\text{Ran}} \rightarrow \mathcal{D}\text{-mod}(\text{Bun}_G)$  is a quotient. Namely one expect  $\text{Poinc}_! \otimes \text{Poinc}_*(\text{unit}_{\text{Whit}}) \xrightarrow{\cong} \Delta_*(\omega_{\text{Bun}_G})$ , but we do not know how to prove this. ■

**Goal 2:** We can look at the diagram

$$\begin{array}{ccc} \text{KL}(G)_{\text{Ran}} & \xleftarrow{\text{FLE}} & \text{IndCoh}^* \left( \text{Op}_{\check{G}}^{\text{mon-free}} \right)_{\text{Ran}} \\ \downarrow \text{Loc}^{\text{cusp}} & & \downarrow \text{Poinc}^{\text{irred}} \\ \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} & \xrightarrow{\mathbb{L}_{G, \text{cusp}}} & \text{QCoh} \left( \text{Op}_{\check{G}}^{\text{mon-free}} \right)_{\text{Ran}} \end{array}$$

We claim that in the above diagram, every category is self-dual and every functor is self-conjugate. In particular, we know  $\mathbb{L}_{G, \text{cusp}}^{\text{conj}} = (\mathbb{L}_{G, \text{cusp}}^{\text{R}})^{\vee}$ , then the self-conjugacy implies  $\mathbb{L}_{G, \text{cusp}}^{\text{R}} \cong (\mathbb{L}_{G, \text{cusp}})^{\vee}$ , as we want. So let us prove for every component. We have already

(i)  $\text{KL} \cong \text{KL}^{\vee}$ ; (ii)  $\mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} \cong \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}}^{\vee}$  due to the self-duality of  $\mathcal{D}\text{-mod}(U)$ ;

(iii)  $\text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}}) \cong \text{QCoh}(\text{LocSys}_{\check{G}}^{\text{irred}})^{\vee}$  due to the Verdier duality in  $\text{QCoh}$ .

(iv) We can cheat (by equivalence of  $\text{FLE}$ ) that (i) implies  $\text{IndCoh}^* \left( \text{Op}_{\check{G}}^{\text{mon-free}} \right)_{\text{Ran}} \cong \text{IndCoh}^* \left( \text{Op}_{\check{G}}^{\text{mon-free}} \right)_{\text{Ran}}^{\vee}$ .  $\text{IndCoh}$  for finite type is self-dual and in general, one would like to see  $\text{IndCoh}^*(\mathcal{Y}) \cong (\text{IndCoh}^!(\mathcal{Y}))^{\vee}$ . It is not hard to see that  $\text{IndCoh}^!(\mathcal{Y})$  acts on  $\text{IndCoh}^*(\mathcal{Y})$ . To obtain the self-duality, we need to find a particular object  $\omega_{\mathcal{Y}}^{\text{fake}} \in \text{IndCoh}_*(\mathcal{Y})$ . For functors, we have already:

(v)  $\text{FLE}$  is self-conjugate; (vi)  $\text{Loc}^{\text{cusp}}$  is self-conjugate. To our purpose, it suffices to show

(vii)  $\text{Poinc}^{\text{irred}}$  is self-conjugate.

$$\begin{array}{ccccc} & \text{Op}_{\check{G}, \text{Ran}}^{\text{mon-free, glob, irred}} & & & \\ & \swarrow \pi' & \wedge & \searrow s' & \\ \text{LocSys}_{\check{G}}^{\text{irred}} & & & & \text{Op}_{\check{G}, \text{Ran}}^{\text{mon-free, glob}} \\ & \searrow \pi & & \swarrow s & \\ & \text{LocSys}_{\check{G}} & & & \text{Op}_{\check{G}, \text{Ran}}^{\text{mon-free}} \end{array}$$

By definition,  $\text{Poinc}^{\text{irred}} = \pi'_* \circ (s')^*$  and thus  $(\text{Poinc}^{\text{irred}})^{\text{conj}} = \pi'_! \circ (s')^!$ . Now  $(s')^!$  is in  $\text{IndCoh}^!$ , but after passing to  $\text{IndCoh}^*$ , it becomes  $(s')^*$ . It remains to compare  $\pi'_!$  and  $\pi'_*$ . We conclude with the following Lemma.

**Lemma 5.3.**  $\pi' : \text{Op}_{\check{G}, \text{Ran}}^{\text{mon-free, glob, irred}} \rightarrow \text{LocSys}_{\check{G}}^{\text{irred}}$  is proper.

We give an idea of proving the Lemma: by definition of an oper, it contains data of  $\check{B}$ -reduction with non-vanishing (of some) sections. The moduli of  $\check{B}$ -reduction is proper, therefore, we only need to deal with non-vanishing condition. In fact, the non-vanishing condition is satisfied automatically over irreducible locus.

Now Lemma 5.3 implies Theorem 5.6 (ii), combine with Theorem 5.6 (i), we have the ambidexterity.

**Goal 3:** We want to describe the stalk of  $\mathcal{A}_{G, \text{cusp}}$ , in fact,  $\mathcal{A}_{G, \text{cusp}}|_{\sigma} = \mathbf{C} \cdot ((\pi')^!(\sigma))$ . We can consider

$$\begin{array}{ccc}
 \text{KL}(G)_{\text{Ran}} & \xrightleftharpoons[\text{FLE}^{-1}]{\text{FLE}} & \text{IndCoh}^* \left( \text{Op}_{\check{G}}^{\text{mon-free}} \right)_{\text{Ran}} \\
 \text{Loc}^{\text{cusp}} \downarrow \uparrow \Gamma^{\text{cusp}} & & \text{Poinc}^{\text{irred}} \downarrow \uparrow \text{coeff}^{\text{irred}} \\
 \mathcal{D}\text{-mod}(\text{Bun}_G)_{\text{cusp}} & \xrightleftharpoons[\mathbb{L}_{G, \text{cusp}}^{\text{R}}]{\mathbb{L}_{G, \text{cusp}}} & \text{QCoh} \left( \text{Op}_{\check{G}}^{\text{mon-free}} \right)_{\text{Ran}}
 \end{array}$$

Using ambidexterity,  $\mathcal{A}_{G, \text{cusp}} \otimes - = \mathbb{L}_{G, \text{cusp}} \circ \mathbb{L}_{G, \text{cusp}}^{\text{L}} \cong \mathbb{L}_{G, \text{cusp}} \circ \mathbb{L}_{G, \text{cusp}}^{\text{R}}$ . According to the above diagram:

$$\begin{aligned}
 \mathbb{L}_{G, \text{cusp}} \circ \mathbb{L}_{G, \text{cusp}}^{\text{R}} &\cong (\mathbb{L}_{G, \text{cusp}} \circ \text{Loc}^{\text{cusp}}) \circ (\Gamma^{\text{cusp}} \circ \mathbb{L}_{G, \text{cusp}}^{\text{R}}) \quad \Gamma^{\text{cusp}} \text{ is fully faithful} \\
 &\cong (\text{Poinc}^{\text{irred}} \circ \text{FLE}) \circ (\text{FLE}^{-1} \circ \text{coeff}^{\text{irred}}) \cong \text{Poinc}^{\text{irred}} \circ \text{coeff}^{\text{irred}}
 \end{aligned}$$

Thus,  $\mathcal{A}_{G, \text{cusp}} \cong \text{Poinc}^{\text{irred}} \circ \text{coeff}^{\text{irred}} \left( \mathcal{O}_{\text{LocSys}_{\check{G}}^{\text{irred}}} \right) = \pi'_! \circ (s')^* \circ s'_* \circ (\pi')^! \left( \mathcal{O}_{\text{LocSys}_{\check{G}}^{\text{irred}}} \right)$ , where we recall

$$\text{LocSys}_{\check{G}}^{\text{irred}} \xleftarrow{\pi'} \text{Op}_{\check{G}, \text{Ran}}^{\text{mon-free, glob, irred}} \xrightarrow{s'} \text{Op}_{\check{G}, \text{Ran}}^{\text{mon-free}}$$

**Proposition 5.2.** We have  $\pi'_! \circ (s')^* \circ s'_* \circ (\pi')^! \cong \pi'_! \circ (\pi')^!$

**Proposition 5.3.** (i) We denote  $\left( \text{LocSys}_{\check{G}}^{\text{irred}} \right)^{\wedge}$  the formal completion, then  $\pi'$  factors as

$$\begin{array}{ccccc}
 \text{Op}_{\check{G}, \text{Ran}}^{\text{mon-free, glob, irred}} & \xrightarrow{\iota} & \left( \text{LocSys}_{\check{G}}^{\text{irred}} \right)^{\wedge} & \xrightarrow{\varpi} & \text{LocSys}_{\check{G}}^{\text{irred}} \\
 \uparrow & & \uparrow & & \downarrow \psi \\
 \text{Op}_{\sigma, \text{Ran}} & \longrightarrow & (\text{Op}_{\sigma, \text{Ran}})_{\text{dR}} & \longrightarrow & \sigma
 \end{array}$$

(ii)  $\text{End}_{\text{IndCoh}^*}(\omega_{\text{Op}_{\sigma, \text{Ran}}}) \cong \text{End}_{\mathcal{D}\text{-mod}}(\omega_{\text{Op}_{\sigma, \text{Ran}}}) = \mathbf{C} \cdot (\mathcal{O}_{\sigma, \text{Ran}})$ .

We can consider a general paradigm: assume  $Z \rightarrow X$  is affine  $\mathcal{D}$ -scheme, for example  $\text{Op}_{\check{G}} \rightarrow \text{B}\check{G}$ , then

$$\pi_! s^* s_* \pi^! \cong \pi_! \pi^! \text{ and } \pi_! \pi^! \cong \pi_{\text{dR}, !} \pi_{\text{dR}}^!$$



where

$$\begin{array}{ccccc}
 (\mathrm{Sect}_{\nabla}(X - \underline{x}, Z)_{\mathrm{Ran}})_{\mathrm{dR}} & \longleftarrow & \mathrm{Sect}_{\nabla}(X - \underline{x}, Z)_{\mathrm{Ran}} & & \\
 \pi_{\mathrm{dR}} \downarrow & \nearrow \pi & & \searrow s & \\
 \mathrm{pt} & & & & \mathcal{L}_{\nabla}(Z)_{\mathrm{Ran}}
 \end{array}$$

#### 5.4 End of the argument(Kevin Lin)

To summarize, we have

$$\begin{array}{ccc}
 \mathcal{D}\text{-mod}(\mathrm{Bun}_G) & \xrightarrow{\mathbb{L}_G} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}}) \\
 \mathrm{coeff}_*^{\mathrm{Vac}} \downarrow & & \downarrow \Gamma \\
 \mathbf{Vect} & \xlongequal{\quad} & \mathbf{Vect}
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 \mathbf{Vect} & \xlongequal{\quad} & \mathbf{Vect} \\
 \mathrm{Poinc}_!^{\mathrm{Vac}} \downarrow & & \downarrow \mathcal{O}_{\mathrm{LocSys}_{\check{G}}} \\
 \mathcal{D}\text{-mod}(\mathrm{Bun}_G) & \xleftarrow{\mathbb{L}_G^L} & \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_{\check{G}})
 \end{array}$$

We denote by  $\mathcal{A}_G$  the algebra in  $\mathrm{QCoh}(\mathrm{LocSys}_{\check{G}})$  producing the monad  $\mathbb{L}_G \circ \mathbb{L}_G^L$ , we need to show that  $\mathcal{O}_{\mathrm{LocSys}_{\check{G}}} \rightarrow \mathcal{A}_G$  is an isomorphism. Its restriction along  $(\iota^{\mathrm{red}})^{\wedge} : (\mathrm{LocSys}_{\check{G}})_{\mathrm{LocSys}_{\check{G}}^{\mathrm{red}}}^{\wedge} \hookrightarrow \mathrm{LocSys}_{\check{G}}$  is known to be okay. It remains to study  $\mathcal{O}_{\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}} \rightarrow \mathcal{A}_{G, \mathrm{cusp}}$ . Some facts about  $\mathcal{A}_{G, \mathrm{cusp}}$ :

**Proposition 5.4.** (1)  $\mathcal{A}_{G, \mathrm{cusp}}$  is a classical vector bundle.

(2) It admits a connection with finite monodromy.

*Proof of Proposition 5.4 (1).* In section 5.2, Dennis explained  $\mathcal{A}_{G, \mathrm{cusp}}$  is a self-dual perfect complex (as a consequence of ambidexterity) and in section 5.3, Lin explained that  $(\iota_{\sigma})^* \mathcal{A}_{G, \mathrm{cusp}} \cong \mathbb{C} \cdot \left( \begin{smallmatrix} \text{generic oper} \\ \text{structure on } \sigma \end{smallmatrix} \right)$  ■

**Remark 5.3.** It was shown by Arinkin that the space of generic oper structure on  $\sigma$  is non-empty then by Beraldo-Kazhdan-Schlank that it is contractible for classical groups.

Next, for simplicity, we assume  $\check{G}$  is simply-connected and  $g \geq 2$ . Under such assumption, we introduce more facts:

**Proposition 5.5.** (1)  $\mathrm{LocSys}_{\check{G}}$  is a classical locally complete intersection (lci) stack. [BD]

(2)  $\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}} \subseteq \mathrm{LocSys}_{\check{G}}$  has  $\mathrm{codim} \geq 2$ ; (3)  $\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}$  is simply connected; (4)  $\Gamma(\mathrm{LocSys}_{\check{G}}, \mathcal{A}_G) = \mathbb{k}$ .

Why does these properties imply that  $\mathcal{O}_{\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}} \cong \mathcal{A}_{G, \mathrm{cusp}}$ ? Proposition 5.4 tells us that  $\mathcal{A}_{G, \mathrm{cusp}}$  is given by some

local system on  $\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}$  Proposition 5.5(3) implies that  $\mathcal{A}_{G, \mathrm{cusp}} \cong \left( \mathcal{O}_{\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}} \right)^{\oplus \mathrm{rk}}$  should be trivial, whose

rank is undetermined. We have thus  $\Gamma(\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}, \mathcal{A}_{G, \mathrm{cusp}}) \cong \Gamma(\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}, \mathcal{O}_{\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}})^{\oplus \mathrm{rk}}$ .

Next, Proposition 5.5(1)+(2) imply that

$$H^0(\mathrm{LocSys}_{\check{G}}, \mathcal{A}_G) \cong H^0(\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}, \mathcal{A}_{G, \mathrm{cusp}}) \cong H^0(\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}, \mathcal{O}_{\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}})^{\oplus \mathrm{rk}}$$

And by Proposition 5.5, it is nothing but  $\mathbb{k}$ . So we know  $\mathrm{rk} = 1$  and  $\mathcal{A}_{G, \mathrm{cusp}} \cong \mathcal{O}_{\mathrm{LocSys}_{\check{G}}^{\mathrm{irred}}}$ .

(Sketch) Proof of Proposition 5.5. For (2) and (3), we can forget the connection on  $\text{LocSys}_{\check{G}}$  to get a  $\check{G}$ -torsor in  $\text{Bun}_{\check{G}}$ , inside of which we have  $\text{Bun}_{\check{G}}^{\text{stable}}$  of stable locus, it is of codim  $\geq 2$  in  $\text{Bun}_{\check{G}}$ . Denote by  $\text{LocSys}_{\check{G}}^{\text{stable}}$  the fiber product, it is an affine fibration over  $\text{Bun}_{\check{G}}^{\text{stable}}$  also it contained in irreducible locus  $\text{LocSys}_{\check{G}}^{\text{irred}}$ :

$$\begin{array}{ccccc} \text{LocSys}_{\check{G}}^{\text{stable}} & \hookrightarrow & \text{LocSys}_{\check{G}}^{\text{irred}} & \hookrightarrow & \text{LocSys}_{\check{G}} \\ \downarrow \text{affine fibration} & & & & \downarrow \\ \text{Bun}_{\check{G}}^{\text{stable}} & \longrightarrow & & & \text{Bun}_{\check{G}} \end{array}$$

$\text{LocSys}_{\check{G}}^{\text{stable}} \rightarrow \text{Bun}_{\check{G}}^{\text{stable}}$  is non-trivial fibration. In fact, the obstruction to  $\mathcal{P}_{\check{G}} \in \text{Bun}_{\check{G}}$  admitting a connection is the Atiyah classes

$$\alpha_{\mathcal{P}_{\check{G}}} \in H^1 \left( X, \Omega_X \otimes (\mathcal{P}_{\check{G}} \times^{\check{G}} \mathfrak{g}) \right) \cong H^0 \left( X, \left( \mathcal{P}_{\check{G}} \times^{\check{G}} \mathfrak{g} \right)^* \right)$$

For (3),  $\text{Bun}_{\check{G}}$  is simply-connected as we assume  $\check{G}$  is simply-connected. By codimension 2 assertion, we get  $\text{Bun}_{\check{G}}^{\text{stable}}$  is also simply-connected. And so does  $\text{LocSys}_{\check{G}}^{\text{stable}}$ , we use again codimension 2 assertion, then we can get  $\text{LocSys}_{\check{G}}^{\text{irred}}$  is simply connected.

For (4), we use definition of  $\mathcal{A}_G$  as monad  $\mathbb{L}_G \circ \mathbb{L}_G^{\mathbb{L}}$ , thus

$$\begin{aligned} \Gamma(\text{LocSys}_{\check{G}}, \mathcal{A}_G) &\cong \text{Maps}_{\text{IndCoh}(\text{LocSys}_{\check{G}})}(\mathcal{O}_{\text{LocSys}_{\check{G}}}, \mathbb{L}_G \circ \mathbb{L}_G^{\mathbb{L}}(\mathcal{O}_{\text{LocSys}_{\check{G}}})) \cong \text{End}_{\mathcal{D}\text{-mod}(\text{Bun}_G)}(\mathbb{L}_G^{\mathbb{L}}(\mathcal{O}_{\text{LocSys}_{\check{G}}})) \\ &\cong \text{End}_{\mathcal{D}\text{-mod}(\text{Bun}_G)}(\text{Poinc}_!^{\text{Vac}}) \end{aligned}$$

Recall  $\text{Poinc}_!^{\text{Vac}}$  is pushforward of certain sheaf on  $\text{Bun}_N^{\Omega_X}$  to  $\text{Bun}_G$ , which factors through  $\text{Bun}_N^{\Omega_X}/T$ .

$$\begin{array}{ccccc} \text{Bun}_N^{\Omega_X} & \longrightarrow & \text{Bun}_N^{\Omega_X}/T & \hookrightarrow & \text{Bun}_G \\ \downarrow & & \downarrow & & \\ \mathbb{A}^r & \xrightarrow{\pi} & \mathbb{A}^r/T & & \\ \downarrow \text{sum} & & & & \\ \mathbb{A}^1 & & & & \end{array}$$

where  $\text{Bun}_N^{\Omega_X} \rightarrow \mathbb{A}^r$  has contractible fibers and  $\text{Bun}_N^{\Omega_X}/T \hookrightarrow \text{Bun}_G$  is locally closed. Then the End group is:

$$\text{End}_{\mathcal{D}\text{-mod}(\mathbb{A}^r/T)}(\pi_! \circ \text{sum}^*(\text{exp}))$$

For  $G = \text{SL}_2$ ,  $\text{Bun}_N^{\Omega_X} = \left\{ \Omega_X^{\frac{1}{2}} \rightarrow \mathcal{E} \rightarrow \Omega_X^{-\frac{1}{2}} \right\} \cong H^1(X, \Omega_X^1) \cong \mathbb{A}^1$ . In this case, we are looking at  $\pi_!(\text{exp}) \cong j_*(\mathbf{k})$ , where  $j : \mathbb{G}_m/\mathbb{G}_m = \text{pt} \hookrightarrow \mathbb{A}^1/\mathbb{G}_m$  and the End group is exactly  $\mathbf{k}$ .  $\blacksquare$

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